Shaking the Traditional Order: Women's Conversion to New Christian Churches in Sub-Saharan Africa*

Pablo Álvarez-Aragón‡, Catherine Guirkinger†, Jean-Philippe Platteau†

‡ *University of Bologna and University of Namur*

† *University of Namur*

September 17, 2026

Abstract

In sub-Saharan Africa, women constitute the majority of new Christian membership (including Evangelical and Pentecostal denominations), and this gender gap exceeds that of any other religion. Existing explanations for conversion to these churches emphasize demand for mutual help or informal insurance. We instead argue that emancipation is central: these churches provide services that support women's economic advancement and likely help them challenge patriarchal norms. Using experimental data from Benin, we show that women randomly offered an economic opportunity become more likely to convert. We argue that this is partly because these churches help counter witchcraft threats, a risk that disproportionately targets economically successful women. Across sub-Saharan Africa, women exposed to local positive economic shocks — using changes in the international prices of locally-suitable crops or the opening of World Bank projects with a gender focus — are more likely to be new Christian, particularly where witchcraft beliefs are widespread. Membership in these churches is also associated with women working more, having fewer children, and exercising greater decision-making power, and with both women and men rejecting traditional beliefs, rituals, and authorities.

Keywords: Religious conversion, Sub-Saharan Africa, Gender, Kinship
JEL Classification: O12, Z12, J16

---

*P. Álvarez-Aragón is affiliated to the University of Bologna, Italy, and to the University of Namur, Belgium. C. Guirkinger and Jean-Philippe Platteau are affiliated to the University of Namur, Belgium. Emails: pablo.alvarezaragon@unibo.it; catherine.guirkinger@unamur.be; jean-philippe.platteau@unamur.be. For useful comments and suggestions, we thank Siwan Anderson, Jean-Marie Baland, Patrick François, Nathan Nunn, Mathilde Sage and participants at seminars and conferences at UCLouvain, Namur, the Vodoun School of Economics Conference at the Université d'Abomey-Calavi, the International Conference in Development Economics at Paris Nanterre University, Simon Fraser University, the III Spanish Workshop in Development Economics, and the CEPR Development Economics annual symposium. Aristophane Aplogan has been of great help during qualitative interviews. We thank Tatiana Patte for excellent research assistance with the classification of World Bank aid projects. Catherine Guirkinger gratefully acknowledges financial support from the European Research Council under the H2020 research and innovation programme/ERC grant agreement 759294. Pablo Álvarez-Aragón acknowledges financial support from an FNRS/FRESH doctoral fellowship and from the European Research Council under the H2020 research and innovation programme/ERC Grant Agreement 101170376.

1. Introduction

The rapid expansion in both poor and rich countries of new Christian churches that include Evangelical and Pentecostal denominations has received widespread press coverage (see, e.g., The Economist, 2021, 2023) and has become the focus of a growing body of scholarly research (Auriol et al., 2020; Hersey, 2024; Alfonsi et al., 2024; Costa et al., 2022). Yet, despite the noticeable gender gap in membership observed in some regions – women are far more likely than men to join these churches in sub-Saharan Africa (SSA) and South East Asia, and the gap is much larger than for any other major religion – it is puzzling how little the gender dynamics of conversion have been explored to date.

In sub-Saharan African countries with a new Christian presence, women are 4 to 5 percentage points more likely than men, on average, to be members of the affiliated churches, which amounts to a 13-20 percent gap when compared to the average male membership rate. This gap is much higher than in any other Christian denomination. Such statistics are especially surprising in light of the fact that, in several countries including the United States, evangelical movements are associated with gender conservatism (Smith et al., 2025: 244; Becker et al., 2025).[1]

The present paper aims to understand the gender gap in conversion in SSA, a region of particular interest because it accounts today for most of the global growth in the membership of Christian churches (Hackett et al., 2025) and where individuals are most likely to stress the important role of religion in their lives (Barro et al., 2025; Lowes et al., 2025). In this endeavor, we assess whether the gender gap in conversion to new Christian churches is at least partly driven by women's search for economic emancipation in the context of a constraining social order.

This emancipation hypothesis stands in contrast with the existing economic literature, which explains conversion through the material vulnerability of converts rather than their pursuit of advancement: it emphasizes the economic services that new Christian churches (NCCs) provide to the poor, insurance and mutual help benefits in particular. In an experimental setting in Ghana, Auriol et al. (2020) thus show that investment in these churches declines when alternative insurance mechanisms expand. Moreover, Hersey (2024) highlights the risk-sharing role of new Christian churches among urban migrants in Ghana while Alfonsi et al. (2024) establish that higher levels of education are associated with lower conversion rates in Kenya. In the same perspective, Costa et al. (2022) find that conversion is counter-cyclical: economic hardship triggers conversion in

[1]Interestingly, the latest estimates show that, in the United States of America, the gender gap in Evangelicalism is smaller than in other brands of Protestantism and roughly similar to what it is in Catholicism (Smith et al., 2025: 98).

Brazil. Taken together, this logic suggests a straightforward explanation for the gender gap in conversion: because women are poorer, more exposed to negative shocks, and more likely to lack insurance than men, they place greater value on the poor-specific services offered by new Christian churches.

A different perspective emerges from research in other social sciences, however. There, non-economic dimensions that matter for individuals seeking economic empowerment are brought to center stage. These include: (i) the rejection of traditional and patriarchal authority structures and rituals (Platteau, 2019; Onyinah, 2009); (ii) the promotion of values aligned with wealth accumulation, such as the doctrine of God-ordained prosperity (Haynes, 2012; Heuser, 2016); and (iii) spiritual services offering healing and protection against "spiritual attacks", which are often attributed to malevolent forces and frequently targeted at economically successful individuals (Ellis and Ter Haar, 2004; Platteau, 2014, 2019), particularly women (Federici, 2018; Miguel, 2005). These observations suggest another possible rationale for the gender gap in membership for new Christian churches: women aspiring to economic emancipation but facing constraints within the prevailing social order may turn to new Christian churches for supportive services, new values and modes of conduct. The implication is that conversion moves are not necessarily countercyclical but may become more prevalent in the presence of new economic opportunities. This said, the two approaches are not mutually exclusive: for women pursuing economic emancipation, the insurance and mutual-help services of new Christian churches may be part of what makes them attractive, alongside the values and protections that support advancement.

We begin by presenting first-hand evidence from a randomized controlled trial (RCT) conducted among 1,009 households in Southern Benin, in which women were randomly offered a pineapple farming opportunity. Women in the treatment group are 6.4 percentage points more likely to have converted to a new religion (a 19% increase over the control-group mean), and conversions to new Christian churches account for most of this effect (+4.7 pp). However, we find no clear advantage of these churches in terms of strictly economic services. Instead, they appear to be particularly valued for their capacity to provide healing and protection against spiritual attacks in a situation where economically successful women are considered especially vulnerable to magical threats. Revealingly, treated women are 7.9 pp more likely to report that their husband opposed — or would oppose — their own conversion to a new Christian church (a 20% increase), and 4 pp more likely to believe that their husband has contacted a witchdoctor to hinder their business activities (a 53% departure from the control mean). When a woman achieves economic success and begins to challenge traditional gender roles, she often encounters household and family resistance, frequently expressed through the

stress and psychosomatic disorder inflicted on her by spells or curses alleged to be magical. Religious conversion supplies a convenient response to these threats insofar as new Christian churches claim to possess effective cures to dispel interpersonal acts of hostility disguised under spiritual attacks.

The broader relevance of our findings for SSA is established on the basis of cross-country data, combining 147 Demographic and Health Survey rounds from 39 SSA countries and covering almost two million respondents. We investigate whether the success of new Christian churches among women can be attributed to improved economic prospects, using two measures of local economic opportunity and difference-in-differences identification at the level of $0.5^\circ \times 0.5^\circ$ grid cells (approximately 55×55 km^2 at the equator), exploiting within-cell variation over time. The first measure exploits exogenous variation in the suitability-weighted international price of locally grown cash crops. A one-standard-deviation increase in the lagged (standardized) cash-crop price index raises the probability of new Christian membership by 1.8 percentage points (about 8% of the sample mean), with the effect larger for women (2.0 pp) than for men (1.5 pp). Crucially, this effect is strongly amplified in regions where witchcraft beliefs are widespread: in grid-cells one standard deviation above the sample mean in witchcraft prevalence, the crop-price effect on women's new Christian affiliation rises to approximately 2.6 pp — more than double the effect in low-witchcraft areas. A similar pattern emerges from our second measure, which exploits the deployment of 1,717 World Bank projects geo-coded across 15,268 SSA sites over 1995–2014, of which nearly 200 explicitly target gender gaps. While these projects have no uniform average effect on new Christian affiliation, their impact is again shaped by the local prevalence of witchcraft beliefs: in high-witchcraft grid-cells, the differential impact of World Bank gender-targeted projects on women relative to men reaches 1.5 pp (6% of the sample mean). These patterns are consistent with the experimental evidence from Benin and point to the spiritual protection offered by new Christian churches as a key mechanism driving women's affiliation with new Christian churches when new economic opportunities arise.

In addition, systematic data from cross-country surveys support the view that new Christian churches facilitate individual economic emancipation, particularly among women. Female members of these churches thus exhibit higher agency through increased labor force participation, lower fertility, and greater decision-making power. It is also striking that more than other believers or members of other Christian denominations, members of new Christian churches tend to reject traditional supernatural beliefs, rituals, and authority structures. Finally, they report a feeling of security under the protection of the Holy Spirit and they embrace a doctrine of God-ordained prosperity.

In addition to the literature on new Christian churches discussed above, our effort

partakes of three main strands of literature. First, it contributes to the body of studies dealing with the (economic) determinants of religious participation and conversion, and their gendered aspects. In line with the literature, we see conversion as a rational decision whereby individuals dynamically compare the costs and benefits of such a change. The costs typically consist of a fixed cost of entry into a religious community often conceived as a club (Iannaccone, 1992, 1994), or the cost of switching and deviating from one's ideal religion (Barro et al., 2010; Barro and McCleary, 2024). As for the benefits, they include economic, spiritual, and psychological aspects (Binzel and Carvalho, 2017, and for reviews, see Aldashev and Platteau, 2014; Iyer, 2016; Carvalho et al., 2019). In their recent survey paper, Becker et al. (2025) review the evidence accounting for between-gender differences in religiosity. Rather than comparing gender gaps across strands of Christianity, they seek to explain differences in religiosity overall — distinguishing Christianity from other religions — with an emphasis on how religious gender patterns shift as gender roles evolve. Our focus is different: we argue that conversion to new Christian churches in Africa may help women escape patriarchal control through the material and spiritual services these churches provide. The dimension of conflict within the family is therefore a core aspect of the (female) conversion dynamic.

Second, we offer insights into the effects of religion on long-term development (growth, education, state formation), an issue which has been largely investigated with reference to the Protestant Reformation in Western Europe (Van Zanden, 2009; Becker and Woessmann, 2008, 2009; Cantoni, 2015; Becker et al., 2016; Iyer, 2016; Cantoni et al., 2018; see also Becker et al., 2021 for a survey and Squicciarini, 2020 for a study of 19th century France). Our work is an addition to the narrow literature devoted to contemporary developing countries (e.g., McCleary and Barro, 2019; Buccione and Mello, 2020; Bernardelli et al., 2025; Solá, 2025). It suggests that the spread of new Christian churches in sub-Saharan Africa may have more in common with the spread of Protestantism in Europe than previously thought and, in particular, it supports the view that new Christianity helps dynamic individuals to prosper economically and to distance themselves from kin-based social structures.

Third, our paper relates to the work of social scientists who attempt to explain surges of witchcraft accusations (see, e.g., Evans-Pritchard, 1937, 1940; Geschiere, 1994; ter Haar, 2007). A dominant explanation is that witchcraft beliefs emerge to provide a culturally acceptable account of misfortune or inequality, frequently rooted in jealousy, envy, and social tensions within closely knit communities. In their recent work, Le Rossignol et al. (2025) show that those who *hold* traditional beliefs are stigmatized. Turning to gender, several studies have also documented that the vast majority of witchcraft attacks target women, who are often marginalized or oppressed members of their community (Miguel,

2005).[2] Our findings resonate with the work of Federici (2018), who, based on historical evidence from Europe and work on contemporary Africa, argues that witchcraft practices aim to preserve gender-based differences in social status, through the intimidation of women willing to change their condition. The magical cover serves to conceal the identity of concrete persons behind the attacks, who are frequently jealous relatives against whom open accusations would have devastating effects on the stability of the local social order.

The outline of the paper follows from the above-described progression of the argument. Section 2 highlights the significance of the shift from traditional to new Christian strands of religion in SSA and the main characteristics of new Christianity. In Section 3, we present the results of the RCT implemented in south Benin and their interpretation. In Section 4, relying on exogenous shocks to women's income, we show that these results are generalizable to the whole SSA region. We then investigate whether membership in new Christian churches is associated with higher levels of women's empowerment and more distancing from the traditional social order in Section 5. Finally, Section 6 concludes and draws broader implications regarding the issue of the role of religion in institutional change.

2. The changing picture of religion in Sub-Saharan Africa

Before proceeding, we define new Christianity and discuss how we measure it. Throughout this paper, "new Christian churches" refers to post-colonial, indigenously led denominations — Evangelical, Pentecostal, Apostolic, Charismatic, African Initiated Churches, and related movements — as distinct from mainline missionary denominations such as Catholic or Protestant churches. Measuring membership of these churches in large-scale survey data is not straightforward. For example, not all DHS country-waves distinguish new Christian churches from mainstream Protestant churches, as the former are closely related to the latter (we discuss differences in Section 2.3). We therefore work with three measures when using DHS data. Our preferred "grouped" measure uses the full DHS sample; in the eight countries where new Christians and Protestants are not separately coded, all respondents in that bundled category are recoded as new Christian. This maximizes coverage but may slightly overstate new Christian membership. As robustness checks, we report appendix results using a "restricted" measure — limiting the sample to country-waves where new Christians are separately identified — and a "raw" measure that retains the full sample without recoding. Full details on variable measures and

[2]For example, in the traditional camps that welcome (and protect) people accused of witchcraft in Northern Ghana, 498 out of the 539 people registered in 2021 were women (The Sanneh Institute, 2021).

"problematic" survey-waves are in Appendix B.

2.1. *The dramatic rise of new Christianity*

Over the past few decades, SSA has undergone a massive movement of religious conversion. It includes both a growing estrangement from African traditional religions and a noticeable shift from "mainline" (e.g., Catholic, Protestant) to new ("post-colonial") Christian churches (e.g., Evangelical, Pentecostal), which some authors refer to as the "New Reformation" (Jenkins, 2002; Thelen, 2017). Thus, when comparing the data reported in all the Demographic and Health Surveys before 2000 with those conducted after 2015, we find that the share of people reporting Catholicism as their main religion has remained rather stable (18.5% before 2000 against 17% after 2015), while the share of adherents to a new Christian church has shot up six-fold (about 3% before 2000 against 19.6% after 2015).[3] A recent study contends that as many as 40% of the continent's Christians identified as new Christians in 2020, not much below the share of mainline Catholic movements (Zurlo and Johnson, 2024). In this regard, Figure A1 in Appendix A shows the predicted probability of belonging to a given denomination/religion (new Christian, Protestant, Catholic, Muslim or Animist) over time.[4]

In a context where conversions are frequent, the existing large-scale survey data do not capture the full dynamic of the religious landscape. Our own data, composed of a representative sample of pineapple producers in Southern Benin (N= 1798, see 3.1 for details) shows that about 40 percent of the population have converted to another religion at some stage of their life. In converting, most of them leave a traditional religion (41% of women and 58% of men) or a mainline missionary church (40% of women and 28% of men). The gender differences in these proportions reflect the important gender gap existing among followers of traditional religions prior to conversion (detailed in the next section). Looking at the other end of conversion, most of the new converts join a new Christian church (57% of women and 54% of men), while a much smaller fraction of them join missionary churches (28% of women and 26% of men).

[3]These figures come from using the raw religious affiliation provided by the DHS data. If we remove country-waves where Protestants and new Christians are likely to be grouped together under the Protestant label, we observe a similar evolution: from 3% before 2000 to 17.5% after 2015.

[4]These estimates come from regressing raw religious affiliation, as provided by the DHS, on survey year with country fixed effects. The sum in each period does not add to one because not all religious categories are included (e.g., "no religion", "orthodox", "other religion" or "Christian"). Appendix A also shows alternative figures that yield very similar conclusions: Figure A2 removes country-waves where new Christians are not separately identified from Protestants; Figure A3 uses data from the World Christian Database (Zurlo and Johnson, 2024). Figure A4 reports religious affiliation by year of birth, using both the "raw" and the "restricted" measures. Finally, Figure A5 shows the evolution of new Christian membership by country (where there is a clearly identified new Christian presence).

2.2. *The gender bias of evangelical church membership*

Motivated by our qualitative research in southern rural Benin and the existing literature on women's prominent role in new Christian churches, this section evaluates whether the rise of these churches masks significant gender disparities in religious affiliation.

Table 1 examines the correlation between gender and religious affiliation. Panel A reports estimates on the full sample of SSA countries. Panel B focuses on countries where new Christians have a non-negligible share of the religious market — i.e., where the share of new Christians (measured with the grouped definition, the same variable used as the outcome in columns 1–2) exceeds 5% of the country sample (see Table C1 in the appendix for average gender gaps by country).[5]

Women's propensity to adhere to a new Christian church is about 4 percentage points higher than men's, corresponding to a 12 percent difference relative to men's average propensity (Panel B, column 1, of Table 1). These patterns remain unchanged when we include DHS cluster fixed effects as well as an extensive set of individual-level covariates such as age, age squared, education, labor market participation, marital status, and urban/rural residence (column 2), suggesting that demographic factors alone do not account for the observed gender gap in religious affiliations.[6]

In stark contrast, women are less likely to adhere to an African traditional religion by about 0.9 percentage points, a reduction of about 30% relative to the male adherence rate (panel B, columns 3 and 4). Similarly, women are less likely to be Muslim (column 9, 0.6 percentage points, a reduction of 3% relative to men) or Catholic (column 5, 1 percentage point, a reduction of 5% relative to men). Finally, women are also more likely to be Protestant than men (by about 0.8-1.1 percentage points), which may be partly explained by the fact that new Christian movements are related to Protestant Christianity.

When we include countries where new Christian churches have only a marginal or negligible presence (Panel A, full sample), the patterns are similar yet absolute gaps are somewhat smaller. This is expected: adding countries where few respondents are new Christians mechanically compresses the gender gap on that outcome, and in several countries in the extended sample a large Muslim share reduces the room for Christian

[5]In both panels, in the eight countries where new Christian churches are not distinguished from Protestant churches in the DHS, the grouped measure considers both as new Christian. See Appendix B for details.

[6]Appendix C.1, Table C2 shows that the patterns displayed in Table 1 are very similar (a bit larger, 5.4 pp, corresponding to a 20% difference relative to men's average) when we exclude from the category "new Christian" respondents from DHS waves where new Christian churches are not clearly distinguished from missionary Protestant churches (Table C3 shows the corresponding results when using the raw religion variables from the DHS).

denominations more broadly.

Table 1: Gender gaps in religious affiliation

	New Christian		Traditional		Catholic		Protestant		Islam	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Full sample (grouped definition)										
Female	2.553***	2.879***	-0.673***	-0.696***	-0.717***	-0.182***	0.841***	1.334***	-0.317***	-1.488***
	(0.0655)	(0.0615)	(0.0313)	(0.0305)	(0.0612)	(0.0606)	(0.0484)	(0.0495)	(0.0591)	(0.0414)
Mean Y men	23.00	23.10	2.603	2.587	18.90	18.87	9.857	9.909	32.98	32.82
R-squared	0.295	0.528	0.104	0.391	0.149	0.365	0.220	0.389	0.502	0.799
N	2219285	2099306	2219285	2099306	2219285	2099306	2219285	2099306	2219285	2099306
Panel B: Countries with positive share of new Christians (grouped definition)										
Female	4.044***	4.357***	-0.853***	-0.895***	-0.961***	-0.429***	0.763***	1.099***	-0.566***	-1.580***
	(0.0971)	(0.0911)	(0.0412)	(0.0395)	(0.0796)	(0.0782)	(0.0588)	(0.0607)	(0.0735)	(0.0490)
Country-round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
DHS cluster FE	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Mean Y men	34.19	34.14	2.961	2.939	20.76	20.67	9.153	9.194	18.33	18.45
R-squared	0.175	0.445	0.127	0.387	0.0997	0.317	0.188	0.336	0.198	0.710
N	1412381	1328788	1412381	1328788	1412381	1328788	1412381	1328788	1412381	1328788

Note. Data: Demographic and Health Surveys. The table reports OLS estimates. The coefficients have been multiplied by 100. Panel A includes the full DHS sample. Panel B restricts the sample to countries with a non-negligible share of new Christians (more than 5%), measured with the grouped definition. Both panels use the grouped definition: the outcome variable equals one if the respondent's religion is New Christian in columns 1 and 2, including Protestants in surveys where new Christians and Protestants are not separately coded. Traditional (columns 3 and 4), Catholic (columns 5 and 6), Protestant (columns 7 and 8), Islam (columns 9 and 10). Individual controls include age, age squared, education, marital status, labor market participation and urban/rural place of residence. Country and year fixed effects are always included. Robust standard errors clustered at the DHS cluster level are reported in parenthesis.

These patterns replicate when different datasets are used (Afrobarometer, PEW Research Center, World Value Surveys, and our own first-hand data): women are significantly more likely than men to report membership of new Christian churches. The difference ranges from 15 percent (in most large-scale datasets) to 70 percent when we compare them to the male average in rural contexts where traditional religions remain dominant (Appendix C.1, Table C4).

These gender differences in religious affiliations suggest that husbands and wives of the same couple may be of different religions, and that wives may convert to new Christian churches before their husbands. Data from Benin (whether our first-hand data or those extracted from DHS) confirm this pattern: in general, spouses have different religions in more than 25% of the couples present in the DHS sample (the proportion for our first-hand data is quite close at 22%). Furthermore, female members of new Christian churches have more often a spouse who is NOT a member (35% in the DHS sample) than male members of these churches (20% only).[7]

2.3. *The new Christian churches as a social game changer*

To better understand the rising success of new Christian churches, we first need to get a better idea of their specific features, especially when we compare them to both tradi-

[7]The corresponding figures are 22 and 4% in the first-hand data.

tional (animist) religions and conventional Christianity. The first thing to note is that, in contrast to the imported, mainstream missionary churches, the new post-colonial Christian churches usually emerge locally, and are mostly founded by Africans. According to Mwaura (2013), new Christian movements are "an attempt by Africans to forge new identities for themselves experimenting with and breaking down traditions. In theological terms they are regarded as processes of inculturation of Christianity in the African context [...]. They can simply be defined as autonomous church groups with an all-African leadership and an all-African membership" (p. 415).

Although there may be subtle differences among existing new Christian movements, they share much in common. They are mostly related to Protestant Christianity and its individualistic ethics. They also emphasize the role of charisma, the importance of a direct relation to God through spontaneous prayer (often accompanied by singing and dancing), and a strong feeling of community backed by a plenitude of congregational activities. Last but not least, they evince a strong belief in the Holy Spirit and its ability to counter traditional evil spirits through the activation of spiritual gifts, mainly divine healing. Finally, they tend to promote individual salvation in the Hereafter and, at the same time, prosperity in this world (Caldwell and Caldwell, 1987; Meyer, 1998; Brown, 2011; Auriol et al., 2020).

An immediate consequence of the indigenization of Christianity in the new Christian churches is their recognition of the existence of spirits. Instead of denying it, they accept the ontological reality of witchcraft and evil spirits, but only to offer protection against their malevolent influence by invoking superior supernatural forces (Le Rossignol et al., 2025). They also stress the importance of supernatural phenomena such as miracles and, sometimes, they put forward a Manichean theology and an apocalyptic vision of warfare between good and evil that match traditional fears of sorcery (Platteau, 2009, p. 679). In the Pentecostal doctrine, the descent of the Holy Spirit is described as a trance, through which it comes to penetrate an individual in order to chase the evil spirit that has taken possession of her (Fancello, 2006).

If, in conformity with the individual ethics inherited from Protestantism, the new Christian churches accord a great importance to the nuclear family, they simultaneously give a central place to community life. This is partly reflected in their liturgy, which typically integrates African music and dances, thereby ensuring the authenticity of the religious experience for all church members.[8] In reality, the church congregation has come to replace the extended family, a major turning point in African social life (Cald-

[8]The new churches can thus be said to have reclaimed the erstwhile charismatic experience which was suppressed by mainline Christianity although it resonated well with African spirituality and its emphasis on healing, the use of African symbolism, and music (Mildnerova, 2014; Mwaura, 2013).

well and Caldwell, 1987: 429; Fancello, 2006: 128). Indeed, unlike the traditional village societies anchored in the local clans, the new religious communities are based on the idea of equal membership in status (all members are "brothers and sisters in Christ"), implying that no distinction should be made between men and women, elders and youngsters, natives and immigrants, masters and slaves, or different castes or social ranks.

The main message is therefore that despite the existence of lines of continuity between the old and the new orders, a fundamental break with the past is occurring under the new wave of religious conversions. What new Christian churches are encroaching upon is no less than the body of tradition and its associated authority structure. Revealingly, necessary conditions to become a member of a new religious community are the renunciation and condemnation of one's past cultural and religious beliefs, as well as disobedience to, or distancing from, the clan elders in charge of implementing the associated rituals. In particular, the customary practice of ancestor worship with demands for abundance of children or good health is replaced by prayers to Jesus and the Christian Saints. Estrangement from the traditional universe also involves bans on the abuse of alcohol and the use of fetishes, as well as condemnation of the waste of time and money in ancient feasts and rituals, the absorption of traditional medicines and visits to traditional healers, and polygamous unions.[9] Represented as the hallmarks of backwardness, these "sinful behaviors" must be abandoned to have access to modernity and prosperity (see, e.g., Mildnerova, 2014). In short, by adhering to a new church, an individual "joins a new family, one bound together by faith" (Caldwell and Caldwell, 1987: 97-8) and becomes a "native outsider" or an "internal stranger" in the community of origin (Hagen, 1975; Kennedy, 1988; Platteau, 2000, 2014).

3. Impact of economic opportunities on female conversion: motivating evidence from an experimental study in Benin

Whether new Christian churches offer complementary economic services or favor the emancipation of women from constraining social structures, we expect new economic opportunities to stimulate conversions to their brand of Christianity. To probe this intuition with facts, we use first-hand data from a randomized controlled trial. The trial is about an intervention designed to incentivize women in Benin to cultivate a profitable cash crop. Although conversion was not a pre-registered outcome, the extensive survey questionnaire and additional qualitative fieldwork provide unique insights into the dynamics of conversion and the motivations driving women to shift their religious

[9]Traditional feasts include puberty rites, traditional weddings, mortuary rituals and the pouring of libation to the ancestors.

allegiance. This exploratory evidence will help to frame and motivate the subsequent cross-country investigation.

3.1. Data and experimental design

The experiment was conducted in Southern Benin in collaboration with the Belgian Development Agency (ENABEL).[10] Consisting of a pro-women agricultural intervention, it is comprised of a business training program and an in-kind subsidy aimed at encouraging the start or the expansion of a pineapple production enterprise. Beneficiary women were randomly selected from a pool of households who had expressed their interest for a previous intervention of the agency (yet had not benefited). The in-kind subsidy consists in the preparation of the soil and the planting of pineapple over an area of 0.5 ha.[11] As for the training program, it was made up of seven sessions (2-4 hours each, every two weeks) dealing with topics such as management and accounting, goal setting, gender issues, and a technical session on pineapple production. Training sessions were organized in set groups of about 8 people, on average, based on the location of respondents.

Randomization was done at the individual level. The selected subjects of the experiment were randomly assigned to the control group or one of two treatment arms with each group comprising one third of the original sample. In one treatment arm, husbands were invited to join their wife to the seven training sessions while in the other arm, husbands were invited only to the session on gender issues. Travel costs for men and women were covered by the program. Randomization and data collection were intended to provide a rigorous measure of the impacts of the program on beneficiary women and their households. In a context where women often rely on their husband for access to land, management of the labor force or negotiation of sale contracts, the two arms were designed to test whether the involvement of husbands helps to leverage their support for the new activity of their wife.

The timeline was as follows. A baseline survey of 1,009 households (877 men and 1,009 women) was carried out in 2020, and it included questions regarding the religion of the respondents. Two years later, in 2022, the randomized assignment to the control and treatment groups was realized and the intervention was implemented. In 2024, a first

[10]The main evaluation of the agricultural intervention, which does not examine the outcomes used in this paper, can be found in Álvarez-Aragón et al. (2026). The RCT was pre-registered in the AEA RCT Registry (AEARCTR-0008898).

[11]It includes the supply of workers for the start-off steps of a new pineapple production (clearing the land, stump removal and plowing) and the provision and planting of pineapple shoots. In addition, a plastic film is delivered to be stretched above the land parcel so as to reduce the amount of labor required at the later stages of production and to accelerate the growth of the crop. Beneficiary women were mandated to make a contribution representing 5 percent of the total cost (50-73$).

follow-up survey was conducted to measure a number of post-intervention outcomes, including questions on religious conversion. Finally, we conducted another follow-up survey in 2025 to gather additional information on the potential mechanisms behind the results uncovered on the basis of the 2024 data. Appendix B.2 includes additional information regarding the details of the intervention, including summary statistics and balance tests, and the analysis of attrition rates. Reassuringly, there are no significant differences between the treatment and control groups at baseline and attrition rates are low at 6.5% (endline 1) and 7.8% (follow-up).

3.2. *Results*

Thanks to the RCT setting, we are able to assess the causal impact of the new, women-focused economic opportunity not only on the latter's proclivity to convert and join a new Christian church, but also on their husband's potential opposition to their conversion drive, and the channels through which opposition may emerge.

Table 2 shows the RCT results. Panel A reports the average treatment effect (intent to treat estimates) of the program on several outcomes: the religious conversion of the targeted woman (column 1); the conversion to a new Christian religion (column 2); the conversion to a mainline (missionary) Christian religion (column 3); opposition against conversion to new Christianity from husbands (column 4) or from the broader family (column 5); and whether the woman believes that her husband has contacted a witch-doctor to hinder her business by resorting to magic means (column 6).[12] In panel B, we distinguish between the two treatment arms: full engagement (participation in all the program's training sessions) versus partial engagement (participation in only one training session) of the husband.[13]

In column 1, the outcome of interest is captured by a binary variable equal to one if the respondent has ever converted. This measure has the advantage of capturing the cases of women who converted but had to later reverse course under the pressures of antagonistic forces, mainly the husband's opposition. Our qualitative work revealed that such cases, far from being anecdotal, have occurred in all the study villages: all the women interviewed are thus aware of situations in which husbands (or the husband's extended family) have opposed their wives' religious conversion. Because of this pushback possibility, our measure of conversion to new Christianity, as shown in column 2, is likely to

[12]Due to the sensitivity of the topic and potential enumerator or social desirability bias, the question about the husband's use of witchcraft was asked in audio computer-assisted self-interview (ACASI) mode.

[13]Point estimates on conversion, resistance and the use of witchcraft, as well as the differences across arms, are larger and more significant if we measure the impact of the intervention on women who actually benefited from the training and subsidy (treatment on the treated, Table C6 and Table C7 in Appendix C.2).

be an underestimation: women who rescinded their initial conversion decision may not have reported new Christianity as their "new religion". All the specifications include a control indicating whether the respondent was a new Christian at baseline.[14]

Results reveal that the targeted women in the aggregate treatment are more likely to have converted to a new religion by more than 6 percentage points corresponding to a 19% increase from the control group mean (column 1). This is true, in particular, for conversions to a new Christian church (column 2, +4.7 percentage points).[15] Targeted women are also more likely to admit to having faced opposition from their husband when they considered conversion to new Christianity, or to say that they would have faced opposition if they had wished to convert. The difference is 8 pp, corresponding to a significant 20% increase from the control group (column 4).[16] Other family members may also oppose conversion, although the estimated impact of the intervention (5 pp) is not statistically significant (column 5). Finally, treated women are more likely to believe that their husband has recourse to supernatural means to throttle their economic activities (a 4 pp increase amounting to a 53% departure from the control group average).[17] These results suggest that, when offered a new economic opportunity, women seek religious conversion, yet face opposition from their husband, implying that the latent demand for conversion coming from women is probably stronger than what observed rates of conversion indicate. Women's acknowledgment of magical tricks used by their husband to counter their plan confirms their perception of a strong resistance from close quarters.

Turning to the difference between treatment arms, we find that, perhaps paradoxically,

[14]In our main specification, we include the following predetermined controls measured at baseline: age, education, marital status, and whether the respondent was a new Christian. Table C5 in Appendix C shows that the results remain virtually the same when only outcome variables at baseline are included as controls (when they are available).

[15]One potential concern is that these effects are driven by a small number of training groups rather than reflecting a diffuse response across the treated population. We address this in three steps. First, Appendix Table C10 shows that conversions are spread evenly across the 107 training groups: the average number of conversions per group is 1.4 (maximum of 6) and the average group-level conversion rate is 23%. Second, Appendix Figure C1 reports 107 leave-one-group-out regressions, each dropping the treated members of one group from the estimation sample. The treatment coefficient is remarkably stable across all jackknife samples, ranging from 0.042 to 0.051 and remaining significant at the 10% level throughout. Third, the intraclass correlation coefficient (ICC) for conversion outcomes among treated women equals 0.014, and a one-way ANOVA F-test cannot reject the null of equal conversion rates across groups ($p = 0.29$), confirming negligible within-group clustering of outcomes. The results are also robust to clustering standard errors at the training-group level, where control respondents—who were not assigned to a training group—are allocated to pseudo-groups of comparable average size based on place of residence (results not shown).

[16]When the respondent had converted, we asked if their husband (initially) opposed conversion whereas when she had not converted we asked whether their husband would oppose her conversion move.

[17]The five-year recall window of the witchdoctor question includes the pre-treatment period (mid-2020 through 2022). Pre-treatment visits are balanced across arms by randomization, adding noise but no signal, so the 4.3 pp ITT likely represents a lower bound on the post-treatment effect.

when husbands were invited to all the sessions of the training program, their opposition to conversion and recourse to magic devices were stronger, and women's conversion rates were actually lower. One possible explanation for this asymmetric reaction between treatment arms is that husbands invited to the training have gained a better understanding of the stakes involved in the external pro-women intervention, meaning the technical and management skills imparted to the beneficiaries as well as the importance of the additional incomes that their acquisition could allow. If participation of husbands in the training module of the program has this effect of raising their awareness, it may well instill in them the fear that their wife will become more independent, prosperous and assertive, thereby undermining erstwhile male predominance in social and economic relations. Their reaction may then consist of sabotaging her involvement in the new economic activity and hampering her religious conversion.

Table 2: Treatment effects: religious conversion, opposition and witchcraft

	Endline 1					Endline 2
	Women's conversion			Opposition by:		Witchcraft
	Ever	To new Xty	To old Xty	Husband	Other family	Used by husband
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Panel A: Aggregated treatment						
Treatment	0.0644**	0.0472*	-0.00502	0.0788**	0.0510	0.0409*
	(0.0325)	(0.0254)	(0.0217)	(0.0321)	(0.0316)	(0.0219)
R-squared	0.0851	0.257	0.0572	0.159	0.149	0.0256
Panel B: By treatment arm						
Partial husband involvement	0.0758**	0.0615**	-0.0199	0.0547	0.0449	0.00927
	(0.0378)	(0.0298)	(0.0245)	(0.0367)	(0.0364)	(0.0239)
Full husband involvement	0.0530	0.0330	0.00988	0.102***	0.0570	0.0716***
	(0.0373)	(0.0290)	(0.0254)	(0.0371)	(0.0364)	(0.0271)
Department FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y control	0.341	0.203	0.115	0.365	0.342	0.0789
R-squared	0.0855	0.258	0.0587	0.161	0.149	0.0325
N	936	936	936	924	924	835

Note. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The sample in column 6 is restricted to married women. The outcome variables are indicators with unitary value if: the respondent has ever converted to any new religion (column 1), to a new Christian religion (column 2), or to an "old" Christian church (column 3). In columns 4 and 5 the outcome variables are indicators equal to one if the husband (column 4) or another family member (column 5) has opposed woman's conversion to a New Christian church. The outcome variable in column 6 is an indicator that equals one if the respondent reports that her husband (in the last 5 years) went to see a witch-doctor to slow down his wife's activities or reduce her resources. Panel A shows treatment effects of the aggregated treatment. In Panel B, "Partial husband involvement" is an indicator equal to one if the woman (or the wife of the man when the sample is restricted to men) has been assigned to the treatment group where business training was reserved for women. "Full husband involvement" is an indicator equal to one if she has been assigned to the treatment group where women attended the business training together with their husband. Controls include age, education and marital status at baseline, whether the respondent was a new Christian at baseline, and department fixed effects. Robust standard errors are in parenthesis.

3.3. Impact of economic opportunities on women's conversion: an interpretation

To interpret our RCT results, it is useful to review the types of benefits that women can expect to receive upon joining a new Christian church. In the following, we successively examine the economic and non-economic benefits of religious conversion for women

and the role of their husbands in the process.

3.3.1. Economic services

A first question to ask is whether women are more attracted to churches supplying economic services that are supportive of their new economic activity. We are thus interested in knowing whether new Christian churches are more likely than old Christian churches to provide such services.

In the followup survey of 2025, we collected data which can help us assess this conjecture since sampled women were queried about the type of services they could avail themselves of in their (current) church. The first three columns of Table 3 display the results of simple regressions indicating the correlation between the religion of the respondents and the availability of three economic services – loans for emergency or production purposes, and access to ROSCAs – in their religious community. In Panel A, the comparison group includes all the women professing to adhere to so-called other religions, while in Panel B the reference category includes members of a missionary church, whether Catholic or Protestant.[18] The category "Non-Christian" includes women belonging to traditional religions (about 10% of all women), Islam (2.7%), and other religions/no religion (3%).

From the correlations shown in Panel A, it is evident that new Christian churches organize rotating credit and savings groups more often than other religious communities. They are also more liable to mediate solutions to intra-household conflicts. On the other hand, they are not more predisposed to grant loans for emergency or investment purposes. When compared to missionary (mainline) churches, however, new Christian churches do not appear to be more ready to supply economic services (Panel B). In contrast, the latter are 6 percentage points more likely than the former to facilitate conflict resolution among household members. This last finding fits well with an important lesson drawn from our RCT study: religious conversion to new Christian churches often involves tensions and conflicts within the close family.

[18]This category also includes the Celestial Church of Christ which, in Benin, was founded in the 1950s and shares much more in common with missionary Christian churches than with the new wave of Christianity that emerged in the late 80s and 90s (Henry, 2008). In fact, some of our enumerators believed it was a branch of Catholicism and classified it as such. In-depth qualitative interviews revealed that, in contrast to the wave of new Christianity and more in-line with the Catholic church, the Celestial church tends to adopt a tolerant attitude vis-à-vis consultations of a traditional sorcerer (the Fa/Bokono) or the recourse to traditional rituals, such as yearly pilgrimages to the seaside where converts communicate with the spirits of the sea. It is also telling that people may decide not to shift between a traditional religion and a Celestial church precisely because, on their own admission, important practices or rituals are rather similar in the two affiliations. It is noteworthy that such testimonies are consistent with systematic accounts according to which Evangelical and Pentecostal churches strongly criticize Celestial churches which they blame for their syncretism (Henry, 2008).

Overall, the results of Table 3 do not support the view that converts to new Christianity prefer this religion to missionary churches because of the (direct) economic advantages in the form of credit, insurance and savings, in particular. It is true that other advantages, such as assistance in the access to manpower, modern inputs and marketing channels, are not featured in the table's list of economic services even though they may have proven critical for cash crop production. However, these were never cited by women when we investigated the matter with them.

Table 3: Supply of economic services and mediation as per religious affiliation

	Current church provides:				
	ROSCAs	Emergency loans	Investment loans	Mediation	Use of ROSCAs
	(1)	**(2)**	**(3)**	**(4)**	**(5)**
Panel A: New Christians compared to all other denominations					
New Christian	0.0488*	0.0280	0.0152	0.116***	-0.00622
	(0.0291)	(0.0227)	(0.0154)	(0.0260)	(0.0259)
R-squared	0.0411	0.0162	0.00982	0.167	0.0611
Panel B: New Christians (and non-Christians) compared to missionary churches					
New Christian	0.00979	0.0161	0.0180	0.0625**	-0.00903
	(0.0316)	(0.0247)	(0.0166)	(0.0271)	(0.0276)
Non Christian	-0.167***	-0.0508*	0.0121	-0.229***	-0.0102
	(0.0349)	(0.0302)	(0.0257)	(0.0462)	(0.0397)
Controls	Yes	Yes	Yes	Yes	Yes
Mean Y	0.232	0.127	0.0534	0.771	0.811
R-squared	0.0558	0.0184	0.0101	0.195	0.0611
N	899	899	899	899	905

Note. Data: First-hand survey data from southern Benin. The outcome variable is an indicator that equals one if the respondent says that her church provides the following services: ROSCA's (Column 1), loan for an emergency (2), loan to invest (3), or mediation in case of conflict within the family (4). In column 5, the outcome variable is an indicator equal to one if the respondent is a member of at least one rotating savings and credit association. Individual controls include age, baseline education and marital status. Department fixed effects are always included. Robust standard errors are reported in parenthesis.

3.3.2. Spiritual protection and the banning of traditional beliefs

The quantitative evidence presented above is in line with the evidence gathered in semi-structured interviews conducted in southern Benin where women were asked about their motivation to convert and about the differences they see between various churches. It is again telling that economic services were not cited as distinctive features of new Christian churches. Instead, two key factors were consistently mentioned: superior protection against "spiritual attacks" (including the necessary healing process that follows the magical spells) and rejection of traditional rituals and beliefs. These aspects deserve careful attention inasmuch as spiritual attacks are closely linked to women's economic eman-

cipation. Successful women may thus be the victims of bad spells and even witchcraft accusations, especially if their business is conspicuously prosperous. A woman fears bad spells or is blamed for being a witch on the ground that she has achieved her position by manipulating natural powers through witchcraft. In other words, economic emancipation makes protection services against (accusations of) witchcraft necessary.[19]

Our survey includes several questions designed to assess whether members of new Christian churches feel protected against "spiritual attacks" and whether they tend to question the power of traditional authorities. We thus asked women whether they feel threatened by spiritual attacks, whether they protect their crops with fetishes to avoid theft, whether they drank sodabi to fight against COVID[20], whether they consulted with a sorcerer (the "Bokono"), and whether they believe that the spirit of ancestors influences their life. Table 4 presents the results.[21]

Compared to women from other religions, members of new Christian churches are much less prone to traditional beliefs and practices. They are less likely to feel threatened by witchcraft (8 pp, 19% of the sample mean), to use fetishes to protect their agricultural fields against theft (17 pp, 50% of the sample mean), to drink a magical potion (called sodabi) to fight against COVID (6 pp, 55% of the sample mean) and to consult with a wizard (called Bokono) (10 pp, 71% of the sample mean). Finally, they are also less likely to believe in the power of ancestors (19 pp, 61% of the sample mean), a defining feature of traditional religions in many African communities (Álvarez-Aragón, 2025).[22] In all these dimensions (except in the probability of consulting with a wizard), the differences remain strong and significant when the comparisons are made with members of missionary churches only.

[19]We initially intended to ask survey questions on witchcraft accusations and the associated fears of women. However, in the pretesting phase of the questionnaire, our survey teams strongly opposed the inclusion of these questions, testifying to the sensitivity of the topic. In their opinion, directly inquiring about the issue of witchcraft would compromise the survey because these questions would raise suspicions that the respondent stands accused of a malevolent role.

[20]Sodabi is a strong homemade alcohol obtained by distilling palm wine. It plays a fundamental role in the Voodoo religion, where it is repeatedly used in rituals.

[21]In this table, we look at the correlations using religious affiliation as recorded at baseline since some outcomes are measured at baseline (e.g., Bokono consultation or sodabi consumption), and reverse causality could potentially exist along other dimensions. For instance, if conversion following the intervention is (at least partly) motivated by the desire for protection against witchcraft, individuals with greater fear of witchcraft in 2025 may be more likely to have joined new Christian churches. This concern is mitigated by considering religion at baseline.

[22]In our sample, membership of an Evangelical church is actually the strongest negative correlate of beliefs in the influence of ancestor spirits on people's lives. Figure C2 in Appendix C.2 displays the correlations between beliefs in the power of ancestors and religion fixed effects (where no religion is the excluded category) as well as other socioeconomic characteristics.

Table 4: New Christian churches and traditional beliefs

	Fearing witchcraft	Using fetishes	Drinking magical potions	Consulting with wizards	Believing in ancestors
	(1)	**(2)**	**(3)**	**(4)**	**(5)**
Panel A: New Christians compared to all other denominations					
New Christian	-0.0770**	-0.167***	-0.0627***	-0.0975***	-0.190***
	(0.0322)	(0.0448)	(0.0213)	(0.0247)	(0.0293)
R-squared	0.148	0.0814	0.0195	0.0610	0.106
Panel B: New Christians (and non-Christians) compared to missionary churches					
New Christian	-0.0930***	-0.167***	-0.0570**	0.00924	-0.147***
	(0.0347)	(0.0483)	(0.0232)	(0.0222)	(0.0317)
Non Christian	-0.0533	0.00192	0.0169	0.313***	0.139***
	(0.0417)	(0.0685)	(0.0318)	(0.0416)	(0.0436)
Controls	Yes	Yes	Yes	Yes	Yes
Mean Y	0.426	0.337	0.110	0.142	0.310
R-squared	0.149	0.0814	0.0199	0.168	0.118
N	924	439	876	767	936

NOTE. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The sample is restricted to women. The sample in column 2 is restricted to pineapple producers. The outcome variables are: an indicator equal to one if the respondent feels threatened by spiritual attacks in column 1 (we interpret it as a proxy for protection against witchcraft), an indicator equal to one if the respondent has used a fetish to protect his/her pineapple field against theft (column 2), if the respondent has drunk sodabi (traditional alcoholic drink) to prevent coronavirus (column 3), if the respondent has consulted a "bokono" (sorcerer) or has used a fetish to prevent coronavirus (column 4), or an indicator that equals one if the respondent believes that the spirits of his/her ancestors influence his/her life (column 5). Individual controls include age, whether the respondent attended school, marital status, whether the respondent cultivated pineapple in 2020, and whether the respondent had electricity at home in 2020. Department fixed effects are always included. Robust standard errors are reported in parenthesis.

To shed further light on the spiritual protection mechanism, Appendix Table C8 shows that the treatment effect on NCC conversion is significantly larger among women who report feeling threatened by witchcraft (columns 3–4): the interaction between treatment and witchcraft threat is positive and significant at the 10% level, indicating that women already exposed to spiritual hostility are more responsive to the economic opportunity in terms of religious conversion. Because witchcraft threat is measured after the intervention, this heterogeneity is correlational rather than causal. Still, the pattern is suggestive that the conversions triggered by the economic opportunity are disproportionately driven by a demand for spiritual protection against the threats that accompany women's economic advancement.

3.4. Wrapping up: conflict brought to center stage

The interpretation of our results, when they are all put together, is at odds with the idea of a strictly economic calculus in which women decide to convert if the expected *economic* benefits exceed the costs. This approach presents two basic limitations. First, it ignores the "spiritual protection" component that is critical in the case of conversion to new Christian churches and, second, it assumes that individuals (women) have full agency in the sense that their decision is unhampered by adversarial forces. In reality, conversions may generate wide repercussions because they disturb the traditional social order. It is therefore not surprising that women's initiatives are often regarded as hostile behavior

by their close relatives (the average values reported at the bottom of Table 2 reveal that as many as 37% and 34% of women declare that they face or would face opposition from their husbands and other family members respectively if they wished to convert). Women may choose to abandon plans to convert, or to rescind a past conversion decision, not because the economic benefits are insufficient, but because they face such a strong marital opposition that the integrity of the family is at risk.

There is an obvious link between a substantial body of mainly anthropological literature and our finding that in Benin many women and men view the new Christian churches as a source of defense against magic threats and witchcraft accusations directed against aspiring women. Dating back to the pioneer work of Evans-Pritchard (1937, 1940) duly followed by multiple case studies (see, in particular, Geschiere, 1994; Meyer, 1998; ter Haar, 2007; Nyaga, 2007), evidence has accumulated to show that witchcraft accusations or magical harassment are often sparked by feelings of envy and jealousy reflecting serious tensions within the (extended) family. Revealing of this situation is the suspicion among people suffering from a witchcraft attack that "some close relative is to blame" (Geschiere, 1994, p.336). The targeting of women seems to be a tactic used to prevent them "from effectively competing in male-dominated areas", such as divination, healing, and cash cropping (Ntloedibe-Kuswani, 2007, p.223). In Ghana, for example, "many women do not want to go beyond their respective (gender) roles, for fear of becoming the object of witchcraft accusations, including by other women who may be jealous of their success" (Akrong, 2007, p.63).

When women try to emancipate from the traditional order by pursuing their own ends distinct from those of their husband and close relatives, they invite the grave criticism that they struggle to cut the ties linking a person with the family (Meyer, 1998). In our survey, 62% of respondents declared that spiritual attacks are more likely against economically successful individuals. In semi-structured interviews, revealingly, women who had converted to new Christian churches explained that they suffered from violent headaches or other physical pains and disturbances which traditional healers as well as modern doctors were unable to cure. The women affected are often said to be bewitched but the key point is that they are in psychological and physical pain. A large literature in medicine and social sciences discusses how health symptoms, in particular mental health symptoms, are often attributed to malevolent spirits in the African context (Ventevogel et al., 2013; Mayston et al., 2020). According to the stories told in Benin, return to normalcy can only be secured by the intervention of a supernatural force deemed more powerful than the ancestors' spirits. In the words of Mwaura (2013, p.422), it is only when "a source of power stronger than the traditional sources, ancestors, spirits and magicians was offered" to women that they could be liberated "from everything that

oppressed them". This force is provided by the new Christian churches in the form of a special intervention of the Holy Spirit acting on behalf of God. It responds to calls for help expressed through intense prayers said in the church by the entire congregation of believers. Here, we cannot miss the parallel with the experiences of conversion to Islam among the Giriama people of coastal Kenya (Parkin, 1972, Chapters 2-5). Kennedy (1988) summarized it vividly:

> "...the conversion of some young entrepreneurs to the Islamic faith might follow a long period of psychological tension and physical illness induced by the possibility of conflict with the elders whose status and power were threatened by the younger men's activities. Such "illness" could be diagnosed as caused by powerful Islamic spirits whose appeasement required nothing less than the religious conversion of those unfortunate enough to become possessed. Once this had occurred, the Islamic ban on the consumption of alcohol and certain foods, as well as the need to follow a partly separate ritual and social life, all provided the opportunity for entrepreneurs to reduce their level of involvement in traditional society. Yet this behavior no longer incurred community displeasure since it was now judged to be religiously determined rather than the result of selfish individualism. (Kennedy, 1988, p.142)".

In sum, to emancipate from the traditional order, dynamic individuals need effective "spiritual protection" which a monotheist religion claiming a superior God can provide (see also Haugerud, 1995, and Sarro, 2009). When such a religion explicitly and forcefully rejects rituals and beliefs liable to hinder the economic success of these individuals, its protective power becomes even more compelling.

4. Economic opportunities and women's affiliation with new Christianity: evidence from SSA

In this section we broaden the perspective and mobilize all available rounds of the DHS surveys conducted in SSA to evaluate whether economic opportunities, for women in particular, trigger their affiliation with new Christian churches and thereby contribute to their rise. We use two measures of economic opportunities: the rise in the international price of locally suitable crops and the presence of World Bank-financed development projects that explicitly target women. In both cases, we rely on difference-in-differences estimations, exploiting variations over time in these indicators of opportunities, and capturing the effect of time-invariant geographical characteristics by including grid-cell fixed effects.

Furthermore, making use of an indicator of witchcraft beliefs, we explore whether the presence of these beliefs amplifies the effect of new economic opportunities on the appeal of new Christian churches. Experimental results from Benin indeed suggest that for women new Christian churches represent a shelter against the hostile use of

magic against them. Because ambitious women are more likely to be victims of magic spells and accused of using witchcraft, we expect the spiritual services offered by these churches to be particularly valuable to them in areas where they are exposed to the risk of traditional aggressiveness.

On the basis of cross-country evidence, we cannot evaluate the relative importance of supply and demand shifts for the overall effect of positive economic shocks on new Christian affiliation. We cannot rule out the possibility that an increase in local incomes – driven, for instance, by favorable cash-crop prices – attracts religious entrepreneurs and shifts the supply of new Christian Churches. What is critical for our argument is that such changes lead to a disproportionate rise in the numbers of new Christians, particularly among women, and that this rise reflects a latent demand for the (women-) specific services these churches provide. In the case of the RCT in Benin, by contrast, we are confident that the conversion decisions of treated women were demand-driven: the intervention was too small in scale and too geographically dispersed to raise local incomes significantly.

4.1. Economic opportunity I: Shocks to crop prices and new Christian churches

We start by examining the effects of a rise in locally suitable crop prices on new Christian membership.

4.1.1. Main estimations

We test this hypothesis by exploiting local exposure to changes in international world prices of agricultural commodities, which have been shown to affect local incomes (McGuirk and Burke, 2020; Berman et al., 2025). We first divide the African continent into cells of $0.5° \times 0.5°$ (approximately 55×55 km^2 at the equator).[23] Then, for each cell, we compute the suitability of its land to crops for which world prices are available, that is, banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, and wheat. The suitability index is extracted from the FAO's Global Agro-Ecological Zones (GAEZ). It is estimated for each crop and cell using information on the location (i.e., temperature, humidity, length of days, sunlight, or rainfall), soil and crop characteristics. Our baseline results use crop suitability rather than actual production since it is exogenous to changes in local conditions and world demand. Figure C3

[23]The choice of 55×55 km^2 cells is standard in the literature (see, for example, Berman et al., 2017; McGuirk and Nunn, 2024; McGuirk and Burke, 2020; McGuirk and Nunn, 2025). Still, we report in Appendix C.3.8 robustness tests to the main results using instead $0.25° \times 0.25°$ cells. For this analysis, we take the cash-crop suitability measure of Roessler et al. (2022), which is measured at that level and uses information on the nine most important export crops: banana, cocoa, coffee, cotton, palm oil, groundnut, sugar, tea, tobacco.

in Appendix C.3.1 shows the spatial distribution of the suitability index across African cells.

Finally, we combine this information with the evolution over time of the international price of each crop, as published by the World Bank. Figure C4 in Appendix C.3.2 shows the evolution of international prices for the crops under concern. Our main treatment variable to proxy for the potential value of cash crops in cell c and year t is given by the following suitability-weighted price index (as in McGuirk and Burke (2020) and Berman et al. (2025)):

$$\text{Cash-crop price index}_{ct} = \sum_{j=1}^{n} (\pi_{jt} \times \text{Suitability}_{jc}),$$

where $j = 1, .., n$ represents a crop in our list. The suitability index suitability_{jc} represents the average cash-crop suitability of crop j in cell c (ranging from 0 to 1) and π_{jt} the global price index for the same crop in year t. This global price index sets the price of the crop to 1 in year 2010.

We then combine the cash-crop price index at the grid-year level with geolocated individual-level information coming from the Demographic and Health Surveys (DHS). We assign to each respondent in the DHS the cash-crop price index in year t-1 of the grid-cell in which their DHS cluster falls.[24] Overall, our full sample includes almost 2 million respondents. The baseline specification we consider is:

$$Y_{i,c,t} = \beta \, \text{Cash-crop price index}_{c,t-1} + FE_c + FE_t + X'_{i,t}\Phi + \epsilon_{i,c,t}, \tag{1}$$

where Y_{ict} is an indicator that equals one if respondent i in cell c and year t is a new Christian, and zero otherwise. To increase coverage, we include countries where the category "new Christian" is not distinct from missionary Protestant churches (and consider them all new Christian in that case).[25] $\text{Cash-crop price index}_{c,t-1}$ is the suitability-weighted price index measured in $t-1$, as described above. FE_c and FE_t denote grid-cell fixed effects and year fixed effects, respectively. Finally, X'_{it} is a vector of individual-level controls such as age, age squared, gender, education, labor market participation, marital status, and rural-urban residence. The inclusion of grid-cell fixed effects implies that the

[24] DHS cluster GPS coordinates are randomly displaced for confidentiality: urban clusters are displaced by up to 2 km and rural clusters by up to 5 km (with 1% of rural clusters displaced by up to 10 km) (Burgert et al., 2013). This may generate classical measurement error in the assignment of clusters to grid cells, potentially attenuating our estimates toward zero. The concern is mitigated by the large size of our baseline $0.5° \times 0.5°$ cells (approximately 55×55 km^2 at the equator), which substantially exceeds the maximum displacement distance.

[25] See Appendix B for details about how the new Christian variable is defined. Tables C16 and C17 in Appendix C.3.6 show the results are robust to using the raw measure of religions provided by the DHS and to the exclusion of country-waves where new Christians and grouped together with Protestants.

coefficient of interest is identified by time variations in the cash-crop price index within cells.[26] We cluster the standard errors at the grid-cell level, which is the level of the treatment.[27]

Columns 1-3 of Table 5 display the basic results while sequentially including control variables. There is a strong, stable, positive and statistically significant relationship between the crop price index and new Christian membership. These results indicate that local exposure to higher crop prices leads to higher rates of affiliation with new Christian churches. The size of the estimated coefficients is meaningful: because the price index has been standardized (mean 0, standard deviation 1), the point estimate in column 3 (1.83 pp) reads directly as the effect of a one-standard-deviation rise in the lagged cash-crop price index — an increase of 1.8 percentage points in the probability of being a new Christian, corresponding to almost 8% of the sample mean.

As expected, the effect of positive shocks to crop prices on new Christian membership is stronger for women than for men (column 4): a one-standard-deviation increase in the (standardized) price index raises the probability of being a new Christian by 1.5 pp for men and by 2.0 pp for women. This gap supports the hypothesis that new Christian churches are particularly suited to women's demands in the face of new economic opportunities.

Finally, we investigate whether the effect of shocks to economic opportunities on the appeal of new Christian churches is amplified in regions where witchcraft beliefs are prevalent. As suggested by our in-depth study of Benin, the demand for spiritual protection rises when new economic opportunities emerge, especially if they come from women who are relatively vulnerable to witchcraft attacks and accusations. Therefore, we expect a stronger effect on new Christian affiliation in places where witchcraft beliefs are widespread, for women in particular.

For this analysis, we use information from the DHS on witchcraft beliefs for about one million respondents across sub-Saharan Africa. To measure witchcraft beliefs, we use the answer given by respondents to the following question: "Do you believe that HIV can be caused by witchcraft?" We build an aggregate measure, using all answers pooled across waves, defined at the grid-cell level corresponding to the (standardized) mean of respondents who answered positively in the $0.5° \times 0.5°$ grid-cell which the DHS cluster of the respondent has been assigned to.[28] We use this measure of witchcraft beliefs as

[26] This identifying variation requires more than one DHS round per country. In our sample, this implies dropping about 89.000/2.000.000 respondents (4.2%).

[27] Appendix C.3.7 reports robustness to the use of Conley standard errors or to a more conservative clustering at the country-year level.

[28] There is significant variation in witchcraft prevalence across space. The sample mean is 0.17, the

our baseline measure as it maximizes coverage and can be defined at a granular level.

Columns 5 and 6 show the results. As hypothesized, the rate of new Christian affiliation is higher in areas where witchcraft beliefs are widespread. Column 5 indicates that, in grid-cells where the prevalence of witchcraft beliefs is one standard deviation above the sample average, a one-standard-deviation increase in the (standardized) cash-crop price index leads to a 2.3 pp increase in the probability of being a new Christian.[29] Column 6 reveals that witchcraft beliefs amplify the affiliation response to cash-crop shocks for women but not for men. Thus the total marginal effect of a one-standard-deviation rise in the (standardized) price on women's new Christian affiliation approximates 2.6 percentage points in areas where witchcraft beliefs are one standard deviation above the mean. This effect is roughly 38% larger than the effect on women in average-witchcraft areas (1.9 pp), and 65% larger than the effect on men in the same high-witchcraft areas (1.6 pp).[30]

---

median is 0.12, and the standard deviation is 0.15 (10th percentile is 0.033; 90th percentile is 0.41).

[29]The 2.3 pp figure is obtained by summing the coefficient on the cash-crop price index (1.82) and the coefficient on its interaction with witchcraft (0.50).

[30]The 2.6 pp figure sums the coefficients on the Cash-Crop Price Index, Cash-Crop PI × Female, Cash-Crop PI × Witchcraft, and Cash-Crop PI × Female × Witchcraft (evaluated one standard deviation above mean witchcraft).

Table 5: Shocks to cash crops and conversion to new Christian churches

	Respondent is New Christian					
	(1)	(2)	(3)	(4)	(5)	(6)
Cash Crop Price Index	0.0168***	0.0168***	0.0183***	0.0150***	0.0182***	0.0162***
	(0.00381)	(0.00382)	(0.00388)	(0.00393)	(0.00409)	(0.00422)
Female		0.0271***	0.0317***	0.0316***	0.0314***	0.0308***
		(0.00159)	(0.00173)	(0.00172)	(0.00182)	(0.00171)
Cash Crop Price Index x Female				0.00491***		0.00280***
				(0.000980)		(0.000940)
Cash-crop PI x witchcraft (cell)					0.00496***	-0.000271
					(0.00111)	(0.00144)
Cash-crop PI x Female x witch.						0.00751***
						(0.00103)
Mean Y	0.243	0.243	0.239	0.239	0.241	0.241
R-squared	0.421	0.422	0.423	0.423	0.429	0.429
Basic controls	No	Yes	Yes	Yes	Yes	Yes
Full controls	No	No	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1966628	1962886	1862101	1862101	1709683	1709683

Note. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). "Cash Crop Price Index" is the sum across fifteen crops (banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, wheat) of their price in t-1 (indexed to 1 in year 2010 for each crop) weighted by their suitability. The index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. "Witchcraft (cell)" is the standardized mean prevalence of witchcraft in a given grid cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable uses the grouped definition: equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. Basic controls include age, age squared and gender. Full controls further include education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the cell level are reported in parenthesis.

Importantly, and also in line with what was found for Benin, the cross-country results appear to be very specific to new Christian churches: an increase in suitability-weighted crop prices leads to a *decrease* in the probability of being a member of a *mainline* Christian denomination (Catholic or Protestant) and has no effect on traditional religion (from a very low average prevalence of 2 percent) (see the results in Appendix C.3.4).

A further question is which side of the price distribution generates our result. Our baseline uses a continuous cash crop price index, whose positive coefficient shows that, on average, rising prices are associated with greater new Christian affiliation. This average, however, does not sign the effect of price falls on its own: a counter-cyclical channel, in which negative shocks raise conversion, could coexist with a dominant positive channel and remain hidden in the net coefficient. To separate the two, Appendix C.3.5 splits the year-on-year change in the lagged price index between positive and negative shocks

$\Delta p_{c,t} = p_{c,t-1} - p_{c,t-2}$ into the magnitude of the rise, $\max(\Delta p_{c,t}, 0)$, and the magnitude of the fall, $\max(-\Delta p_{c,t}, 0)$, each rescaled by the standard deviation of $\Delta p_{c,t}$ so that the two coefficients are directly comparable. Positive shocks increase the probability of new Christian affiliation, while negative shocks have a precise null effect (Table C15). We thus find no evidence of a counter-cyclical channel operating beneath the dominant positive one.

Robustness. The stronger effect of positive shocks where witchcraft beliefs are more prevalent echoes our Benin results, in which protection against witchcraft attacks emerges as an important motive for conversion. A concern is that witchcraft beliefs capture other grid-cell characteristics whose interaction with the shock drives conversion. To address this possibility, we add controls likely correlated with witchcraft, together with their interactions with the price shock.[31]. The witchcraft interaction retains its magnitude and significance (Table C22), suggesting the interaction is not merely picking up these confounds. Appendix C.3 further shows that the results are robust to: i) using the raw measure (table C16) or the restricted measure of new Christians (table C17, where we exclude country-waves where Protestants are likely clubbed together with new Christians); ii) clustering the standard errors at the country-year level instead of at the cell-level (table C19) or using Conley standard errors (table C18; iii) using $0.25^\circ \times 0.25^\circ$ grid-cells — approximately 28x28km at the equator (table C20); iv) using an alternative measure of cash crop potential — M3-CROP dataset from Monfreda et al. (2008) (table C21). Finally, we verify in Appendix C.3.3, Table C11, that positive shocks to the crop price index translate into a positive local income shock, as other papers have documented (McGuirk and Burke, 2020; Berman et al., 2025). Positive shocks to the cash crop price index are associated with increases in the wealth index as reported in the DHS or with increases in nightlights.[32]

4.1.2. Alternative measure of witchcraft prevalence

Our baseline measure of witchcraft prevalence — the cell-level share of DHS respondents who believe that HIV/AIDS can be caused by witchcraft — is limited in the sense that

[31]The variables used in the interactions are education, rurality, traditional religion, employment, health conditions proxied by child mortality, healthcare access proxied by using modern contraception, and women's decision-making capacity

[32]We conduct both analyses at the cell-year level, focusing on cells within countries for which we have DHS data. Wealth measures are computed using responses from all individuals within a given cell-year. The DHS creates a wealth score using principal component analysis of household asset ownership within each country-year survey. The nightlights data come from the adjusted luminosity series of Chiovelli et al. (2026).

it only captures respondents who both hold witchcraft beliefs *and* mobilize them to explain the etiology of one specific disease, a narrow framing shown to understate the true prevalence of such beliefs (Sievert, 2025; Le Rossignol et al., 2025). We therefore build a broader measure at the regional level from Pew Research Center (2009), a survey of 25,091 respondents in 188 regions across 19 sub-Saharan African countries. It asks whether the respondent believes in "the evil eye, or that certain people can cast curses or spells that cause bad things to happen to someone". This question is broader and likely less prone to under-reporting, since the question sits among many other items on supernatural beliefs in a survey devoted to religion, a context in which admitting such beliefs is arguably less stigmatizing. For each of the 188 Pew regions we compute the share of positive answers and assign it to every DHS respondent whose cluster falls within the region's boundaries (see Appendix C.3.11 for details).

Table C23 confirms that the conclusions are unchanged: the effect of a positive cash-crop price shock on women's new Christian affiliation remains significantly amplified where witchcraft beliefs are widespread.

4.1.3. Women's involvement in cash-cropping

Because we expect women to be in special need of the sort of support provided by new Christian churches when their *own* economic prospects improve, we further investigate whether cash-crop price shocks have a greater effect in situations where women are more involved in cash-crop production. We use two different measures. The first is contemporaneous but potentially endogenous; the second is historical but available for a sub-sample. First, we exploit information from the DHS and compute the proportion of women who earn cash or in-kind payments from their work *and* whose primary occupation is agriculture in each grid cell. The concern is that unobserved time-varying characteristics but also prices themselves may simultaneously affect women's average involvement in cash-cropping and religious affiliation. To mitigate this possibility, we pool data across waves to compute the measure. Second, following Alesina et al. (2013), we turn to a pre-determined proxy of women's participation in farming, drawn from the Ethnographic Atlas (Murdock, 1967) and assigned to DHS respondents through the ethnic group to which they declare belonging. This proxy is an indicator equal to one if, in the respondent's group, women were reported to perform appreciably more agricultural work than men (or to be the only ones to farm). Its main drawback is coverage: it requires the survey to record the respondent's ethnicity and the group to be matched with an Atlas society. We are thus left with 927,000 respondents (666,800 of them women) belonging to 149 ethnic groups in 25 of our 33 countries, and 32 percent of them are

members of a group in which women historically dominated agriculture. Appendix C.3.13 describes the variable, the matching procedure, and the remaining sample in detail.

We re-estimate Equation 1 by interacting the cash-crop price index with the measures of women's involvement in cash-crop production, separately for women and men.[33] We therefore exploit variations in the cash crop price index to examine the heterogeneous effects of crop prices on women's new Christian affiliation, depending on their level of involvement in cash crop production (Table C24 in Appendix C.3.12 and Table C25 in Appendix C.3.13). The main finding is that the positive impact of crop prices on women's new Christian affiliation (but not men's) appears to be especially significant in cells where women play a significant role in cash crop production. Among women, a one-standard-deviation rise in the (standardized) cash-crop price index raises new Christian affiliation by an additional 0.45 percentage points in cells one standard deviation above the mean in women's cash-crop participation (column 3 of Table C24 , about 19% of the base effect of 2.34 pp). Similarly, the effect of the cash-crop price index on new Christian membership is markedly larger in the ethnic groups where women historically did most of the farming: for a one-standard-deviation rise in the (standardized) price index, it goes from 2.0 to 3.3 percentage points (column 3), or from about 9 to 15 percent of the sample mean, suggesting that new economic opportunities seized by women specifically give rise to higher rates of new Christian affiliation among them. However, while column 4 confirms that a positive price shock raises women's new Christian affiliation most in cells where witchcraft beliefs are widespread, the triple interaction between the price index, witchcraft beliefs, and women's involvement in cash crops does not add a further increase (its coefficient is slightly negative). The double interactions are both strongly positive: the shock's effect is larger where women are involved (rising prices then bear more on women's own prospects) and where witchcraft beliefs are salient (the churches' protective services are then more valuable). That these effects do not compound may be related to a bounded conversion response: when both forces are already strong, there is little room for one to add further to the other. The combined effect nonetheless remains largest where both are high.

4.2. Economic opportunity II: Development aid projects

Main estimation A second setting allows us to test whether new economic opportunities for women drive their affiliation with new Christian churches. It exploits the deployment in certain areas of development aid projects targeted on them. Towards this

[33] We split the sample by gender to avoid quadruple interactions.

purpose, we rely on geo-coded data on development aid projects funded by the World Bank during the years 1995–2014 (AidData, 2020). There are as many as 1,717 such projects distributed in 15,268 geo-coded project sites for the SSA region.

A gender-related or women-focused project is one whose main theme is gender according to the World Bank classification. The official definition of "gender theme" is as follows: "[it] encompasses World Bank Group activities that – irrespective of sector – address and/or close gaps between males and females (for example: gaps in financial inclusion, agricultural productivity, occupational sex segregation, gaps in participation in local service delivery management, etc.) and other gaps that may be identified in the systematic country diagnostic at the country level (e.g., low female labor force participation relative to male labor force participation)" (The World Bank Group, 2016, p. 57). Almost 200 project sites have gender as their main theme (1.25%).

To measure the influence of exposure to women-focused development projects on new Christian membership, we combine the data on development projects with the geo-located individual-level information from the DHS, using $0.5^\circ \times 0.5^\circ$ grid-cells. We consider an individual treated if at least one women-focused World Bank project is localized in the grid cell where the DHS cluster of the respondent lay at the time of the survey.[34] In our final sample, 6% (6.3% for men and 6% for women) of the respondents lived in grid-cells with varying exposure to World Bank women-focused projects (116,000 respondents in 64 cells).

Results are displayed in Table 6. The central message is clear: income-earning opportunities created by World Bank gender projects promote membership of new Christian churches specifically for women in places where witchcraft beliefs are widespread. Consistent with a mechanism that operates through the demand for protection, these projects have no detectable effect on average — the response is concentrated where witchcraft beliefs are prevalent, and diluted elsewhere. In a cell one standard deviation above the mean in witchcraft beliefs, the differential effect of a gender project on women compared to men amounts to approximately 1.5 percentage points (column 4, 6% of the sample mean).

[34] Our preferred specification is at the $0.5^\circ \times 0.5^\circ$ (approximately 55x55km at the equator), following existing papers that also look at the impact of World Bank projects (McGuirk and Nunn, 2024). We show in Appendix C.4 that our results are robust to alternative definitions of grid-cell sizes such as $0.25^\circ \times 0.25^\circ$ (28x28km at the equator).

Table 6: New Christians and World Bank gender-related projects, 55x55km cells

	New Christian (grouped def.)			
	(1)	(2)	(3)	(4)
Female	0.0282***	0.0302***	0.0296***	0.0326***
	(0.00176)	(0.00169)	(0.00172)	(0.00181)
Female x Witchcraft (cell)			0.00882***	0.00896***
			(0.00194)	(0.00188)
WB gender project	-0.00898	0.0294***	0.00336	0.00509
	(0.0228)	(0.0103)	(0.00897)	(0.00885)
WB gender project x Female	0.00575	0.00134	-0.000403	-0.000781
	(0.00759)	(0.00754)	(0.00440)	(0.00423)
WB gndr project x Witch. (cell)			0.0423***	0.0412***
			(0.00684)	(0.00652)
WB gndr x Female x Witch. (cell)			0.0139***	0.0153***
			(0.00495)	(0.00474)
Total number of WB projects	0.00822***	-0.000429	-0.00104	-0.000941
	(0.00181)	(0.000985)	(0.000954)	(0.000972)
Mean Y	0.250	0.250	0.249	0.247
R-squared	0.287	0.434	0.444	0.441
Country and year FE	Yes	Yes	Yes	Yes
Grid-cell FE	No	Yes	Yes	Yes
Individual	No	No	No	Yes
N	1784632	1784632	1650755	1580922

Note. Data: Demographic and Health Surveys and AidData research lab (2020). "WB gender project" is an indicator that equals one if there is a WB gender-related project before the month of the interview in the grid-cell where the individual lives. "Witchcraft (cell)" is the (standardized) mean prevalence of witchcraft in a given grid cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable uses the grouped definition: "New Christian" equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. The total number of World Bank projects before the month of the interview in a given cell always included as a control. Individual controls include age, age squared, gender, education, labor market participation, marital status and urban/rural residence. Country and year fixed effects are always included. Robust standard errors clustered at the cell level are reported in parenthesis.

Robustness. Appendix C.4 shows that the results are robust to: i) using the raw measure (table C26) or the restricted measure of new Christians (table C27, where we exclude country-waves where Protestants are likely clubbed together with new Christians); ii) clustering the standard errors at the country-year level instead of at the cell-level (table C28) or using Conley standard errors (table C29); iii) using $0.25^\circ \times 0.25^\circ$ grid-cells — approximately 28x28km at the equator (table C31); and iv) broadening the definition of gender-related projects to include those whose main theme is not officially "gender" but

which nonetheless provide income-earning opportunities for women, based on our own inspection of project documents (table C33). Reassuringly, under this broader definition the differential effect of a gender project on women (relative to men) in high-witchcraft cells remains large and significant — if anything, larger than under the baseline definition. Appendix C.4.5 describes the construction and validation of this alternative measure in detail.

Alternative measure of witchcraft prevalence. The results carry through when we replace the DHS cell-level measure of witchcraft with a regional-level measure constructed from the PEW dataset.[35] As Table C30 in Appendix C.4 shows, the triple interaction remains positive and significant (0.028, column 4): in a region one standard deviation above the mean in witchcraft beliefs, the differential effect of a gender project on women (relative to men) amounts to about 2.8 percentage points, if anything larger than the 1.5 pp obtained with the DHS-based measure.

5. New Christianity, women's empowerment and escape from the traditional order: empirical regularities in SSA

The previous section showed that positive economic shocks raise women's conversion to new Christian churches, especially where witchcraft beliefs prevail. We now investigate whether affiliation itself is associated with outcomes in line with the paper's central claim: that women turn to new Christian churches not (only) to protect their livelihoods but to support their economic independence and personal autonomy. We examine two dimensions in turn. The first is women's empowerment: whether membership goes together with greater female labor supply, lower fertility, and more decision-making power. The second is distancing from the traditional order: whether it goes together with the rejection of traditional beliefs, rituals, and authorities. We cannot infer causality here—these churches may confer greater agency on their female members, or simply attract women already determined to resist the constraints of husband and community. In the present context, we need not be too worried about this: the two possibilities are likely at play simultaneously, and either way, the key point is that new Christian churches emerge as a space where women can pursue economic independence and contest customary constraints

The reader is referred to Appendix B for details regarding the datasets used, the total number of surveys and respondents in each survey, and the way the main variables were

[35]The PEW measure is the share of respondents in a given sub-national region of the 19 PEW countries who believe in the "evil eye" or that certain people can cast curses or spells that cause bad things to happen — see Appendix C.3.11 for details.

precisely constructed, including how we define membership of a "new Christian" church in the different datasets.

5.1. New Christian churches and women's empowerment

To explore the relationship between new Christianity and women's empowerment, we look at labor market and reproductive outcomes, two pivotal dimensions of personal autonomy, and at participation of women in decision-making, both at home and in the church. Based on DHS (columns 1-5) and PEW (columns 6 and 7) data, Table 7 reports the results. Panel A shows that, compared to women professing any other religion, women in new Christian churches exhibit better labor market and reproductive outcomes: they are 5.4 percentage points more likely to work, 5 pp more likely to use contraceptives, and they report a smaller number of children, whether ideal (-0.6) or actual (-0.4).[36] These correlations remain significant when we compare new Christians to women belonging to missionary Christian churches (Panel B), even if the differences are smaller in size.[37]

Column 5 highlights the correlation between women's participation in decision-making and new Christianity. The outcome variable is an index computed as the simple average of all the non-missing answers to questions related to four matters: decisions about the way women's earnings are used, and about healthcare, large household purchases, and visits to relatives. For each dimension, we construct an indicator equal to one if the woman takes part in the decision, alone or jointly with the husband. New Christian women score 6.9 pp higher on this index than women of any other religion (12% of the sample mean of 0.58), which — since the index is the average across four decision domains — corresponds to participating in roughly 0.28 additional decisions out of four. The gap remains sizable but smaller (2.2 pp, 4% of the mean) when they are compared to women in missionary churches. Although we only report the index, positive relationships are observed in virtually all components of the index. Disaggregated results by

[36] In the context of Benin, we were told that in many cases contraceptives are supplied by the new Christian churches themselves.

[37] Tables C34 and C35 in Appendix C.5 check how these correlations depend on the way we classify new Christians in the DHS: the first uses the raw DHS categories, while the second drops the country-waves in which new Christians and Protestants are lumped together in a single "Protestant" category. The correlations with labor participation, contraceptive use, decision-making, and the two PEW outcomes survive both alternatives. The reproductive outcomes are more sensitive: under the restricted definition, the coefficients on ideal number of children and children ever born shrink markedly in Panel A and even reverse sign in Panel B. The negative correlation between new Christian membership and fertility reported in the main text should therefore be interpreted with caution — most likely reflecting negative selection into new Christian churches on fertility-related dimensions.

category are available in Appendix C.5, Table C36.[38,39]

Table 7: New Christian churches and women's labor and reproductive outcomes, and decision-making

	Labor market	Reproductive			Decision-making		
					At home	In the church	
	Working status	Contraceptives	Ideal nr of children	Children ever born	Decision-making Index	Respect for women	Female leadership
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: New Christians compared to all other denominations							
New Christian	0.0527***	0.0480***	-0.572***	-0.388***	0.0693***	0.0968***	0.185***
	(0.00205)	(0.00130)	(0.0117)	(0.0142)	(0.00178)	(0.00570)	(0.00686)
R-squared	0.0921	0.111	0.244	0.155	0.270	0.0859	0.197
Panel B: New Christians (and non-Christians) compared to missionary churches							
New Christian	0.0127***	0.0105***	-0.0841***	-0.0570***	0.0215***	0.0234***	0.0571***
	(0.00203)	(0.00129)	(0.00953)	(0.0138)	(0.00166)	(0.00795)	(0.00975)
Non Christian	-0.0969***	-0.0932***	1.254***	0.801***	-0.112***	-0.113***	-0.197***
	(0.00233)	(0.00133)	(0.0133)	(0.0151)	(0.00192)	(0.00917)	(0.0107)
Country-round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y	0.765	0.217	5.155	5.562	0.578	0.791	0.460
R-squared	0.0979	0.117	0.268	0.165	0.279	0.0921	0.210
N	626816	1554236	1392334	443637	965130	24605	23839

Note. Data: All available DHS in columns 1-5, and PEW for columns 6 and 7. The sample is restricted to women. In column 1, the sample is restricted to women over 30 years old and in column 4 to women over 35 years old to approximate completed fertility. The table reports OLS estimates. "New Christian" uses the grouped definition: it equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. Outcome variables are: probability to work (column 1), use of contraceptives at the time of the survey (column 2), ideal number of children (column 3), total number of children ever born (column 4), decision-making index (column 5). This index is calculated as the average of all non-missing responses to four yes/no questions (indicators) that equal one if the respondent has some say – either the woman decides together with her partner/husband or she decides alone – in the following matters: (1) how a woman's earning are used, (2) healthcare, (3) large household purchases, and (4) visiting relatives. In column 6, the outcome variable is an indicator that equals one if the respondent associates attitudes of respect for women with Christianity. In column 7, the outcome variable is an indicator equal to one if the respondent thinks that women should be allowed to serve in religious leadership roles, such as pastor, priest or imam. Country-wave fixed effects always included. Controls include age, age squared, and rural/urban residence. Robust standard errors clustered at the DHS cluster level are reported in parentheses when using the DHS data.

Finally, columns 6 and 7 use the PEW data to test whether members of new Christian churches support a strong role for women in the church and whether they link up respect for women with the new creed. More specifically, respondents were asked whether "women should be allowed to serve in religious leadership roles, such as pastor, priest, or imam", and whether they associate attitudes of respect for women with Christianity.[40] Compared to Christian respondents from missionary churches, new Christian women appear more likely to associate Christianity with respect for women (column 6 panel B, +2.3 pp, 3% of the sample mean), and to think that women should be allowed to serve

[38]These results do not change when using alternative definitions of the decision-making variables. For example, Lowes (2022) constructs an index that takes on three values: 1 when the respondent's partner or another person makes the decision, 2 when both the respondent and the partner decide, and 3 when the respondent alone is responsible for the decision. These results are reported in Appendix C.5, Table C37.

[39]Women affiliated with a new Christian church have a greater participation in decisions made about healthcare, large household purchases and visits to their family, although no relationship is observed when a woman's earnings must be allocated. More probing nevertheless reveals that the latter result is explained by a decrease in the probability of autonomous decisions by women, itself compensated by an increase in the probability of joint conjugal decisions (See Table C38 in Appendix C.5).

[40]The first question imposed a dichotomous choice between two opinions: "women should be allowed to serve in religious leadership roles, such as pastor, priest, or imam" (in which case the outcome indicator is equal to one) versus "only men should be able to serve in religious leadership roles, such as pastor, priest, or imam" (the indicator is equal to zero).

in religious leadership roles (column 7, +6 pp, 14% of the sample mean). Here lies a radical difference between new Christianity, on the one hand, and social norms ruling in the traditional patriarchal order or prescriptions imposed in mainline Christian churches (where women are not allowed to act as leaders), on the other hand.

The correlation between women's agency and new Christian churches reflects the latter's promotion of equality in all its forms, including gender equality. In the Pentecostal tradition, for example, men and women alike are allowed to experience the power of the Holy Spirit, a gift which endows them with the ability to dream, see visions, prophesy, preach, teach, exorcise, and even heal (Mwaura, 2013, p.–422; Lugazia, 2017, p.–308). In some countries, women are increasingly involved in the foundation of Pentecostal churches, acting as bishops, evangelists and pastors, and filling special ministries such as counseling, service to the community, and mentoring fellow women in distress (Audu and Eregare, 2020). Prominent among other gender-equalizing mechanisms operating in new Christian churches are prohibitions against, or condemnation of, different well-anchored practices such as alcohol consumption, polygamy, and mobility restrictions for women. These rules are welcomed by many women because they are often harmed by the targeted practices: alcohol is mainly consumed by men and has been shown to be a strong predictor of domestic violence (e.g., Abramsky et al., 2011; Theiss, 2024), polygamy creates room for intra-household conflict and exacerbates unequal gender relations (e.g., Thiam, 1978; Barr et al., 2019), and mobility restrictions prevent them from leaving the area traditionally earmarked for them, which may make it impossible for them to travel to the new churches (as much as they want or at the appropriate times).

The data confirm that prohibitions like the use of (excessive) alcohol consumption and the ban of polygyny are stronger in new Christian churches. Table C39 in Appendix C.6, thus shows that, compared to mainline denominations, new Christian women are more likely to say that their husbands do not drink alcohol (column 1: -6.7 pp, 19% of the sample mean) while new Christian male respondents are less likely to report drinking alcoholic beverages (column 2: -14.7 pp, 34% of the sample mean).[41] Regarding polygyny, we learn from Table C40 in Appendix C.6 that being a new Christian slightly decreases the probability of being in a polygamous union.[42]

[41] In line with the DHS data, our own data from Benin confirm these findings since new Christians are less likely to say that they consume sodabi (a traditional alcoholic beverage).

[42] To explain the small magnitude of the coefficients associated with polygyny, we may bear in mind that, in the context of some Evangelical churches in Benin, for example, the ban only applies *after* an individual has become a member of the church. In other words, it is possible to join a Christian church while being in a polygamous union, but it is forbidden to become polygamous after joining.

5.2. *New Christian churches as escapes from the traditional order*

To better understand the spiritual and social dynamics surrounding the rise of new Christian churches, we now turn to the relationship between the new religion and (1) traditional beliefs, (2) trust in customary authorities, and (3) practices such as miracles and exorcism. The positioning of these churches vis-à-vis traditional beliefs may appear paradoxical: for one thing, they strongly reject traditional animist rituals and the authority structures which sustain them but, for another thing, they seem to reckon with the ontological reality of witchcraft insofar as they claim to protect members against magic attacks with the help of collective prayers.

Rather than blending the two systems, new Christian churches occupy the same spiritual terrain as traditional religion while displacing it. They address the concerns that animate traditional spirituality (healing, deliverance from malevolent forces, an expressive relation to the divine through dancing and singing) but they channel them into an exclusively Christian frame: the "rituals through which family links are symbolized and confirmed" (Meyer, 1998, p.324) are banished rather than accommodated, and replaced by new rituals centered on the cult of the Christian God.[43,44]

5.2.1. *New and old religions vis-à-vis traditional beliefs*

We start by examining the relationship between religious affiliation and traditional beliefs and rituals, using data from the PEW research center on more than 20,000 respondents from 19 SSA countries. Table 8 reveals that, in comparison to mainline denominations, followers of new Christianity are significantly less likely to: 1) participate in rituals to honor ancestors (4.3 pp, 17% of sample mean); 2) believe that sacrifices to ancestors can prevent disturbances from happening (7 pp, 21% of sample mean); 3) participate in initiation rituals (3.1 pp, 12% of sample mean); and 4) have recourse to traditional religious healers (4 pp, 10% of sample mean). These findings confirm the previous result obtained on the basis of our first-hand data from Benin: the rejection of traditional beliefs

[43]Descriptively, our first-hand data clearly show how the accommodation of the traditional frame of mind is much stronger in mainstream missionary churches than what is observed in new Christian churches. For instance, 60 percent of those whose second religion is Voodoo report Catholicism as their main religion, while only 15 percent report belonging to an Evangelical church. This is consistent with the local saying that people are "Catholics during the day, but followers of Voodoo at night".

[44]At first sight, this claim may appear to sit uneasily with the finding of Le Rossignol et al. (2025) that, in their corpus of DRC sermons, traditional beliefs are discussed with a strongly negative sentiment in *all* denominations, and not only in Evangelical churches. An important remark is that the convergence in sermon content is itself a symptom of the dynamic we describe. For example, in the context of Benin, faced with the rapid growth of the new Christian churches, mainline denominations have recently responded by opening charismatic sections *within* their own structures — the Catholic Charismatic Renewal being the most prominent instance — which borrow the idiom of spiritual warfare, divine healing and deliverance from their competitors in order to retain their flock.

and rituals is a defining feature of new Christian churches, especially when compared to missionary churches.

At the same time, new Christian churches stick to beliefs in the intervention of supernatural agents (the Holy Spirit, in particular), thereby providing a dimension of continuity between the old and the new worldviews. Thus, members of these churches are much more likely to have experienced or witnessed a divine intervention to heal an illness or an injury (Column 5), or the expulsion of the devil and evil spirits from a possessed person (Column 6). They are also more likely to believe that God can directly intervene to ensure the material success and good health of the worshipers.[45] When they are compared with mainline denominations (Panel B), new Christians are 15.4 percentage points more likely to have experienced or witnessed a miracle (a 32% increase from the sample mean), 11 pp more likely to have experienced or witnessed an exorcism (a 26% increase from the sample mean), and 5 pp more likely to believe in God's ability to grant them material success and healthy condition if they worship him. These results are consistent with the extensive literature highlighting the role of divine healing in new Christian churches (Cox, 2011; Mildnerova, 2014).

[45]More precisely, the respondent is asked to choose between two alternative statements: "God will grant wealth and good health to all believers who have enough faith", and "God does not always give wealth and good health to believers who have deep faith".

Table 8: New Christians and attitudes toward tradition and supernatural power

	Traditional beliefs and practices				Beliefs in supernatural power		
	Rituals	Sacrifices	Initiation	Trad. healing	Miracles	Exorcism	God-ordained prosperity
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: New Christians compared to all other denominations							
New Christian	-0.0492***	-0.0874***	-0.0534***	-0.0748***	0.205***	0.173***	0.0548***
	(0.00611)	(0.00643)	(0.00617)	(0.00691)	(0.00702)	(0.00696)	(0.00712)
R-squared	0.108	0.164	0.101	0.104	0.0747	0.0687	0.0718
Panel B: New Christians (and non-Christians) as compared to missionary churches							
New Christian	-0.0434***	-0.0704***	-0.0310***	-0.0407***	0.154***	0.108***	0.0525***
	(0.00886)	(0.00944)	(0.00901)	(0.00982)	(0.0100)	(0.0101)	(0.0102)
Non Christian	0.00882	0.0258**	0.0343***	0.0523***	-0.0778***	-0.0993***	-0.00354
	(0.00979)	(0.0104)	(0.00997)	(0.0108)	(0.0109)	(0.0109)	(0.0113)
Mean Y	0.263	0.329	0.267	0.424	0.477	0.420	0.571
R-squared	0.108	0.165	0.101	0.105	0.0766	0.0719	0.0718
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	24204	23458	24132	23587	24369	23694	23562

Note. Data: PEW Research Center. The table reports OLS estimates. "New Christian" is an indicator equal to one if the respondent reports one of the new Christian religions as their main religion. The outcome variable, also an indicator, is equal to one if the respondent has ever participated in an African traditional ceremony or performed rituals to honor ancestors (column 1), if s/he believes that sacrifices to ancestors can prevent disturbances from happening (column 2), if s/he has ever participated in an initiation ritual (column 3), if s/he has ever visited a traditional healer (column 4). In column 5, the binary variable is equal to one if the respondent answered "yes" to the question "have you ever experienced or witnessed a divine healing of an illness or injury?". In column 6, value one is obtained equal if s/he answered "yes" to the question "have you ever experienced or witnessed a situation where the devil or evil spirits were driven out of a person?". In column 7, the outcome indicator equals one if the respondent thinks that "God will grant wealth and good health to all believers who have enough faith in Him" rather than "God doesn't always give wealth and good health even to believers who have deep faith in Him". Individual controls include age, urban/rural place of residence, marital status, education, and whether the respondent has recently experienced a money shortage. Robust standard errors are reported in parentheses.

Finally, we want to know whether a gender divide exists with respect to the sort of beliefs and behavior patterns mentioned above. From Table C41 in Appendix C.7, it is evident that women are significantly less likely than men to hold "traditional" beliefs or follow "traditional" modes of conduct. For example, women are about 3 pp less likely than men to believe that sacrifices to ancestors can avoid bad outcomes (8% of the sample mean), or to participate in an initiation ritual (12% of the sample mean). The same difference is found in our first-hand data (Table C42 in Appendix C.7), where in addition to being less prone to report adherence to a traditional religion, women are 15 pp less likely to declare a belief in the influence of ancestral spirits (40% of the sample mean), 5.4 pp less likely to consult with a traditional sorcerer, and 10 pp less likely to drink highly alcoholic traditional beverages to fight against COVID. Such a strong gender difference is not coincidental. It is in line with the idea that the beliefs and rituals promoted by patriarchal structures tend to be called into question or rejected by women whose scope for autonomy is severely restricted under the rules of the existing order. New Christian churches provide a community in which members – both men and women – distance themselves from these structures (see Table C43 in appendix for the absence of systematic gender gaps *among members of these churches*).

Taken together the results suggest that, since missionary churches (such as mainline Catholic or Protestant denominations) are generally more syncretic in the sense that they are not radical in their rejection of traditional rituals and beliefs, women tend to have a special attraction for the new Christian churches where they are able to participate in religious life and find an inclusive space to express themselves.

5.2.2. *Religious affiliation and trust in traditional authorities*

A second essential feature of new Christian churches highlighted in the specialized qualitative literature is their stress on the necessity to break away from the traditional clan-based society and its associated patriarchal structure (Caldwell and Caldwell, 1987; Meyer, 1998; Mildnerova, 2014). Using data from Afrobarometer, Table 9 examines the relationship between religious affiliation and trust in traditional leaders (male elders) or the family.[46] We see that new Christians tend to profess less trust in traditional leaders and in the family than members of mainline Christian denominations and other religions.[47] These differences are modest, amounting to approximately 2 to 3 percent of the average trust level observed in the sample.[48]

It is worth emphasizing that, while trust in the family and traditional authorities is lower among members of new Christian churches, this shift is accompanied by the formation of new communal ties expressed through vibrant celebrations, more frequent religious services, and higher donations to charity or the church.[49] From Table 9 (columns 4-6, panel B), it is apparent that, when compared with missionary churches, members of new Christian churches are 12 pp more likely to participate in religious festivities (a 45% increase from the average), 16 pp more likely than other religious groups to attend religious services more than once a week (a 33% increase from the average), and 14 pp more

[46]Trust is measured as a discrete variable with four distinct values: 0 (no trust at all), 1 (just a little trust), 2 (a reasonable amount of trust) and 3 (a lot of trust).

[47]The new Christian indicator using data from Afrobarometer is based on a broader measure including people who declared to be "Christian" when asked about their main religion. Importantly, respondents who declared themselves to be Catholic or Protestant are excluded from this measure. This is because the Afrobarometer survey distinguishes between Catholic (or Protestant) and Christian believers, suggesting that the latter category might be capturing new Christians.

[48]Interestingly, the negative correlation between new Christianity and the trust in traditional leaders and the family is even stronger for women (see Table C44 in appendix).

[49]Gershman (2016) interprets the latter as a measure of prosociality, showing that holding witchcraft beliefs is negatively correlated with it. The strong, positive correlation between social attitudes and behaviors and new Christianity highlights the potentially high level of cooperation and solidarity within the church, which characterizes religions with moralizing high gods (Norenzayan and Shariff, 2008), and chimes well with previous research documenting strong sharing norms in Pentecostal churches (Kapepula et al., 2022).

likely to make donations to charity or the church (an 18% increase from the average).[50]

Table 9: New Christian churches and community identity: old vs new

	Afrobarometer data		PEW data		
	Trad. leaders	Family	Singing	High attendance	Charitable giving
	(1)	(2)	(3)	(4)	(5)
Panel A: New Christians as compared to all other denominations					
New Christian	-0.0932***	-0.0693***	0.121***	-0.0175**	-0.0567***
	(0.00554)	(0.00566)	(0.00623)	(0.00700)	(0.00620)
R-squared	0.128	0.122	0.0418	0.0746	0.104
Panel B: New Christians (and non-Christians) as compared to missionary churches					
New Christian	-0.0345***	-0.0335***	0.134***	0.193***	0.142***
	(0.00616)	(0.00619)	(0.00754)	(0.00818)	(0.00973)
Non Christian	0.154***	0.0984***	0.0237***	0.387***	0.355***
	(0.00707)	(0.00678)	(0.00873)	(0.00978)	(0.00920)
Mean Y	1.838	2.361	0.265	0.469	0.801
R-squared	0.130	0.123	0.0421	0.126	0.163
Controls	Yes	Yes	Yes	Yes	Yes
Country-Wave FE	Yes	Yes	Yes	Yes	Yes
N	197268	141462	24291	24273	21466

Note. Data: Afrobarometer Surveys 3,4,5 and 9 in columns 1-2 and PEW survey in columns 3-5. The outcome variables in columns 1 and 2 show the degree of trust in traditional leaders or in the family, respectively. They take value 0 (not at all), 1 (a little), 2 (a lot) and 3 (very much). In column 3, the outcome variable is an indicator equal to one if the respondent participates in religious chanting or singing gatherings and ceremonies at least once a week. In column 4, unitary value is obtained if s/he attends religious services more than once a week whereas in column 5, this is so if s/he donates a set percentage of his/her wealth to a charity or the church/mosque. In both datasets, controls include age, age squared, gender, and urban/rural place of residence. The explanatory variable is an indicator that equals one if the respondent's religion is New Christian. Country fixed effects are always included. With the Afroberometer data, wave FE are always included. Robust standard errors are reported in parentheses.

6. Conclusion: religious conversion and institutional change

The historical experience of Europe demonstrates how religious transformation can act as a powerful catalyst for social change, particularly in dismantling kinship-based structures and promoting women's emancipation (Goody, 1973). The Catholic Church's reforms — which prohibited cousin marriage and traditional inheritance practices that had reinforced patriarchal control — gradually reshaped family norms and property rights, contributing to the emergence of a more individualistic society (see, for example, Goody, 1983; Schulz, 2022; Fukuyama, 2012: 231-8; Guirkinger and Platteau, 2020: 492-4; Henrich, 2020: 164-86). Crucially, these changes were not imposed by political authority but

[50]Interestingly, when compared to all other religions the correlation with high attendance and charitable giving is negative (panel A), because these aspects are even stronger among Muslims (hence the contrast with Panel B).

emerged through the Church's ability to establish cultural norms. As a result of its actions, however, "a far more individualistic European society was already in place during the Middle Ages, before the process of state-building began, and centuries before the Reformation, the Enlightenment and the Industrial Revolution" (Fukuyama, 2012: 239).

A process that partly resembles the above may be presently unfolding in SSA. Under the impulse of local and endogenous new Christian churches, a growing number of Africans choose to follow a set of rules and practices radically different from those prescribed by the traditional clan-based order. These churches actively challenge customary norms centered on obligations to clan elders and the cult of ancestors, offering alternative spiritual frameworks that emphasize personal agency and divine protection. While rejecting traditional rituals, they maintain belief in supernatural forces capable of bringing healing and prosperity and countering magic spells, often perceived as obstacles to women's emancipation.

In this context, women emerge as central agents of change. As primary beneficiaries of the shift away from patriarchal and spiritual control, they form the majority of new converts and actively participate in religious practices that promise both spiritual empowerment and social liberation. Much like in Europe, where religious reform helped to loosen the grip of kinship structures and enabled female emancipation, the rise of new Christian churches in Africa appears to be fostering a similar transformation – one in which cultural change, rooted in spiritual renewal, may play a pivotal role in broader processes of development.

Our conclusion should not be read as contradicting findings from other settings where the expansion of Neo-evangelical or Pentecostal churches has been associated with regressions in women's status or increases in far-right political orientations (for Brazil, see Buccione and Mello, 2020; Solá, 2025). The reason is that these studies examine contexts in which religious competition primarily occurs between new Christian churches and mainline monotheist denominations – settings where both types of churches tend to operate as centralized, hierarchical organizations. This configuration differs fundamentally from the environment we study, where new Christianity coexists with traditional religions and where their appeal hinges on a distinct set of spiritual services.

A fascinating question is whether the bottom-up movement toward women's emancipation observed in SSA today will have more immediate effects than the top-down reforms of the Catholic Church in Western Europe. The effects of the latter played out over several centuries due to the slow enforcement of new cultural norms. This is attested by the following finding: the significant negative correlation between the spread of Christianity and the absence of clans and lineages can only be observed over a long period (Korotayev, 2003).

References

Abramsky, T, C. H Watts, C Garcia-Moreno, K Devries, L Kiss, M Ellsberg, H AFM Jansen, and L Heise, "What factors are associated with recent intimate partner violence? findings from the WHO multi-country study on women's health and domestic violence," *BMC Public Health*, 2011, *109* (11).

AidData, "WorldBank Geocoded Research Release Level1 v1.4.2 Geocoded Dataset," 2020.

Akrong, Abraham, *A Phenomenology of Witchcraft in Ghana*, Asmara, Eritrea: Africa World Press,

Aldashev, Gani and Jean-Philippe Platteau, "Chapter 21 - Religion, Culture, and Development," in Victor A. Ginsburgh and David Throsby, eds., *Handbook of the Economics of Art and Culture*, Vol. 2 of *Handbook of the Economics of Art and Culture*, Elsevier, 2014, pp. 587–631.

Alesina, Alberto, Paola Giuliano, and Nathan Nunn, "On the Origins of Gender Roles: Women and the Plough," *The Quarterly Journal of Economics*, 2013, *128* (2), 469–530.

Alfonsi, Livia, Michal Bauer, Julie Chytilová, and Edward Miguel, "Human capital affects religious identity: Causal evidence from Kenya," *Journal of development economics*, 2024, *167*, 103215.

Álvarez-Aragón, Pablo, "Paths *to* the Rainforests: Ancestral Beliefs and Fertility in Sub-Saharan Africa," *Working Paper*, 2025.

_ , Catherine Guirkinger, Anna Jolivet, and Angela Lulle, "Husbands in the room: The Impact of Joint vs. Women-Only Training on Household Dynamics in Benin," *Working Paper*, 2026.

Audu, Sunday Didam and Emmanuel Orihentare Eregare, *Women and Christianity in Africa*, Cham: Springer International Publishing,

Auriol, Emmanuelle, Julie Lassébie, Amma Panin, Eva Raiber, and Paul Seabright, "God Insures those Who Pay? Formal Insurance and Religious Offerings in Ghana," *The Quarterly Journal of Economics*, 2020, *135* (4), 1799–1848.

Barr, Abigail, Marleen Dekker, Wendy Janssens, Bereket Kebede, and Berber Kramer, "Cooperation in Polygynous Households," *American Economic Journal: Applied Economics*, April 2019, *11* (2), 266–83.

Barro, Robert J and Rachel M McCleary, "Religious conversion across countries," Technical Report, National Bureau of Economic Research 2024.

_ , Edgard Dewitte, and Laurence Iannaccone, "Looking Backward: Long-Term Religious Service Attendance in 66 Countries," Technical Report, National Bureau of Economic Research 2025.

Barro, Robert, Jason Hwang, and Rachel McCleary, "Religious conversion in 40 countries," *Journal for the Scientific Study of Religion*, 2010, *49* (1), 15–36.

Becker, Sascha O. and Ludger Woessmann, "Luther and the Girls: Religious Denomination and the Female Education Gap in Nineteenth-century Prussia," *The Scandinavian Journal of Economics*, 2008, *110* (4), 777–805.

_ **and** _ , "Was Weber Wrong? A Human Capital Theory of Protestant Economic History*," *The Quarterly Journal of Economics*, 05 2009, *124* (2), 531–596.

Becker, {Sascha O.}, Jared Rubin, and Ludger Woessmann, *Religion in economic history: a survey*, 1st ed., United States of America: Academic Press, Publisher Copyright: © 2021 Elsevier Inc. All rights reserved.

Becker, Sascha O, Jeanet Bentzen, and Chun Chee Kok, "Gender and Religion: A Survey," 2025.

Becker, Sascha O., Steven Pfaff, and Jared Rubin, "Causes and consequences of the Protestant Reformation," *Explorations in Economic History*, 2016, *62*, 1–25.

Berman, Nicolas, Lorenzo Rotunno, and Roberta Ziparo, "Sweet Child of Mine: Parental Income, Child Health and Inequality," *The Economic Journal*, 05 2025, *135* (672), 2455–2482.

_ **, Mathieu Couttenier, Dominic Rohner, and Mathias Thoenig**, "This Mine Is Mine! How Minerals Fuel Conflicts in Africa," *American Economic Review*, June 2017, *107* (6), 1564–1610.

Bernardelli, L.V., E. Michellon, M.A. Kortt, and Charles M.B., "The Association Between Protestantism and Economic Development in Brazil," *Int J Lat Am Relig*, 2025.

Binzel, Christine and Jean-Paul Carvalho, "Education, Social Mobility and Religious Movements: The Islamic Revival in Egypt," *The Economic Journal*, 04 2017, *127* (607), 2553–2580.

Boserup, Ester, *Woman's Role in Economic Development*, London: George Allen and Unwin, 1970.

Brown, Candy Gunther, *Global Pentecostal and Charismatic Healing*, Oxford University Press, 01 2011.

Buccione, Giulia and Marcela Mello, "Religious Media, Conversion, and Its Socio-Economic Consequences: The Rise of Pentecostals in Brazil," *Working Paper*, 2020.

Burgert, Clara R., Josh Colston, Thea Roy, and Blake Zachary, "Geographic Displacement Procedure and Georeferenced Data Release Policy for the Demographic and Health Surveys," DHS Spatial Analysis Reports No. 7, ICF International, Calverton, Maryland, USA 2013.

Caldwell, JC and P Caldwell, "The cultural context of high fertility in sub-Saharan Africa," *Popul Dev Rev*, 1987, (13), 409.

Cantoni, Davide, "The Economic Effects of the Protestant Reformation: Testing the Weber Hypothesis in the German Lands," *Journal of the European Economic Association*, 08 2015, *13* (4), 561–598.

_ **, Jeremiah Dittmar, and Noam Yuchtman**, "Religious Competition and Reallocation: the Political Economy of Secularization in the Protestant Reformation*," *The Quarterly Journal of Economics*, 06 2018, *133* (4), 2037–2096.

Carvalho, J.P., S. Iyer, and J. Rubin, *Advances in the Economics of Religion*, Palgrave Macmillan, 2019.

Chiovelli, Giorgio, Stelios Michalopoulos, Elias Papaioannou, and Tanner Regan, "Illuminating the Global South*," *The Economic Journal*, 01 2026, p. ueaf134.

Costa, Francisco, Angelo Marcantonio, and Rudi Rocha, "Stop Suffering! Economic Downturns and Pentecostal Upsurge," *Journal of the European Economic Association*, 06 2022, *21* (1), 215–250.

Cox, H, "Foreword," in "Global Pentecostal and Charismatic Healing," Oxford University Press, 01 2011.

Ellis, Stephen and Gerrie Ter Haar, *Worlds of Power: Religious Thought and Political Practice in Africa* Series in Contemporary History and World Affairs; 1, New York / London: Oxford University Press, 2004.

Evans-Pritchard, E. E, *Witchcrat , Oracles and Magic among the Azande*, Oxford and New York : Oxford University Press, 1937.

_ , *The Nuer: A Description of the Modes of Livelihood and Political Institutions of a Nilotic People*, Oxford: Clarendon Press, 1940.

Fancello, Sandra, *Les Aventuriers du Pentecôtisme Ghanéen*, Editions Karthala, Paris, 2006.

Federici, Silvia, *Witches, witch-hunting, and women*, PM Press, 2018.

Fukuyama, Francis, *The Origins of Political Order: From Prehuman Times to the French Revolution*, Cambridge University Press, 2012.

Gershman, Boris, "Witchcraft beliefs and the erosion of social capital: Evidence from Sub-Saharan Africa and beyond," *Journal of Development Economics*, 2016, *120*, 182–208.

Geschiere, P, "Parenté et argent dans une société lignagère," in J. F. Bayart, ed., *Handbook of the Economics of Art and Culture*, La réinvention du capitalisme, Paris : Karthala, 1994, p. 87 –113.

_ , *Witchcraft, Intimacy, and Trust: Africa in Comparison*, Chicago, IL: University of Chicago Press, 2013.

Goody, Jack, "Strategies of Heirship," *Comparative Studies in Society and History*, 1973, *15* (1), 3–20.

_ , *The Development of the Family and Marriage in Europe* Past and Present Publications, Cambridge University Press, 1983.

Guirkinger, Catherine and Jean-Philippe Platteau, *13. The Dynamics of Family Systems: Lessons from Past and Present Times*, Princeton: Princeton University Press,

Hackett, Conrad, Marcin Stonawski, Yunping Tong, Stephanie Kramer, Anne Shi, and Dalia Fahmy, "How the Global Religious Landscape Changed From 2010 to 2020," 2025.

Hagen, E. E, *The economics of development*, New York, NY: Richard D. Irwin, 1975.

Haugerud, Angelique, *The Culture of Politics in Modern Kenya* African Studies, Cambridge University Press, 1995.

Haynes, Naomi, "Pentecostalism and the morality of money: prosperity, inequality, and religious sociality on the Zambian Copperbelt," *Journal of the Royal Anthropological Institute*, 2012, *18* (1), 123–139.

Henrich, J. P, *The WEIRDest people in the world: how the West became psychologically peculiar and particularly prosperous*, New York, Farrar, Straus and Giroux, 2020.

Henry, Christine, "Le sorcier, le visionnaire et la guerre des Églises au Sud-Bénin," *Cahiers d'études africaines*, 2008, *189-190*, 101–130.

Hersey, Timothy, "Migration, Networks, and Religious Choice," *Working Paper*, 2024.

Heuser, Andreas, "Charting African prosperity gospel economies," *HTS: Theological Studies*, 2016, *72* (4), 1–9.

Iannaccone, Laurence R., "Sacrifice and Stigma: Reducing Free-riding in Cults, Communes, and Other Collectives," *Journal of Political Economy*, 1992, *100* (2), 271–291.

_ , "Why Strict Churches Are Strong," *American Journal of Sociology*, 1994, *99* (5), 1180–1211.

Iyer, Sriya, "The New Economics of Religion," *Journal of Economic Literature*, June 2016, *54* (2), 395–441.

Jenkins, Philip, *The Next Christendom: The Coming of Global Christianity*, Oxford University Press, 05 2002.

Kapepula, Gilbert Tshiebue, Max Mbosho Konshi, and Jonathan L. Weigel, "Prosociality and Pentecostalism in the D.R. Congo," *Religion, Brain & Behavior*, 2022, *12* (1-2), 150–170.

Kennedy, P, *African Capitalism: The Struggle for Ascendency*, Cambridge University Press, Cambridge, 1988.

Korotayev, Andrey V., "Unilineal Descent Organization and Deep Christianization: A Cross-Cultural Comparison," *Cross-Cultural Research*, 2003, *37* (1), 133–157.

Lowes, Sara, "Kinship Structure and the Family: Evidence from the Matrilineal Belt," Working Paper 30509, National Bureau of Economic Research 2022.

_ **, Benjamin Marx, and Eduardo Montero**, "Religion in Emerging and Developing Regions," Technical Report, National Bureau of Economic Research 2025.

Lugazia, Faith K., ""Theology of Presence" in African Christianity: A Transforming Missiological Factor for Women in Contemporary Pentecostal Churches in Africa," *International Review of Mission*, 2017, *106* (2), 307–321.

Mayston, Rosie, Souci Frissa, Bethlehem Tekola, Charlotte Hanlon, Martin Prince, and Abebaw Fekadu, "Explanatory models of depression in sub-Saharan Africa: Synthesis of qualitative evidence," *Social science & medicine*, 2020, *246*, 112760.

McCleary, Rachel M. and Robert J. Barro, *Protestants and Catholics and Educational Investment in Guatemala*, Cham: Springer International Publishing,

McGuirk, Eoin and Marshall Burke, "The Economic Origins of Conflict in Africa," *Journal of Political Economy*, 2020, *128* (10), 3940–3997.

McGuirk, Eoin F and Nathan Nunn, "Transhumant Pastoralism, Climate Change, and Conflict in Africa," *The Review of Economic Studies*, 03 2024, *92* (1), 404–441.

_ **and** _ , "Development Mismatch? Evidence from Agricultural Projects in Pastoral Africa," *Working Paper*, 2025.

Meyer, Birgit, "'Make a Complete Break with the past.' Memory and Post-Colonial Modernity in Ghanaian Pentecostalist Discourse," *Journal of Religion in Africa*, 1998, *28* (3), 316–349.

Miguel, Edward, "Poverty and Witch Killing," *The Review of Economic Studies*, 10 2005, *72* (4), 1153–1172.

Mildnerova, Kateřina, "African Independent Churches in Zambia (Lusaka)," *Ethnologia Actualis*, 12 2014, *14*.

Monfreda, Chad, Navin Ramankutty, and Jonathan A. Foley, "Farming the planet: 2. Geographic distribution of crop areas, yields, physiological types, and net primary production in the year 2000," *Global Biogeochemical Cycles*, 2008, *22* (1).

Murdock, George Peter, *Ethnographic Atlas*, Pittsburgh: University of Pittsburgh Press, 1967.

Mwaura, Philomena Njeri, "Chapter Sixteen Gender and Power in African Christianity : African Instituted Churches and Pentecostal Churches," in "in" 2013.

Norenzayan, Ara and Azim F. Shariff, "The Origin and Evolution of Religious Prosociality," *Science*, 2008, *322* (5898), 58–62.

Ntloedibe-Kuswani, Gomang Seratwa, *Witchcraft as a Challenge to Batswana Ideas of Community and Relationships*, Asmara, Eritrea: Africa World Press,

Nyaga, Stephen Nyoka, *The Impact of Witchcraft Beliefs and Practices on the Socio-Economic Development of the Abakwaya in Musoma-Rural District, Tanzania*, Asmara, Eritrea: Africa World Press,

Onyinah, Opoku, "Deliverance as a Way of Confronting Witchcraft in Contemporary Africa: Ghana as a Case Study," in Veli-Matti K"akrkk"ainen, ed., *The Spirit in the World*, Cambridge: William B. Eerdmans Publishing Company, 2009, pp. 181–202.

Parkin, D. J, *Palm, Wine and Witnesses*, London : Intertext Books, 1972.

Pew Research Center, "Overview: Pentecostalism in Africa," Spirit and Power: A 10-Country Survey of Pentecostals, Pew Forum on Religion & Public Life, Washington, DC 2006. Accessed 2026-09-09.

_ , "Tolerance and Tension: Islam and Christianity in Sub-Saharan Africa," 2009.

Platteau, Jean-Philippe, *Institutions, social norms and economic developmen*, Harwood Academic, 2000.

_ , "Institutional obstacles to African economic development: State, ethnicity, and custom," *Journal of Economic Behavior & Organization*, 2009, *71* (3), 669–689. The Economic Performance of Civilizations: Roles of Culture, Religion and the Law.

_ , *Redistributive Pressures in Sub-Saharan Africa: Causes, Consequences, and Coping Strategies*, Cambridge University Press,

_ , *The Economics of Religious Conversion in Sub-Saharan Africa*, Emerald Publishing Limited,

Roessler, Philip, Yannick I. Pengl, Robert Marty, Kyle Sorlie Titlow, and Nicolas van de Walle, "The cash crop revolution, colonialism and economic reorganization in Africa," *World Development*, 2022, *158*, 105934.

Rossignol, Etienne Le, Sara Lowes, and Nathan Nunn, "Culture at a Crossroads: Traditional Beliefs in Contemporary Africa," *Working Paper*, 2025.

Sarro, R, *The politics of religious change on the Upper Guinea coast. Iconoclasm done and undone*, Edinburgh: Edinburgh University Press, 2009.

Schulz, Jonathan F, "Kin Networks and Institutional Development," *The Economic Journal*, 04 2022, *132* (647), 2578–2613.

Sievert, Clara, "Supernatural Beliefs about Illness and the Use of Modern Medicine: Evidence from the DR Congo," *Working Paper*, 2025.

Smith, Gregory A, Alan Cooperman, Becka A Alper, Besheer Mohamed, Chip Rotolo, P Tevington, J Nortey, A Kallo, J Diamant, and D Fahmy, "Decline of Christianity in the US has slowed, may have leveled off," *PEW Research Center*, 2025.

Solá, Daniela, "Brother Votes for Brother: The Effects of Pentecostal Political Influence in Brazil," *Working Paper*, 2025.

Squicciarini, Mara P., "Devotion and Development: Religiosity, Education, and Economic Progress in Nineteenth-Century France," *American Economic Review*, November 2020, *110* (11), 3454–91.

ter Haar, Gerrie, *Imagining Witchcraft: Witchcraft Beliefs and Accusations in Contemporary Africa*, Asmara, Eritrea:Africa World Press, 2007.

The Economist, "What is an evangelical Christian?," *The Economist*, 2021. Accessed: 2025-09-13.

_ , "Evangelicals may soon rival Catholics in Latin America," *The Economist*, 2023. Accessed: 2025-09-13.

The Sanneh Institute, "Report on Alleged Witches' Camps in Ghana," 2021.

The World Bank Group, "Theme Taxonomy and Definitions," 2016.

Theiss, Katherine, "Time-of-day Effects on Disclosure of Intimate Partner Violence," *Working Paper*, 2024.

Thelen, F. M, "The Explosive Growth of Pentecostal-Charismatic Christianity in the Global South, and its Implications for Catholic Evangelization," 2017.

Thiam, Awa, *La Parole aux négresses*, Paris: Denoël/Gonthier, 1978.

Ventevogel, Peter, Mark Jordans, Ria Reis, and Joop de Jong, "Madness or sadness? Local concepts of mental illness in four conflict-affected African communities," *Conflict and health*, 2013, *7* (1), 3.

Zanden, J. Luiten Van, *The Long Road to the Industrial Revolution. The European Economy in a Global Perspecti-ve, 1000-1800*, Brill, 2009.

Zurlo, Gina A. and Todd M. Johnson, "World Christian Database," *Leiden/Boston: Brill*, 2024.

Online Appendix

Shaking the Traditional Order: Women's Conversion to New Christian Churches in Sub-Saharan Africa

September 17, 2026

Pablo Álvarez-Aragón
University of Bologna

Catherine Guirkinger
University of Namur

Jean-Philippe Platteau
University of Namur

Appendix Contents

A The New Reformation in Sub-Saharan Africa **1**

B Data Description and Variable Definitions **6**
B.1 Large-scale surveys across Sub-Saharan Africa 6
B.2 Randomized Controlled Trial in Benin 11

C Additional Figures and Tables **16**
C.1 Gender gaps in religious affiliation 16
C.2 RCT Benin 18
C.3 Economic opportunity I: Shocks to crop prices and new Christian churches 22
C.4 Economic opportunity II: Development aid projects 41
C.5 New Christian churches and decision-making 50
C.6 Prohibitions in new Christian churches: alcohol and polygyny 53
C.7 New Christian churches vis-à-vis traditional beliefs and practices 54

List of Appendix Figures

A1 Religious affiliation over time in Sub-Saharan Africa, raw measure 1
A2 Religious affiliation over time in Sub-Saharan Africa, restricted measure . . . 2
A3 Religious affiliation over time in Sub-Saharan Africa, World Christian Database 3
A4 Religious affiliation by year of birth in Sub-Saharan Africa, raw and restricted measures 3
A5 New Christian church membership by country and survey year 4
A6 Husband's religion given wife's 4
A7 Wife's religion given husband's 5
C1 Leave-one-group-out stability of the treatment effect on conversion to new Christianity 21
C2 Correlation between belief in ancestral spirits and religion 21
C3 Cash crop suitability across Africa 22
C4 Price indices for agricultural commodities (Base year: 2010) 23
C5 Crop suitability, M3 database 33
C6 Women-related words by type of project, different definitions 48

List of Appendix Tables

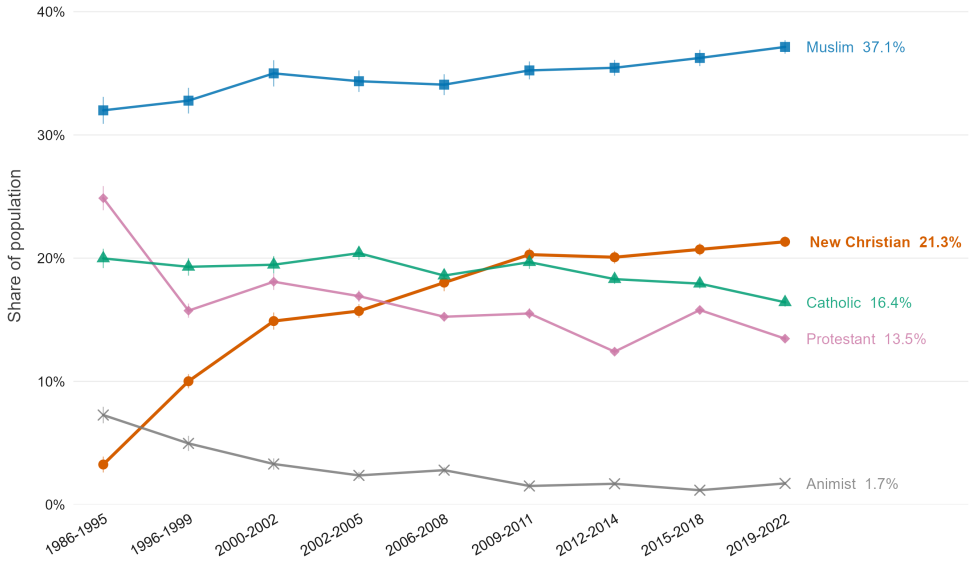
B1 DHS country-waves in which New Christians are (likely) grouped with Protestants 9
B2 RCT Benin, baseline characteristics 12
B3 RCT Benin, baseline characteristics in the endline 1 sample 13
B4 RCT Benin, baseline characteristics in the follow-up sample 14
B5 Differential attrition across treatment and control groups 15
C1 Average religious affiliation by gender, raw data 16

C2 Religious affiliation and gender, restricted definition . . . 17
C3 Religious affiliation and gender, raw definition . . . 17
C4 Religious affiliation and gender, different datasets . . . 18
C5 Treatment effects: religious conversion, opposition and witchcraft . . . 18
C6 Treatment on the Treated: religious conversion, opposition and witchcraft . . . 19
C7 Treatment on the Treated: religious conversion, opposition and witchcraft. No controls . . . 19
C8 Witchcraft, spiritual protection and NCC conversion . . . 20
C9 New Christianity and protection by husbands . . . 20
C10 Distribution of conversions across training groups . . . 21
C11 Effect on wealth and income . . . 24
C12 Shocks to crops and conversion to/from different religions . . . 24
C13 Shocks to crops and conversion to/from different religions, 25° × 25° cells (Roessler et al., 2022) . . . 25
C14 Positive vs. negative shocks to cash-crop prices and conversion to new Christian churches . . . 26
C15 Positive vs. negative cash-crop price shocks and conversion to new Christian churches . . . 27
C16 Shocks to cash crops and conversion to New Christian churches, raw definition 28
C17 Shocks to cash crops and conversion to New Christian churches, restricted definition . . . 29
C18 Shocks to cash crops and conversion to new Christian churches — Conley standard errors . . . 30
C19 Shocks to cash crops and conversion to new Christian churches, country-year clustering . . . 31
C20 Shocks to cash crops and conversion to New Christian churches, 0.25° × 0.25° cells (Roessler et al., 2022) . . . 32
C21 Shocks to cash crops and conversion to New Christian churches, M3 crop database . . . 34
C22 Cash crop price shocks and new Christian conversion: robustness to alternative baseline interactions . . . 35
C23 Shocks to cash crops and conversion to new Christian churches, witchcraft from PEW . . . 37
C24 Shocks to cash crops, women participation in cash crop cultivation and conversion . . . 38
C25 Shocks to cash crops, historical participation of women in agriculture and conversion . . . 40
C26 New Christians and World Bank gender-related projects, raw definition (55x55km) . . . 41
C27 New Christians and World Bank gender-related projects, restricted definition (55x55km) . . . 42
C28 New Christians and World Bank gender-related projects, country-year clustering . . . 43

C29 New Christians and World Bank gender-related projects — Conley standard errors . . . 44
C30 New Christians and World Bank gender-related projects, witchcraft from PEW 45
C31 New Christians and World Bank gender-related projects, 28x28km . . . 46
C32 Women-related words and World Bank projects . . . 48
C33 New Christians and World Bank gender-related projects, alternative WB gender project measure (55x55km) . . . 49
C34 New Christian churches, labor and reproductive outcomes, and decision-making, raw definition . . . 50
C35 New Christian churches, labor and reproductive outcomes, and decision-making, restricted definition . . . 50
C36 New Christian churches and women's participation in decision-making, different components . . . 51
C37 New Christian churches and women's participation in decision-making, 1-3 outcome . . . 52
C38 New Christianity and decisions about the use of women's earnings . . . 52
C39 New Christian churches and alcohol consumption . . . 53
C40 New Christian churches and polygny . . . 53
C41 Traditional beliefs and gender in SSA . . . 54
C42 Traditional beliefs or practices and gender in South Benin . . . 54
C43 New Christians and attitudes toward tradition and supernatural power, by gender . . . 55
C44 New Christianity and trust in traditional leaders and the family . . . 56
C45 Treatment effects on protection against witchcraft, by NCC conversion . . . 57

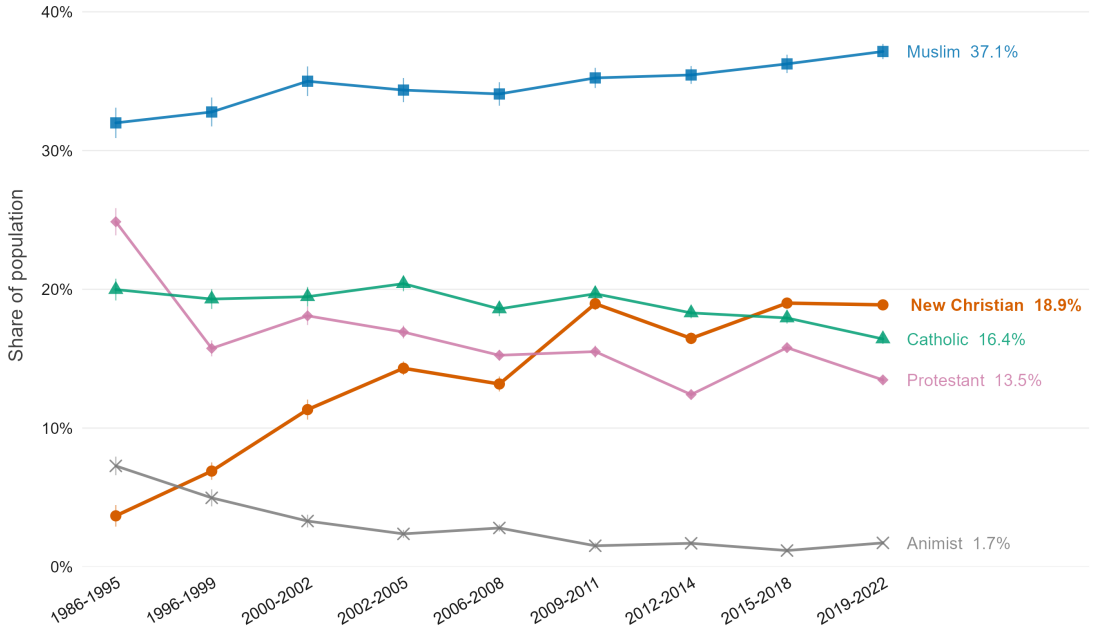
A. The New Reformation in Sub-Saharan Africa

Figure A1: Religious affiliation over time in Sub-Saharan Africa, raw measure



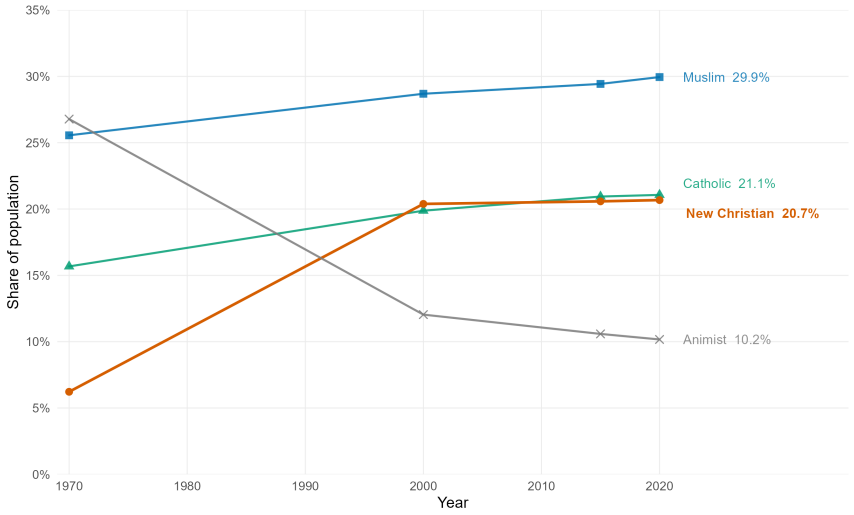
Note: Data come from 147 Demographic and Health Surveys (DHS) in 39 Sub-Saharan African countries. The figure shows country-fixed-effects-adjusted shares of the population belonging to each religious denomination by survey-year period, estimated via OLS with country fixed effects and standard errors clustered at the DHS cluster level. Declared religion is measured at the time of the survey. "New Christian" (NCC) denotes Pentecostals, Charismatics, Evangelicals, African Independent Churches, Assemblies of God, Apostolic churches, *Églises de réveil*, and related movements. This figure uses the *raw* measure: DHS religion categories are used as reported. Countries include: Angola, Benin, Burkina Faso, Burundi, Cameroon, Central African Republic, Chad, Comoros, Congo, DRC, Côte d'Ivoire, Eswatini, Ethiopia, Gabon, Ghana, Guinea, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sao Tome & Principe, Senegal, Sierra Leone, South Africa, Sudan, Tanzania, The Gambia, Togo, Uganda, Zambia, Zimbabwe.

Figure A2: Religious affiliation over time in Sub-Saharan Africa, restricted measure



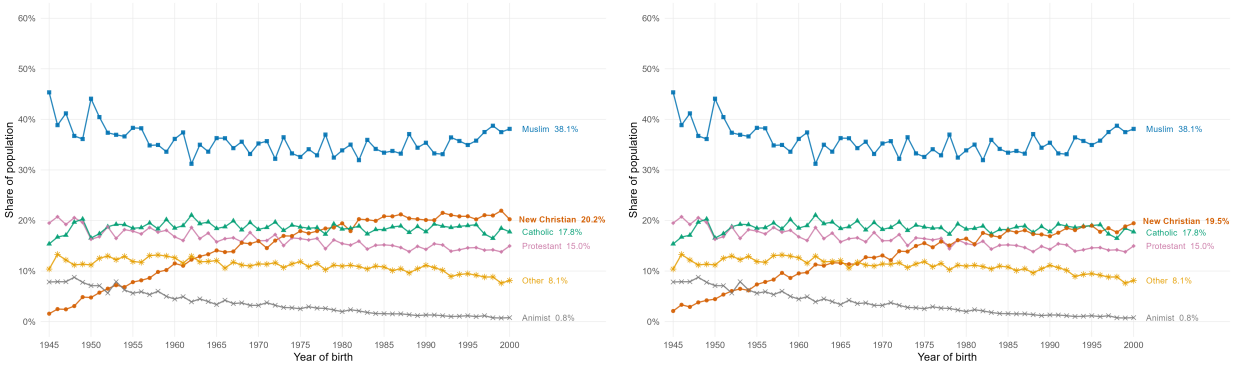
Note: Data come from 120 Demographic and Health Surveys (DHS) in 35 Sub-Saharan African countries. The figure shows country-fixed-effects-adjusted shares of the population belonging to each religious denomination by survey-year period, estimated via OLS with country fixed effects and standard errors clustered at the DHS cluster level. Declared religion is measured at the time of the survey. "New Christian" (NCC) denotes Pentecostals, Charismatics, Evangelicals, African Independent Churches, Assemblies of God, Apostolic churches, *Églises de réveil*, and related movements. This figure uses the *restricted* measure: the 27 country-waves in which NCC cannot be separately identified from Protestants are excluded from the sample, namely: Angola (2011), Central African Republic (1994), Côte d'Ivoire (1994, 1998), Ethiopia (2000, 2005, 2010, 2016, 2019), Kenya (2003, 2008, 2014), Liberia (1986, 2007, 2013, 2019), Nigeria (1990, 2008, 2013, 2018, 2024), Togo (1988), and Zambia (1996, 2001, 2007, 2013, 2018); these waves represent 23.5% of the full sample. Four countries are entirely absent from the restricted sample as all their DHS waves are flagged (Central African Republic, Ethiopia, Liberia, Zambia). Countries include: Angola, Benin, Burkina Faso, Burundi, Cameroon, Chad, Comoros, Congo, DRC, Côte d'Ivoire, Eswatini, Gabon, Ghana, Guinea, Kenya, Lesotho, Madagascar, Malawi, Mali, Mauritania, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sao Tome & Principe, Senegal, Sierra Leone, South Africa, Sudan, Tanzania, The Gambia, Togo, Uganda, Zimbabwe.

Figure A3: Religious affiliation over time in Sub-Saharan Africa, World Christian Database



Note: Data come from the World Christian Database. Countries include: Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Central African Republic, Chad, Comoros, Congo, DRC, Côte d'Ivoire, Djibouti, Equatorial Guinea, Eritrea, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mauritius, Mayotte, Mozambique, Namibia, Niger, Nigeria, Rwanda, Réunion, Saint Helena, Sao Tome & Principe, Senegal, Seychelles, Sierra Leone, Somalia, South Africa, South Sudan, Tanzania, Togo, Uganda, Zambia, Zimbabwe.

Figure A4: Religious affiliation by year of birth in Sub-Saharan Africa, raw and restricted measures

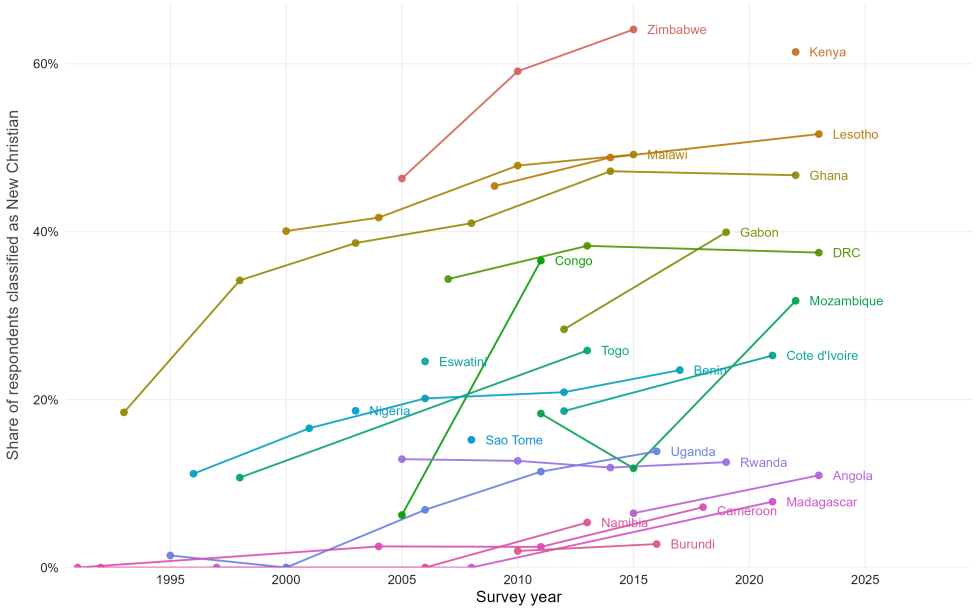


(a) Raw measure

(b) Restricted measure

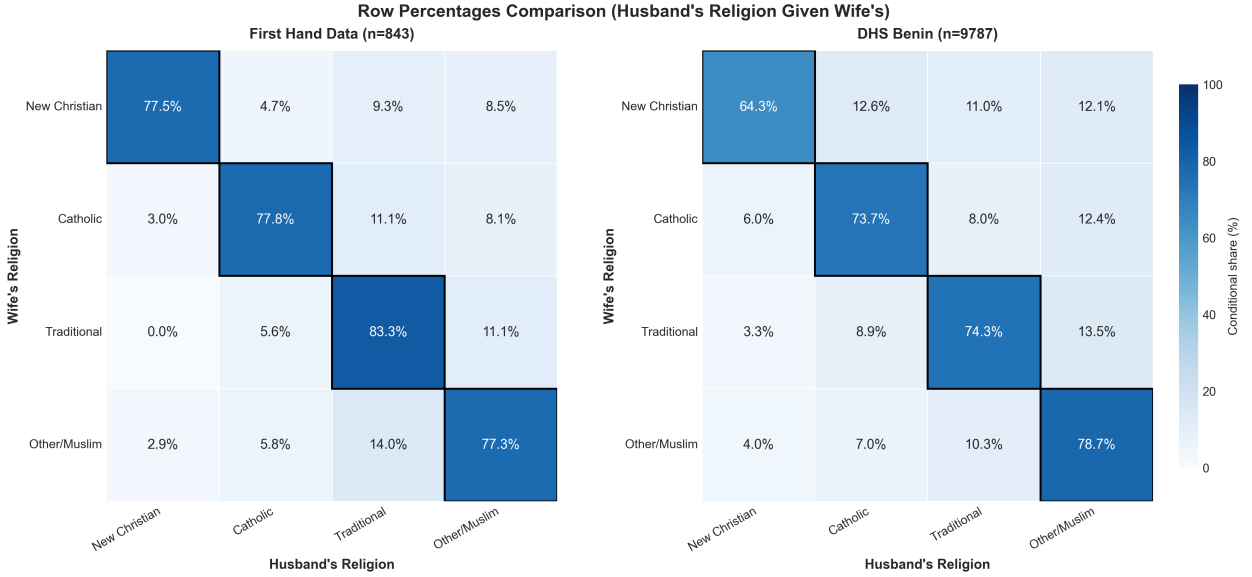
Notes: Authors' calculations using data from Demographic and Health Surveys (DHS) in Sub-Saharan Africa. The figure shows the share of people belonging to different religious denominations by year of birth. Declared religion is measured at the time of the survey. "New Christian" (NCC) denotes Pentecostals, Charismatics, Evangelicals, African Independent Churches, Assemblies of God, Apostolic churches, *Églises de réveil*, and related movements. Panel A uses the *raw* measure (147 surveys, 39 countries): DHS religion categories are used as reported; NCC and Protestants are kept separate in survey waves where the questionnaire distinguishes them, and pooled otherwise; in the pooled waves, the NCC share is mechanically zero or artificially low. Panel B uses the *restricted* measure (120 surveys, 35 countries): the 27 country-waves in which NCC cannot be separately identified from Protestants are excluded, namely: Angola (2011), Central African Republic (1994), Côte d'Ivoire (1994, 1998), Ethiopia (2000, 2005, 2010, 2016, 2019), Kenya (2003, 2008, 2014), Liberia (1986, 2007, 2013, 2019), Nigeria (1990, 2008, 2013, 2018, 2024), Togo (1988), and Zambia (1996, 2001, 2007, 2013, 2018); four countries (Central African Republic, Ethiopia, Liberia, Zambia) are entirely absent from the restricted sample as all their DHS waves are flagged. Raw-measure countries (39): Angola, Benin, Burkina Faso, Burundi, Cameroon, Central African Republic, Chad, Comoros, Congo, DRC, Côte d'Ivoire, Eswatini, Ethiopia, Gabon, Ghana, Guinea, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sao Tome & Principe, Senegal, Sierra Leone, South Africa, Sudan, Tanzania, The Gambia, Togo, Uganda, Zambia, Zimbabwe. Restricted-measure countries (35): same list excluding Central African Republic, Ethiopia, Liberia, and Zambia.

Figure A5: New Christian church membership by country and survey year



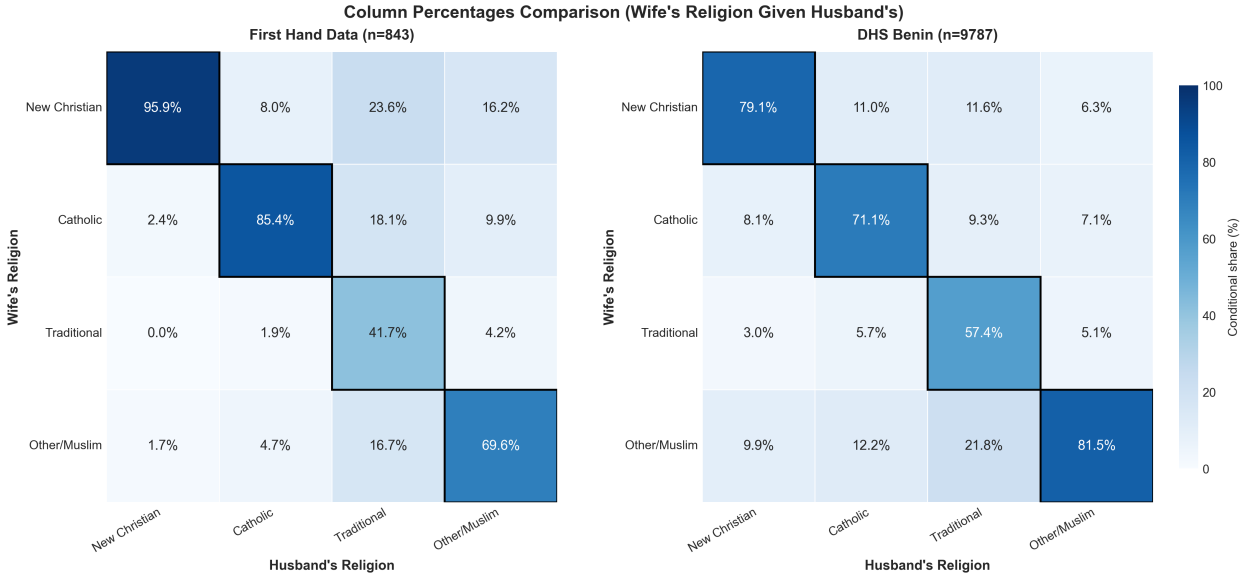
Source: Authors' calculations using data from the Demographic and Health Surveys (DHS). The figure shows raw survey-wave shares of respondents classified as new Christian, by country and survey year. Only country-waves where new Christian denominations are separately identified in the DHS are included. "New Christian" equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination, using DHS categories as reported without recoding. Countries shown are those where the new Christian share exceeds 2 percent in at least one survey wave: Angola, Benin, Congo, Côte d'Ivoire, D.R. Congo, Eswatini, Gambia, Ghana, Kenya, Lesotho, Madagascar, Malawi, Mozambique, Togo, Uganda, Zimbabwe.

Figure A6: Cross-tabulation of husbands' and wives' religions (row proportions sum to 100%)



Notes: Each cell shows the percentage of husbands with a given religion conditional on their wife's religion (rows sum to 100%). First Hand Data come from the 2025 survey conducted in Benin (N = 843 couples). DHS data come from the Demographic and Health Survey for Benin (2021). Religious categories are mutually exclusive: "New Christian" includes Evangelical, Pentecostal, Apostolic, Charismatic, and related denominations, "Catholic" includes Roman Catholics, "Traditional" includes Vodoun and other indigenous religions, and "Other/Muslim" includes Muslims and other unclassified religions.

Figure A7: Cross-tabulation of husbands' and wives' religions (column proportions sum to 100%)



Notes: Each cell shows the percentage of wives with a given religion conditional on their husband's religion (columns sum to 100%). First Hand Data come from the 2025 survey conducted in Benin (N = 843 couples). DHS data come from the Demographic and Health Survey for Benin (2021). Religious categories are mutually exclusive: "New Christian" includes Evangelical, Pentecostal, Apostolic, Charismatic, and related denominations, "Catholic" includes Roman Catholics, "Traditional" includes Vodoun and other indigenous religions, and "Other/Muslim" includes Muslims and other unclassified religions.

B. Data Description and Variable Definitions

B.1. Large-scale surveys across Sub-Saharan Africa

B.1.1. Demographic and Health Surveys

We use data from 147 DHS surveys from 39 Sub-Saharan African countries. These are nationally representative surveys which, in addition to detailed information on the religious affiliation of individuals at the denomination level, contain useful information on sociodemographic characteristics, fertility preferences, or contraceptive methods. In total, we have information on about 2,300,000 individuals living in more than 65,000 clusters.

Countries and survey years.– Angola (2011, 2015, 2023), Benin (1996, 2001, 2006, 2011, 2017), Burkina Faso (1993, 1999, 2003, 2010, 2021), Burundi (2010, 2016), Cameroon (1991, 2004, 2011, 2018), Central African Republic (1994), Chad (1996, 2004, 2014), Comoros (2012), Congo Brazzaville (2005, 2011), Democratic Republic of Congo (2007, 2013, 2023), Côte d'Ivoire (1994, 1998, 2012, 2021), Eswatini (2006), Ethiopia (2000, 2005, 2011, 2016, 2019), Gabon (2012, 2019), Gambia (2013, 2019), Ghana (1993, 1998, 2003, 2008, 2014, 2022), Guinea (1999, 2005, 2012, 2018), Kenya (2003, 2008, 2014, 2022), Lesotho (2004, 2009, 2014, 2023), Liberia (1986, 2007, 2013, 2019), Madagascar (1997, 2008, 2021), Malawi (1992, 2000, 2004, 2010, 2015), Mali (1995, 2001, 2006, 2012, 2018, 2023), Mauritania (2020), Mozambique (2011, 2015, 2022), Namibia (1992, 2000, 2006, 2013), Niger (1992, 1998, 2006, 2012, 2021), Nigeria (1990, 2003, 2008, 2013, 2018, 2024), Rwanda (2005, 2010, 2014, 2019), Sao Tome and Principe (2008), Senegal (1986, 1992, 1997, 2005, 2010, 2012, 2014, 2015, 2016, 2017, 2018, 2019, 2023), Sierra Leone (2008, 2013, 2019), South Africa (1998, 2016), Sudan (1990), Tanzania (1992, 1996, 1999, 2007, 2010, 2015, 2022), Togo (1988, 1998, 2013), Uganda (1995, 2000, 2006, 2011, 2016), Zambia (1996, 2001, 2007, 2013, 2018), Zimbabwe (1994, 1999, 2005, 2010, 2015).

Identifying New Christian Churches (NCC) in the DHS

The measurement problem.– In several DHS country-waves, adherents of new Christian Churches (NCC) — Pentecostals, Charismatics, Evangelicals, African Independent Churches (AIC), *Églises de réveil*, Assemblies of God, Apostolic churches, P'ent'ay, and related movements — are not reported as a separate category. They are instead pooled with mainline (missionary) Protestants or, in some waves, absorbed into a broader *other Christian* category. Because our analysis hinges on the NCC share, we need a transparent rule for deciding when this pooling is a substantive problem, when it is not, and how to correct for it. The list of country-waves we flag as problematic is reported in Table B1.

Classification criterion.– A country-wave is flagged as "problematic" in Table B1 only when *both* of the following conditions hold:

(i) **DHS coding.** The DHS questionnaire in that wave does not separately identify NCC denominations from mainline Protestants (or absorbs them into a broad *other Christian* category); *and*

(ii) **External evidence.** Independent estimates — primarily the Pew Research Center's country-level figures on the Pentecostal and Charismatic ("renewalist") population (Pew Research Center, 2006) — indicate a non-negligible NCC presence in the country.

This joint criterion ensures that our correction addresses a genuine measurement problem rather than mechanically reclassifying respondents who are, in all likelihood, mainline Protestants.

Country-waves not *flagged despite the absence of a distinct NCC category.*– In several country-waves the DHS reports Christians only under a mainline "Protestant" label (or, at most, a generic "Christian" label), yet we do *not* pool them with NCC because Pew and complementary sources (Pew Research Center, 2006) document a very low or negligible renewalist presence in these countries. In such cases the respondents recorded as Protestant are overwhelmingly mainline Protestants rather than misclassified NCC, and pooling would introduce spurious NCC where essentially none exists. This applies to:

- **Predominantly Muslim countries with a small Christian minority** (Burkina Faso, Chad, Comoros, Gambia, Guinea, Mali, Mauritania, Niger, Senegal, Sierra Leone, Sudan), where the residual Christian population is overwhelmingly Catholic or historical (mainline) Protestant and Pew reports negligible renewalist shares.
- **Countries with a documented low renewalist share**, such as Madagascar (placed by Pew Research Center (2006) in the lower-end of the new Christian presence distribution), and Namibia (mostly missionary Protestants and Lutherans; a few Seventh-day Adventists appear separately from 2013). In Cameroon, zero NCC are recorded in 1991 and a small share is properly identified from 2004 onwards.
- **Countries with no historical NCC penetration in early rounds** (Tanzania before 2007, Uganda before 2006), where NCC first appear as small shares in later waves and are correctly reported as separate categories from the moment they become non-negligible.

Three alternative NCC measures.– To assess the robustness of the results to this measurement issue, the paper uses three alternative measures of the NCC share. Each is built

from the same underlying list of NCC denominations and differs only in how it treats the problematic country-waves of Table B1.

Baseline NCC denomination list. Across all three measures, an individual is coded as "new Christian" if her main declared religion is one of: *other Christian, Evangelical, Pentecostal, Apostolic, Salvation Army/Armée du Salut, Charismatic, Assembly of God, Église du Réveil, Seventh-day Adventist, Kimbanguist, Other Protestant, African Initiated Church, Revival church*. The three measures differ only in how respondents coded as *Protestant* are treated in the country-waves of Table B1.

1. Grouped measure. For every country in which at least one survey wave fails to distinguish NCC from Protestants (i.e. any country appearing in Table B1), NCC and Protestants are collapsed into a single *Protestant + NCC* category. This guarantees that the religion variable is measured consistently over time within each country, without dropping a significant share of the sample. Concretely, the "new Christian" indicator equals one for the baseline denomination list *plus* anyone coded as Protestant in a country listed in Table B1.

2. Raw measure. The raw DHS categories are used as reported, with no adjustment. NCC and Protestants are kept separate whenever the questionnaire distinguishes them, and pooled whenever it does not. This maximizes the use of the information available in each wave but implies that, in the country-waves listed in Table B1, the measured NCC share is mechanically zero (or artificially low) even when Pew evidence (Pew Research Center, 2006) suggests a non-negligible NCC presence. The "new Christian" indicator equals one for the baseline denomination list only.

3. Restricted measure. The country-waves listed in Table B1 are *dropped* from the sample. NCC and Protestants are kept as separate categories in the remaining waves. The "new Christian" indicator equals one for the baseline denomination list only. This yields the cleanest measure of the NCC share, at the cost of a reduced sample size: the Restricted sample contains about 1.66 million observations, against 2.22 million under the Grouped measure — a drop of roughly one quarter (consistent with the 23.52% figure reported in the last row of Table B1). Correspondingly, the measured NCC share falls from 22.96% under the Grouped measure to 15.87% under the Restricted measure.

The Restricted measure should be viewed as a lower bound rather than a neutral benchmark: in country-waves where the pooled *Protestant + NCC* category is in fact dominated by NCC, dropping the wave discards observations that the Grouped measure

would have classified (approximately) correctly. Ethiopia is a clear illustration: the World Christian Database (2005) estimates the P'ent'ay share at roughly 16% of the population, and the DHS *Protestant* category in the problematic Ethiopian waves likewise records about 16% of respondents, indicating that essentially all DHS "Protestants" in Ethiopia are in fact NCC. Excluding such waves therefore removes genuine NCC from the analysis rather than reducing measurement error.

Table B1: DHS country-waves in which New Christians are (likely) grouped with Protestants

Country	Problematic wave(s)	% of sample	Rationale
Angola	2011	0.36	Only "Protestant" category appears. *Pew evidence:* upper-end country (>20% Pentecostal/Charismatic).
Central African Republic	1994	0.32	Only "Protestant" category appears. *Pew evidence:* intermediate country (10–20% Pentecostal/Charismatic).
Côte d'Ivoire	1994, 1998	0.61	*DHS coding:* in 1994 and 1998, only the Protestant category appears. In subsequent rounds, the "Protestant" category disappears and instead "Methodist", "Evangelical" and "Other Christian" emerge.
Ethiopia	2000, 2005, 2010, 2016, 2019	4.43	*World Christian Database:* P'ent'ay comprises ≈16% of the population. In the DHS, the religion variable includes "Protestant", covering 16.4% of the pooled sample.
Kenya	2003, 2008, 2014	2.83	Only "Protestant" category appears. *DHS coding:* Before 2022, only the "Protestant" label is included in the survey. In 2022, when the denominations inside the "Protestant" category are dissaggreated, "Evangelical" and "AIC" emerge while the "Protestant" share declines.
Liberia	1986, 2007, 2013, 2019	1.84	Only "Protestant/Christian" category appears. *Pew evidence:* intermediate country (10–20% Pentecostal/Charismatic).
Nigeria	1990,2008, 2013, 2018, 2024	9.22	Only "Protestant" category appears (1990). Only "Other Christian" category appears (2008, 2013, 2018, 2024). *Pew evidence:* upper-end country (>20% Pentecostal/Charismatic)
Togo	1988	0.14	Only "Protestant" category appears. *DHS coding:* in 1998, "Presbyterian" includes 6.64% of the sample, and "other Christian" includes 10.7%. In 1988, only the "Protestant" category appears, including 11% of the sample.
Zambia	1996, 2001, 2007, 2013, 2018	3.78	Only "Protestant" category appears. *Pew evidence:* upper-end country (>20% Pentecostal/Charismatic).
Total		23.52	

Notes: A country-wave is classified as "problematic" when *both* (i) the DHS religion variable does not separately identify new Christian Churches from mainline Protestants (or absorbs them into a broader *other Christian* category), and (ii) external evidence (primarily Pew Research Center) indicates a non-negligible NCC population in the country. Country-waves in which the entire Christian population is reported under a single *Christian* label (e.g. Zimbabwe 1994, 1999) raise a distinct measurement problem that is not addressed by the three NCC definitions and are therefore excluded from this table. The "% of sample" column reports the share of observations in the full pooled DHS sample (N = 2,391,043) contributed by the listed country-waves.

B.1.2. PEW Research Forum

We complement the demographic surveys with individual-level information on contemporary supernatural beliefs from 25,000 respondents in 19 SSA countries. The information comes from a survey conducted between December 2008 and April 2009 by the Pew Research Center's Forum on Religion and Public Life (Pew Research Center, 2009). In addition to detailed information on religious affiliation, this database provides individual-level information on religious beliefs, practices, and knowledge for all religions including African traditional religions, and it also provides information about the goods and services offered by the different churches.

Countries.– Botswana, Cameroon, Chad, Democratic Republic of the Congo, Djibouti, Ethiopia, Ghana, Guinea Bissau, Kenya, Liberia, Mali, Mozambique, Nigeria, Rwanda, Senegal, South Africa, Tanzania, Uganda, Zambia.

Variable definitions.– PEW specifically includes a question aimed at identifying new Christian movements. We code the variable "new Christian church" as equal to one if the respondent is Christian *and* mentions that s/he identifies with one of the following churches: born-again Christian, evangelical Christian, Pentecostalist Christian, or charismatic Christian.

B.1.3. Afrobarometer

We also use information from Afrobarometer, which deals with more than 300,000 respondents from 40 countries. As before, this dataset contains information on both gender and (detailed) religious affiliation.

Countries and survey year.– Angola (2019, 2022), Benin (2005, 2008, 2011, 2014, 2016, 2020, 2022), Botswana (2003, 2005, 2008, 2012, 2014, 2017, 2019, 2022), Burkina Faso (2008, 2012, 2015, 2017, 2019, 2022), Burundi (2012, 2014), Cape Verde (2002, 2005, 2008, 2011, 2014, 2017, 2019, 2022), Cameroon (2013, 2018, 2021, 2022), Côte d'Ivoire (2013, 2014, 2017, 2019, 2021), Eswatini (2013, 2015, 2018, 2021, 2022), Ethiopia (2020), Gabon (2015, 2017, 2020, 2021), Gambia (2018, 2021, 2022), Ghana (2002, 2005, 2008, 2012, 2014, 2017, 2019, 2022), Guinea (2013, 2015, 2017, 2019, 2022), Kenya (2003, 2005, 2008, 2011, 2014, 2016, 2019, 2021), Lesotho (2003, 2005, 2008, 2012, 2014, 2017, 2020, 2022), Liberia (2008, 2012, 2015, 2018, 2020, 2022), Madagascar (2005, 2008, 2013, 2014, 2018, 2022), Malawi (2003, 2005, 2008, 2012, 2014, 2016, 2019, 2022), Mali (2002, 2005, 2008, 2012, 2014, 2017, 2020, 2022), Mauritania (2022), Mauritius (2012, 2014, 2017, 2020, 2022), Mozambique (2002, 2005, 2008, 2012, 2015, 2021), Namibia (2003, 2006, 2008, 2012, 2014, 2017, 2019, 2021), Niger (2013, 2015, 2018, 2020, 2022), Nigeria (2003, 2005, 2008, 2012, 2014, 2017, 2020, 2022), Senegal (2002, 2005, 2008, 2013, 2014, 2017, 2020, 2022), Seychelles (2022),

Sierra Leone (2012, 2015, 2018, 2020, 2022), South Africa (2002, 2006, 2008, 2011, 2015, 2018, 2021, 2022), Sudan (2013, 2015, 2018, 2021, 2022), Sao Tome and Principe (2015, 2018, 2022), Tanzania (2003, 2005, 2008, 2012, 2014, 2017, 2021, 2022), Togo (2012, 2014, 2017, 2020, 2022), Tunisia (2013, 2015, 2018, 2020, 2022), Uganda (2002, 2005, 2008, 2012, 2015, 2016, 2019, 2022), Zambia (2003, 2005, 2009, 2013, 2014, 2017, 2020, 2022), Zimbabwe (2004, 2005, 2009, 2012, 2014, 2017, 2021, 2022).

Variable definitions.– We now define "new Christian church" as an indicator variable that equals one if the respondent's religion is Other Christian, Apostolic, Church of Christ, Evangelical, African Initiated Church, Jehovah's Witness, Pentecostal, Seventh-day Adventist, Zionist Christian Church, or Christian (as a different category than Catholic or Protestant).

B.1.4. World Value Surveys

Finally, we use the World Value Surveys, which are large-scale demographic surveys supplying information on gender and religious affiliation for more than 45,000 respondents from 12 Sub-Saharan African countries.

Countries and survey year.– Ethiopia (2007, 2020), Ghana (2007, 2012), Kenya (2021), Mali (2007), Nigeria (1990, 1995, 2000, 2012, 2018), Rwanda (2007, 2012), South Africa (1982, 1990, 1996, 2001, 2006, 2013), Zimbabwe (2001, 2012, 2020), Uganda (2001), Tanzania (2001), Burkina Faso (2007), Zambia (2007).

Variable definitions.– We define "new Christian church" as an indicator variable equal to one if the respondent's main religious affiliation is Evangelical, Jehovah's Witness, Pentecostal, Charismatic, Other Christian, African Initiated Church, or Zion Christian Church.

B.2. Randomized Controlled Trial in Benin

Summary Statistics.– Table B2 examines the balance in some baseline variables between treatment and control. There are no significant differences between the treatment and control groups, neither for demographic variables nor for socioeconomic and gender-related outcomes, as we expected from computerized randomization.

Table B2: RCT Benin, baseline characteristics

	Mean / (SD)			Difference / [p-value]		
	Control	Couple	Individual	Couple-Control	Individual-Control	Obs.
New Christian	0.363	0.371	0.357	0.008	-0.006	1009
	(0.482)	(0.484)	(0.480)	[0.834]	[0.873]	
Age woman	37.802	37.531	36.793	-0.271	-1.010	1002
	(10.389)	(10.125)	(9.602)	[0.733]	[0.193]	
Number children	4.244	4.151	4.131	-0.093	-0.113	1009
	(2.315)	(2.241)	(2.418)	[0.598]	[0.536]	
Mother produced pineapple	0.102	0.083	0.101	-0.018	-0.000	1005
	(0.303)	(0.277)	(0.302)	[0.410]	[0.990]	
Father produced pineapple	0.171	0.150	0.169	-0.021	-0.002	999
	(0.377)	(0.358)	(0.375)	[0.471]	[0.946]	
Bank account	0.167	0.125	0.140	-0.042	-0.027	1009
	(0.373)	(0.331)	(0.347)	[0.122]	[0.336]	
Has mobile money	0.426	0.433	0.363	0.008	-0.062*	1009
	(0.495)	(0.496)	(0.482)	[0.842]	[0.098]	
Owns TV	0.375	0.362	0.333	-0.013	-0.042	1009
	(0.485)	(0.481)	(0.472)	[0.728]	[0.259]	
Phone	0.667	0.691	0.664	0.025	-0.003	1009
	(0.472)	(0.463)	(0.473)	[0.493]	[0.935]	
Attended school	0.399	0.421	0.393	0.023	-0.006	1009
	(0.490)	(0.495)	(0.489)	[0.553]	[0.875]	
Pineapple not for women	0.170	0.205	0.179	0.035	0.009	1009
	(0.376)	(0.404)	(0.384)	[0.244]	[0.761]	
Extension area	0.500	0.472	0.473	-0.028	-0.027	1009
	(0.501)	(0.500)	(0.500)	[0.465]	[0.488]	
Isolated	0.027	0.039	0.054	0.012	0.027*	1009
	(0.162)	(0.193)	(0.226)	[0.391]	[0.077]	
Polygamous	0.372	0.386	0.423	0.014	0.051	1009
	(0.484)	(0.487)	(0.495)	[0.714]	[0.181]	
Pineapple women	0.402	0.383	0.414	-0.019	0.012	1009
	(0.491)	(0.487)	(0.493)	[0.614]	[0.754]	
Pineapple men	0.741	0.733	0.725	-0.008	-0.017	877
	(0.439)	(0.443)	(0.447)	[0.817]	[0.649]	
IGA women	0.661	0.700	0.693	0.040	0.033	1009
	(0.474)	(0.459)	(0.462)	[0.272]	[0.365]	
Woman applied individually	0.021	0.015	0.009	-0.006	-0.012	1009
	(0.143)	(0.121)	(0.094)	[0.557]	[0.203]	
Empowerment: free to visit family	0.144	0.179	0.192	0.035	0.048	969
	(0.351)	(0.384)	(0.394)	[0.222]	[0.102]	
Empowerment: free to buy land	0.168	0.188	0.211	0.020	0.042	969
	(0.375)	(0.392)	(0.408)	[0.504]	[0.169]	

NOTE. Sample: female respondents from the baseline survey. *Age woman* is the woman's age in years. *Number children* is the total number of children born to the woman. *Mother/Father produced pineapple* equals one if the respondent reports that her mother/father grew pineapple. *Bank account* and *Has mobile money* equal one if the woman has, respectively, a bank account or a mobile-money account. *Owns TV* equals one if the household owns a television. *Phone* equals one if the woman owns a mobile phone. *Attended school* equals one if the woman ever attended formal schooling. *Pineapple not for women* equals one if the respondent says that producing pineapple is a male activity, unfit for women. *New Christian* equals one if the woman's main religious affiliation is a new-Christian church. *Extension area* equals one if the respondent lives in a region where pineapple production is uncommon (as compared to regions where pineapple production has been traditionally widespread). *Isolated* equals one if the household is geographically isolated. *Polygamous* equals one if the husband has more than one wife. *Pineapple women* and *Pineapple men* equal one if, respectively, the woman and the husband produce pineapple. *IGA women* is the number of income-generating activities the woman participates in. *Woman applied individually* equals one if the woman applied to the program on her own. *Empowerment: free to visit family* equals one if the respondent says that she can go and visit her family if she wants. *Empowerment: free to buy land* equals one if the respondent says that she can buy land with her own money if she wants.

Attrition.– Tables B3 and B4 show that the respondents' characteristics remain balanced at baseline after endline-1 and follow-up attritors, respectively, have been removed.

Table B3: RCT Benin, baseline characteristics in the endline 1 sample

	Mean / (SD)			Difference / [p-value]		
	Control	Couple	Individual	Couple-Control	Individual-Control	Obs.
New Christian	0.368	0.384	0.361	0.016	-0.007	943
	(0.483)	(0.487)	(0.481)	[0.680]	[0.862]	
Age woman	37.682	37.486	36.723	-0.196	-0.959	936
	(10.470)	(9.984)	(9.550)	[0.811]	[0.233]	
Number children	4.329	4.238	4.112	-0.091	-0.217	943
	(2.302)	(2.247)	(2.423)	[0.618]	[0.251]	
Mother produced pineapple	0.111	0.089	0.103	-0.022	-0.008	939
	(0.315)	(0.285)	(0.305)	[0.356]	[0.736]	
Father produced pineapple	0.174	0.158	0.173	-0.016	-0.001	934
	(0.380)	(0.365)	(0.379)	[0.589]	[0.979]	
Bank account	0.147	0.121	0.143	-0.026	-0.003	943
	(0.354)	(0.326)	(0.351)	[0.342]	[0.907]	
Has mobile money	0.410	0.425	0.358	0.015	-0.052	943
	(0.493)	(0.495)	(0.480)	[0.706]	[0.180]	
Owns TV	0.358	0.349	0.333	-0.009	-0.025	943
	(0.480)	(0.477)	(0.472)	[0.813]	[0.511]	
Phone	0.664	0.692	0.667	0.028	0.002	943
	(0.473)	(0.462)	(0.472)	[0.463]	[0.954]	
Attended school	0.381	0.410	0.396	0.028	0.015	943
	(0.486)	(0.493)	(0.490)	[0.469]	[0.709]	
Pineapple not for women	0.166	0.216	0.178	0.050	0.011	943
	(0.373)	(0.412)	(0.383)	[0.115]	[0.705]	
Extension area	0.489	0.448	0.467	-0.041	-0.021	943
	(0.501)	(0.498)	(0.500)	[0.307]	[0.594]	
Isolated	0.026	0.038	0.056	0.012	0.030*	943
	(0.160)	(0.192)	(0.230)	[0.396]	[0.059]	
Polygamous	0.381	0.400	0.424	0.019	0.043	943
	(0.486)	(0.491)	(0.495)	[0.630]	[0.278]	
Pineapple women	0.404	0.400	0.424	-0.004	0.020	943
	(0.491)	(0.491)	(0.495)	[0.921]	[0.616]	
Pineapple men	0.748	0.745	0.723	-0.003	-0.026	819
	(0.435)	(0.436)	(0.449)	[0.942]	[0.501]	
IGA women	0.661	0.708	0.695	0.047	0.033	943
	(0.474)	(0.455)	(0.461)	[0.211]	[0.370]	
Woman applied individually	0.020	0.016	0.009	-0.004	-0.010	943
	(0.139)	(0.125)	(0.096)	[0.729]	[0.283]	
Empowerment: free to visit family	0.137	0.182	0.188	0.044	0.051*	905
	(0.345)	(0.386)	(0.391)	[0.137]	[0.090]	
Empowerment: free to buy land	0.161	0.195	0.205	0.034	0.044	905
	(0.368)	(0.397)	(0.404)	[0.273]	[0.162]	

Note. Sample: respondents from the baseline survey who also appear in endline 1 (no attrition between baseline and endline 1). *Age woman* is the woman's age in years. *Number children* is the total number of children born to the woman. *Mother/Father produced pineapple* equals one if the respondent reports that her mother/father grew pineapple. *Bank account* and *Has mobile money* equal one if the woman has, respectively, a bank account or a mobile-money account. *Owns TV* equals one if the household owns a television. *Phone* equals one if the woman owns a mobile phone. *Attended school* equals one if the woman ever attended formal schooling. *Pineapple not for women* equals one if the respondent says that producing pineapple is a male activity, unfit for women. *New Christian* equals one if the woman's main religious affiliation is a new-Christian church. *Extension area* equals one if the respondent lives in a region where pineapple production is uncommon (as compared to regions where pineapple production has been traditionally widespread). *Isolated* equals one if the household is geographically isolated. *Polygamous* equals one if the husband has more than one wife. *Pineapple women* and *Pineapple men* equal one if, respectively, the woman and the husband produce pineapple. *IGA women* is the number of income-generating activities the woman participates in. *Woman applied individually* equals one if the woman applied to the program on her own. *Empowerment: free to visit family* equals one if the respondent says that she can go and visit her family if she wants. *Empowerment: free to buy land* equals one if the respondent says that she can buy land with her own money if she wants.

Table B4: RCT Benin, baseline characteristics in the follow-up sample

	Mean / (SD)			Difference / [p-value]		
	Control	Couple	Individual	Couple-Control	Individual-Control	Obs.
New Christian	0.373	0.381	0.361	0.008	-0.012	931
	(0.484)	(0.486)	(0.481)	[0.837]	[0.759]	
Age woman	37.731	37.374	36.719	-0.357	-1.012	924
	(10.405)	(10.078)	(9.545)	[0.666]	[0.211]	
Number children	4.330	4.190	4.188	-0.140	-0.142	931
	(2.308)	(2.215)	(2.398)	[0.443]	[0.456]	
Mother produced pineapple	0.113	0.089	0.106	-0.024	-0.007	927
	(0.317)	(0.285)	(0.308)	[0.328]	[0.776]	
Father produced pineapple	0.186	0.151	0.177	-0.035	-0.009	922
	(0.390)	(0.359)	(0.383)	[0.249]	[0.783]	
Bank account	0.158	0.124	0.137	-0.035	-0.021	931
	(0.366)	(0.330)	(0.345)	[0.217]	[0.463]	
Has mobile money	0.419	0.422	0.348	0.003	-0.071*	931
	(0.494)	(0.495)	(0.477)	[0.938]	[0.071]	
Owns TV	0.366	0.349	0.329	-0.017	-0.037	931
	(0.483)	(0.477)	(0.471)	[0.658]	[0.332]	
Phone	0.677	0.686	0.665	0.009	-0.012	931
	(0.469)	(0.465)	(0.473)	[0.808]	[0.751]	
Attended school	0.386	0.410	0.396	0.023	0.010	931
	(0.488)	(0.493)	(0.490)	[0.553]	[0.799]	
Pineapple not for women	0.168	0.210	0.169	0.041	0.001	931
	(0.375)	(0.408)	(0.376)	[0.192]	[0.973]	
Extension area	0.479	0.444	0.463	-0.034	-0.015	931
	(0.500)	(0.498)	(0.499)	[0.396]	[0.704]	
Isolated	0.030	0.038	0.054	0.008	0.025	931
	(0.170)	(0.192)	(0.227)	[0.566]	[0.129]	
Polygamous	0.389	0.400	0.428	0.011	0.039	931
	(0.488)	(0.491)	(0.496)	[0.789]	[0.330]	
Pineapple women	0.422	0.400	0.425	-0.022	0.002	931
	(0.495)	(0.491)	(0.495)	[0.572]	[0.950]	
Pineapple men	0.750	0.746	0.730	-0.004	-0.020	807
	(0.434)	(0.436)	(0.445)	[0.923]	[0.606]	
IGA women	0.673	0.692	0.700	0.019	0.026	931
	(0.470)	(0.462)	(0.459)	[0.616]	[0.481]	
Woman applied individually	0.020	0.013	0.010	-0.007	-0.010	931
	(0.140)	(0.112)	(0.098)	[0.485]	[0.291]	
Empowerment: free to visit family	0.150	0.178	0.179	0.029	0.029	893
	(0.357)	(0.383)	(0.384)	[0.347]	[0.336]	
Empowerment: free to buy land	0.170	0.195	0.196	0.025	0.026	893
	(0.376)	(0.397)	(0.398)	[0.437]	[0.417]	

Note. Sample: respondents from the baseline survey who also appear in the follow-up (no attrition between baseline and follow-up). *Age woman* is the woman's age in years. *Number children* is the total number of children born to the woman. *Mother/Father produced pineapple* equals one if the respondent reports that her mother/father grew pineapple. *Bank account* and *Has mobile money* equal one if the woman has, respectively, a bank account or a mobile-money account. *Owns TV* equals one if the household owns a television. *Phone* equals one if the woman owns a mobile phone. *Attended school* equals one if the woman ever attended formal schooling. *Pineapple not for women* equals one if the respondent says that producing pineapple is a male activity, unfit for women. *New Christian* equals one if the woman's main religious affiliation is a new-Christian church. *Extension area* equals one if the respondent lives in a region where pineapple production is uncommon (as compared to regions where pineapple production has been traditionally widespread). *Isolated* equals one if the household is geographically isolated. *Polygamous* equals one if the husband has more than one wife. *Pineapple women* and *Pineapple men* equal one if, respectively, the woman and the husband produce pineapple. *IGA women* is the number of income-generating activities the woman participates in. *Woman applied individually* equals one if the woman applied to the program on her own. *Empowerment: free to visit family* equals one if the respondent says that she can go and visit her family if she wants. *Empowerment: free to buy land* equals one if the respondent says that she can buy land with her own money if she wants.

Table B5 examines whether the characteristics of attritors differ by treatment category. Although attrition is slightly lower in the individual treatment arm, the characteristics of respondents are not correlated with attrition rates.

Table B5: Differential attrition across treatment and control groups

	P(Attrition at endline 1)			P(Attrition at follow-up)		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	-0.0313*		-0.111	-0.0313*		-0.147
	(0.0177)		(0.103)	(0.0189)		(0.116)
Individual		-0.0417**			-0.0298	
		(0.0190)			(0.0213)	
Couple		-0.0210			-0.0329	
		(0.0204)			(0.0211)	
New Christian x T			-0.00527			0.0476
			(0.0381)			(0.0426)
Age woman x T			0.00124			0.000514
			(0.00237)			(0.00252)
Number children x T			0.00279			-0.00192
			(0.00857)			(0.01000)
Mother produced pineapple x T			0.128***			0.0341
			(0.0478)			(0.0419)
Father produced pineapple x T			-0.117**			0.0204
			(0.0563)			(0.0453)
Bank account x T			-0.0942			-0.0138
			(0.0728)			(0.0687)
Has mobile money x T			-0.0300			0.00887
			(0.0394)			(0.0454)
Owns TV x T			-0.0360			-0.0153
			(0.0438)			(0.0455)
Phone x T			0.0650			0.0664
			(0.0402)			(0.0468)
Attended school x T			-0.0157			-0.0163
			(0.0390)			(0.0427)
Pineapple not for women x T			-0.0684			-0.0138
			(0.0529)			(0.0562)
Extension area x T			0.0525			-0.0251
			(0.0461)			(0.0531)
Polygamous x T			-0.0257			0.0152
			(0.0429)			(0.0434)
Pineapple women x T			-0.0362			0.0168
			(0.0436)			(0.0445)
Pineapple men x T			0.0880			-0.00528
			(0.0574)			(0.0612)
IGA women x T			0.00164			0.0677
			(0.0418)			(0.0462)
Woman applied individually x T			0.0345			-0.0456
			(0.0770)			(0.0946)
Empowerment visit family x T			-0.0141			0.0944
			(0.0639)			(0.0582)
Empowerment buy land x T			-0.0671			-0.0311
			(0.0603)			(0.0653)
Mean Y control	0.0863	0.0863	0.0830	0.0982	0.0982	0.104
R-squared	0.00357	0.00473	0.0736	0.00306	0.00308	0.0772
Observations	1009	1009	859	1009	1009	859

NOTE. Sample: respondents from the baseline survey. The table reports OLS estimates of an attrition indicator on treatment status and pre-treatment characteristics. The outcome variable in columns 1–3 is an indicator equal to one if the respondent is missing at endline 1; in columns 4–6 it is an indicator equal to one if the respondent is missing at the follow-up. *Treatment* pools the two treatment arms (Individual and Couple); *Individual* and *Couple* report the differential attrition of each arm relative to the Control group. Interaction terms (*x T*) report the differential association between each baseline characteristic and attrition in the treatment group relative to the control group. The baseline characteristics used are: *New Christian* (woman's main religious affiliation is a new-Christian church), *Age woman* (woman's age in years), *Number children* (total number of children born to the woman), *Mother/Father produced pineapple* (respondent's mother/father grew pineapple), *Bank account* and *Has mobile money* (woman has a bank account / a mobile-money account), *Owns TV* (household owns a television), *Phone* (woman owns a mobile phone), *Attended school* (woman ever attended formal schooling), *Pineapple not for women* (respondent says that producing pineapple is a male activity, unfit for women), *Extension area* (lives in a region where pineapple production is uncommon), *Isolated* (household is geographically isolated), *Polygamous* (husband has more than one wife), *Pineapple women* and *Pineapple men* (woman/husband produces pineapple), *IGA women* (number of income-generating activities the woman participates in), *Woman applied individually* (woman applied to the program on her own), and *Empowerment visit family / Empowerment buy land* (woman reports she can visit her family / buy land with her own money if she wants). Robust standard errors in parentheses.

C. Additional Figures and Tables

C.1. Gender gaps in religious affiliation

Table C1: Average religious affiliation by gender, raw data

	Share of New Christians			Share of Catholics		
Country	Men	Women	Female - male	Men	Women	Female - male
Zimbabwe	48.2	64.4	16.2	9.6	8.4	-1.2
Gambia	25.0	39.9	14.9	37.6	35.8	-1.8
Eswatini	19.2	29.0	9.7	5.1	4.7	-0.5
Lesotho	42.5	50.5	7.9	40.7	39.0	-1.8
Cote d'Ivoire	19.3	27.1	7.8	16.8	18.2	1.4
Kenya	63.4	69.8	6.4	21.9	18.6	-3.3
Congo	31.7	37.2	5.5	33.4	31.4	-2.0
Angola	7.2	12.7	5.5	36.2	37.6	1.4
Ghana	43.2	48.1	4.9	12.1	11.8	-0.3
D.R. Congo	36.3	41.1	4.8	28.3	26.5	-1.7
Mozambique	15.6	20.3	4.7	30.4	29.5	-0.8
Benin	17.6	21.5	3.9	29.8	29.6	-0.3
Togo	19.1	22.6	3.5	29.9	27.8	-2.1
Malawi	42.5	45.5	3.1	21.4	20.7	-0.7
Uganda	6.5	9.4	2.9	42.5	41.0	-1.6
Madagascar	7.1	8.5	1.4	32.0	32.5	0.4

Source: Author's calculations based on data from the Demographic and Health Surveys (DHS). The restricted definition is used: the sample includes only country-waves where new Christian churches are separately identified in the DHS.

Table C2: Religious affiliation and gender, restricted definition

	New Christian		Traditional		Catholic		Protestant		Islam	
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**	**(10)**
Panel A: Full sample										
Female	2.792***	2.879***	-0.673***	-0.696***	-0.717***	-0.182***	1.098***	1.549***	-0.317***	-1.488***
	(0.0694)	(0.0696)	(0.0313)	(0.0305)	(0.0612)	(0.0606)	(0.0525)	(0.0532)	(0.0591)	(0.0414)
Mean Y men	15.45	15.47	2.603	2.587	18.90	18.87	15.20	15.28	32.98	32.82
R-squared	0.268	0.437	0.104	0.391	0.149	0.365	0.305	0.458	0.502	0.799
N	1657801	1600698	2219285	2099306	2219285	2099306	2219285	2099306	2219285	2099306
Panel B: Countries with positive share of new Christians (restricted definition)										
Female	5.423***	5.197***	-0.936***	-1.014***	-1.081***	-0.462***	1.249***	1.558***	-0.601***	-1.620***
	(0.123)	(0.122)	(0.0488)	(0.0465)	(0.0932)	(0.0910)	(0.0760)	(0.0773)	(0.0746)	(0.0552)
Country-round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
DHS cluster FE	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Mean Y men	27.63	27.50	3.548	3.515	23.69	23.57	12.91	12.97	18.60	18.75
R-squared	0.146	0.337	0.132	0.391	0.0781	0.308	0.163	0.324	0.201	0.700
N	850897	830180	1164795	1101255	1164795	1101255	1164795	1101255	1164795	1101255

Note. Data: Demographic and Health Surveys. The table reports OLS estimates. The coefficients should be divided by 100. Panel A includes the full DHS sample. Panel B restricts the sample to countries with a non-negligible share of new Christians (more than 5%), measured with the restricted definition (new Christian equals missing in country-waves where these denominations are not separately coded). The outcome variable in columns 1 and 2 uses the restricted definition. The outcome variable equals Traditional (columns 3 and 4), Catholic (columns 5 and 6), Protestant (columns 7 and 8), Islam (columns 9 and 10). Individual controls include age, age squared, education, marital statis, labor market participation and urban/rural place of residence. Country and year fixed effects always included. Robust standard errors clustered at the DHS cluster level in parenthesis.

Table C3: Religious affiliation and gender, raw definition

	New Christian		Traditional		Catholic		Protestant		Islam	
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**	**(10)**
Panel A: Full sample										
Female	2.295***	2.665***	-0.673***	-0.696***	-0.717***	-0.182***	1.098***	1.549***	-0.317***	-1.488***
	(0.0628)	(0.0589)	(0.0313)	(0.0305)	(0.0612)	(0.0606)	(0.0525)	(0.0532)	(0.0591)	(0.0414)
Mean Y men	17.65	17.72	2.603	2.587	18.90	18.87	15.20	15.28	32.98	32.82
R-squared	0.266	0.512	0.104	0.391	0.149	0.365	0.305	0.458	0.502	0.799
N	2219285	2099306	2219285	2099306	2219285	2099306	2219285	2099306	2219285	2099306
Panel B: Countries with positive share of new Christians (raw definition)										
Female	4.045***	4.489***	-0.856***	-0.934***	-0.951***	-0.425***	1.091***	1.441***	-0.598***	-1.638***
	(0.104)	(0.0970)	(0.0455)	(0.0436)	(0.0862)	(0.0845)	(0.0702)	(0.0717)	(0.0793)	(0.0533)
Country-round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
DHS cluster FE	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Mean Y men	29.83	29.78	3.297	3.270	21.55	21.45	11.70	11.76	20.35	20.49
R-squared	0.143	0.429	0.129	0.390	0.0987	0.322	0.174	0.333	0.189	0.711
N	1270736	1196782	1270736	1196782	1270736	1196782	1270736	1196782	1270736	1196782

Note. Data: Demographic and Health Surveys. The table reports OLS estimates. The coefficients should be divided by 100. Panel A includes the full DHS sample. Panel B restricts the sample to countries with a non-negligible share of new Christians (more than 5%), measured with the raw definition. The outcome variable in columns 1 and 2 uses the raw definition: equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination, using DHS categories as reported without recoding. The outcome variable equals Traditional (columns 3 and 4), Catholic (columns 5 and 6), Protestant (columns 7 and 8), Islam (columns 9 and 10). Individual controls include age, age squared, education, marital statis, labor market participation and urban/rural place of residence. Country and year fixed effects always included. Robust standard errors clustered at the DHS cluster level in parenthesis.

Table C4: Religious affiliation and gender, different datasets

	Afrobarometer		PEW		WVS		South Benin		North Benin	
	Trad.	new Christian	Trad.	new Christian	Trad.	new Christian	Trad.	new Christian	Trad.	Christian
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Female	-0.623***	2.898***	-0.893***	5.450***	0.00660	2.339***	-6.507***	7.189***	-19.96***	20.16***
	(0.0437)	(0.151)	(0.161)	(0.562)	(0.0637)	(0.270)	(1.482)	(2.367)	(4.829)	(4.667)
Country and round FE	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	Yes
Mean Y men	0.0187	0.306	0.0216	0.458	0.00413	0.106	0.141	0.478	0.610	0.291
R-squared	0.0459	0.194	0.0477	0.219	0.0436	0.287	0.0110	0.00516	0.0383	0.0403
N	307149	307149	25091	24943	40726	40726	1783	1783	430	430

Note. Data: Afrobarometer, PEW Research Center, World Values Survey (WVS), and first-hand data from southern and northern Benin. The table reports OLS estimates with no individual-level controls. The outcome variable is an indicator equal to one if the respondent's religion is Traditional — Vodoun in the case of southern Benin (columns 1, 3, 5, 7, 9) — or "New" Christian (columns 2, 4, 6, 8, 10). In column 10, the underlying survey does not disaggregate Christian denominations, so the outcome captures Christian affiliation of any kind. The "Female" coefficient is expressed in percentage points. Country and survey-round fixed effects are included where indicated. The Benin samples cover a single country. Robust standard errors are reported in parentheses.

C.2. RCT Benin

Table C5: Treatment effects: religious conversion, opposition and witchcraft

	Endline 1					Endline 2
	Women's conversion			Opposition by:		Witchcraft
	Ever	To new Xty	To old Xty	Husband	Other family	Used by husband
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Panel A: Aggregated treatment						
Treatment	0.0644**	0.0425*	-0.00314	0.0754**	0.0457	0.0409*
	(0.0322)	(0.0254)	(0.0217)	(0.0336)	(0.0327)	(0.0215)
R-squared	0.0792	0.252	0.0489	0.0625	0.0649	0.00939
Panel B: By treatment arm						
Partial husband involvement	0.0805**	0.0567*	-0.0148	0.0548	0.0411	0.0101
	(0.0374)	(0.0295)	(0.0245)	(0.0384)	(0.0377)	(0.0236)
Full husband involvement	0.0481	0.0280	0.00874	0.0959**	0.0502	0.0712***
	(0.0372)	(0.0289)	(0.0254)	(0.0390)	(0.0379)	(0.0269)
Department FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y control	0.342	0.202	0.117	0.366	0.343	0.0784
R-squared	0.0800	0.252	0.0498	0.0636	0.0650	0.0161
N	943	943	943	931	931	842

Note. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The sample in column 6 is restricted to married women. The outcome variables are indicators with unitary value if: the respondent has ever converted to any new religion (column 1), to a new Christian religion (column 2), or to an "old" Christian church (column 3). In columns 4 and 5 the outcome variables are indicators equal to one if the husband (column 4) or another family member (column 5) has opposed woman's conversion to a New Christian church. The outcome variable in column 6 is an indicator that equals one if the respondent reports that her husband (in the last 5 years) went to see a witch-doctor to slow down his wife's activities or reduce her resources. Panel A shows treatment effects of the aggregated treatment. In Panel B, "Partial husband involvement" is an indicator equal to one if the woman (or the wife of the man when the sample is restricted to men) has been assigned to the treatment group where business training was reserved for women. "Full husband involvement" is an indicator equal to one if she has been assigned to the treatment group where women attended the business training together with their husband. In columns 1-3, being New Christian at baseline is included as a control. Department fixed effects included. Robust standard errors are in parenthesis.

Table C6: Treatment on the Treated: religious conversion, opposition and witchcraft

	Endline 1					Endline 2
	Women's conversion			Opposition by:		Witchcraft
	Ever	To new Xty	To old Xty	Husband	Other family	Used by husband
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Panel A: Aggregated treatment						
Treatment	0.0825**	0.0605*	-0.00643	0.101**	0.0625	0.0475*
	(0.0415)	(0.0325)	(0.0278)	(0.0416)	(0.0409)	(0.0281)
R-squared	0.0655	0.241	0.0555	0.110	0.0929	0.0181
Panel B: By treatment arm						
Partial husband involvement	0.0980**	0.0795**	-0.0257	0.0708	0.0582	0.0117
	(0.0488)	(0.0384)	(0.0317)	(0.0475)	(0.0472)	(0.0306)
Full husband involvement	0.0673	0.0419	0.0124	0.131***	0.0728	0.0900***
	(0.0472)	(0.0367)	(0.0321)	(0.0473)	(0.0464)	(0.0342)
Department FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y control	0.341	0.203	0.115	0.365	0.342	0.0789
N	936	936	936	924	924	835

Note. Data: First-hand data collected in southern Benin. The table reports Treatment on the Treated (ToT) estimates using 2SLS, where actual treatment receipt is instrumented by random assignment. The sample in column 6 is restricted to married women. The outcome variables are indicators with unitary value if: the respondent has ever converted to any new religion (column 1), to a new Christian religion (column 2), or to an "old" Christian church (column 3). In columns 4 and 5 the outcome variables are indicators equal to one if the husband (column 4) or another family member (column 5) has opposed woman's conversion to a New Christian church. The outcome variable in column 6 is a indicator that equals one if the respondent reports that her husband (in the last 5 years) went to see a witch-doctor to slow down his wife's activities or reduce her resources. Panel A shows ToT effects of the aggregated treatment. In Panel B, "Partial husband involvement" is the ToT effect for women assigned to receive business training alone. "Full husband involvement" is the ToT effect for women assigned to attend business training together with their husband. Controls include age, education and marital status at baseline, whether the respondent was a new Christian at baseline, and department fixed effects. Robust standard errors are in parenthesis.

Table C7: Treatment on the Treated: religious conversion, opposition and witchcraft. No controls

	Endline 1					Endline 2
	Women's conversion			Opposition by:		Witchcraft
	Ever	To new Xty	To old Xty	Husband	Other family	Used by husband
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Panel A: Aggregated treatment						
Treatment	0.0825**	0.0544*	-0.00402	0.0961**	0.0548	0.0467*
	(0.0413)	(0.0324)	(0.0278)	(0.0435)	(0.0424)	(0.0276)
R-squared	0.0596	0.236	0.0472	-0.00134	-0.00233	0.00204
Panel B: By treatment arm						
Partial husband involvement	0.104**	0.0732*	-0.0190	0.0710	0.0531	0.0127
	(0.0484)	(0.0381)	(0.0316)	(0.0497)	(0.0488)	(0.0301)
Full husband involvement	0.0611	0.0356	0.0110	0.122**	0.0640	0.0893***
	(0.0470)	(0.0366)	(0.0321)	(0.0498)	(0.0484)	(0.0338)
Department FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y control	0.342	0.202	0.117	0.366	0.343	0.0784
N	943	943	943	931	931	842

Note. Data: First-hand data collected in southern Benin. The table reports Treatment on the Treated (ToT) estimates using 2SLS, where actual treatment receipt is instrumented by random assignment. The sample in column 6 is restricted to married women. The outcome variables are indicators with unitary value if: the respondent has ever converted to any new religion (column 1), to a new Christian religion (column 2), or to an "old" Christian church (column 3). In columns 4 and 5 the outcome variables are indicators equal to one if the husband (column 4) or another family member (column 5) has opposed woman's conversion to a New Christian church. The outcome variable in column 6 is a indicator that equals one if the respondent reports that her husband (in the last 5 years) went to see a witch-doctor to slow down his wife's activities or reduce her resources. Panel A shows ToT effects of the aggregated treatment. In Panel B, "Partial husband involvement" is the ToT effect for women assigned to receive business training alone. "Full husband involvement" is the ToT effect for women assigned to attend business training together with their husband. Only outcome at baseline included as a control. Department fixed effects included. Robust standard errors are in parenthesis.

Table C8: Witchcraft, spiritual protection and NCC conversion

	Treatment effects on conversion		Heterogeneity: Witchcraft threat	
	(1)	(2)	(3)	(4)
Treatment	0.0472*		0.0116	
	(0.0254)		(0.0358)	
Individual treatment		0.0615**		0.0288
		(0.0298)		(0.0416)
Couple treatment		0.0330		-0.00463
		(0.0290)		(0.0405)
Treat × Witchcraft threat			0.101*	
			(0.0520)	
Individual × Witchcraft threat				0.0938
				(0.0608)
Couple × Witchcraft threat				0.107*
				(0.0591)
Mean control	0.203	0.203	0.203	0.203
R-squared	0.257	0.258	0.273	0.273
N	936	936	905	905

NOTE. Data: First-hand survey data from southern Benin. The sample is restricted to women. The outcome is an indicator equal to one if the respondent has ever converted to a new Christian church. Columns 1–2 report average treatment effects (ITT). Columns 3–4 report heterogeneous effects for women who feel threatened by witchcraft (*Witchcraft threat*). Because witchcraft threat is measured after the intervention, the interactions in columns 3–4 are correlational rather than causal. Controls include age, education at baseline, NCC membership at baseline, and marital status. Department fixed effects are included. Robust standard errors in parenthesis.

Table C9: New Christianity and protection by husbands

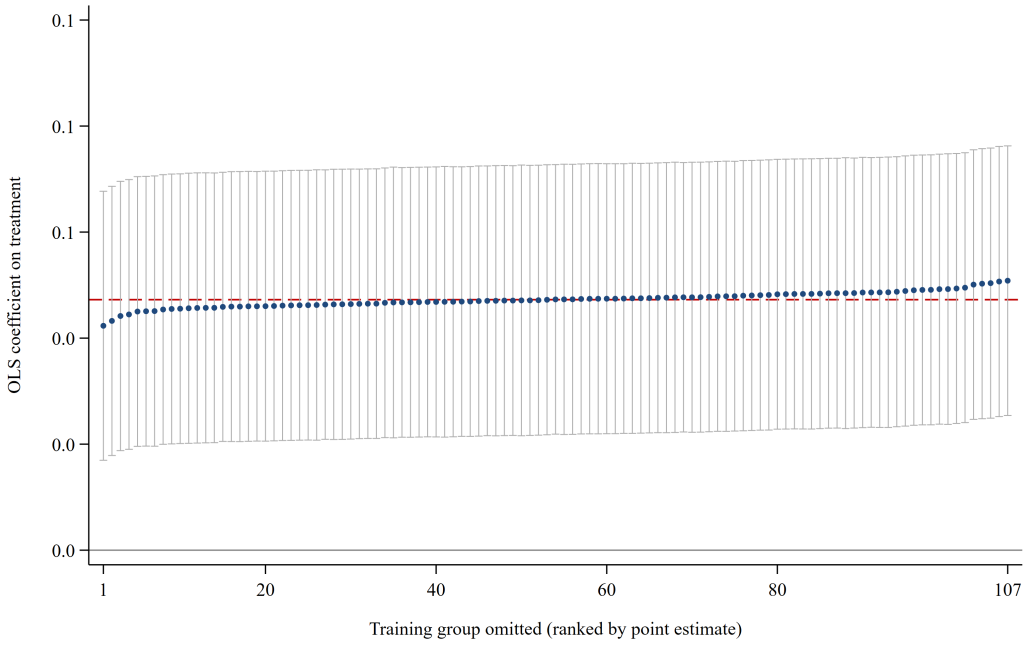
	Husband offers protection		
	(1)	(2)	(3)
Catholic	0.0274		-0.0369
	(0.0355)		(0.0389)
New Christian		-0.158***	-0.137***
		(0.0335)	(0.0371)
Mean Y	0.570	0.570	0.570
Controls	No	No	Yes
R-squared	0.000636	0.0238	0.174
Controls	931	931	924
New Christian=Catholic			0.00922

NOTE. Data: First-hand survey data from southern Benin. The outcome variable is a dummy variable equal to one if the female respondent said that her husband protects her against witchcraft attacks. Age, education, marital status, whether the respondent grows pineapple, and department fixed effects are included as controls. Robust standard errors are in parenthesis.

Table C10: Distribution of conversions across training groups

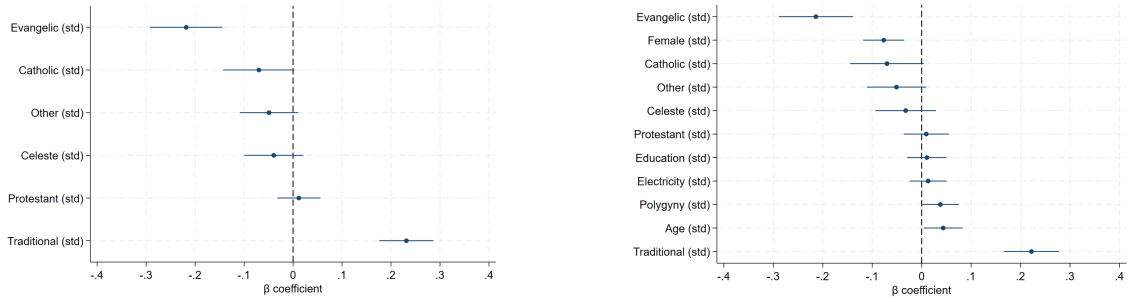
	N	Mean	SD	Min	p25	Median	p75	Max
Number of conversions		1.449	1.268	0.000	0.000	1.000	2.000	6.000
Conversion rate		0.235	0.199	0.000	0.000	0.200	0.375	0.667

Figure C1: Leave-one-group-out stability of the treatment effect on conversion to new Christianity



Notes: Each point is the OLS estimate of the treatment effect on ever having converted to a new Christian church by 2025, obtained by dropping all treated members of one training group from the estimation sample in turn. Bars represent ± 1 standard error. Estimates are ranked in ascending order. The dashed red line is the full-sample estimate. Controls: age, schooling, pre-existing Christianity at baseline, polygamy, and marital status. Department fixed effects. Robust standard errors.

Figure C2: Correlation between belief in ancestral spirits and religion



(a) Religion fixed effects (FE)

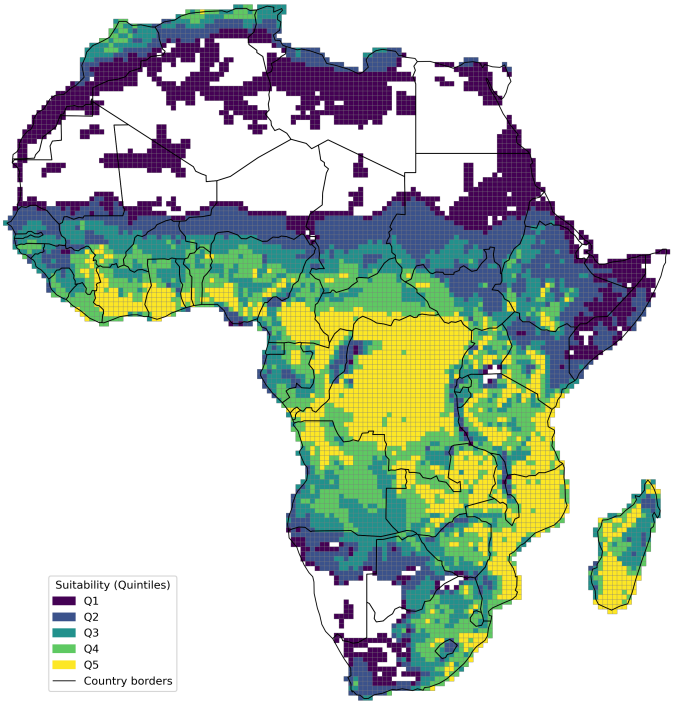
(b) Religion FE and controls

Notes: The figure shows the coefficients obtained by regressing, on religion fixed effects alone (Panel a) or on religion fixed effects together with a set of control variables (Panel b), an indicator variable equal to one if the respondent believes that the spirits of her ancestors have an influence on her life.

C.3. Economic opportunity I: Shocks to crop prices and new Christian churches

C.3.1. Crop suitability

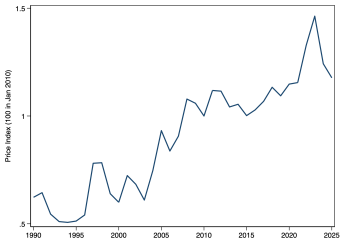
Figure C3: Cash crop suitability across Africa



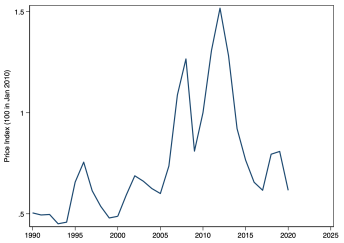
Source: Data come from GAEZ (FAO). The figure shows the spatial distribution of cash crop suitability. The unit of observation is a $0.5° \times 0.5°$ grid-cell. Suitability is measured as the sum of the suitability score across the fifteen crops used in the analysis: banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, and wheat.

C.3.2. International crop prices

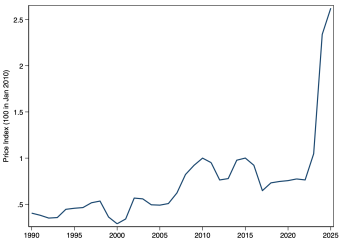
Figure C4: Price indices for agricultural commodities (Base year: 2010)



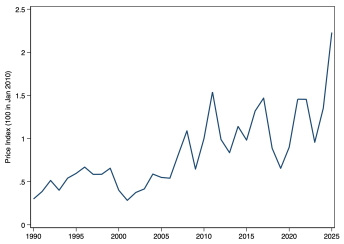
(a) Banana



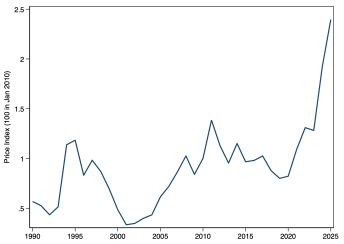
(b) Barley



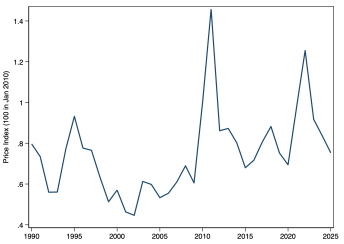
(c) Cocoa



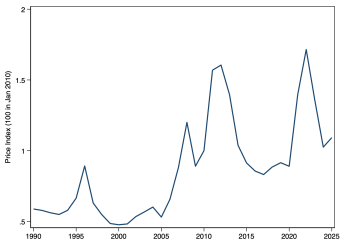
(d) Coconut



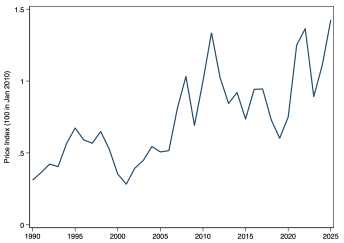
(e) Coffee



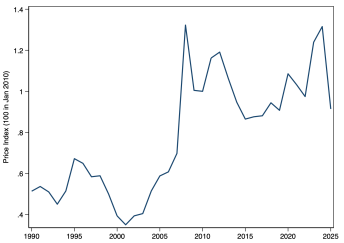
(f) Cotton



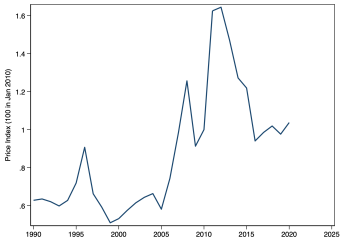
(g) Maize



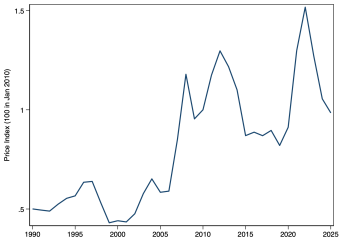
(h) Oil Palm



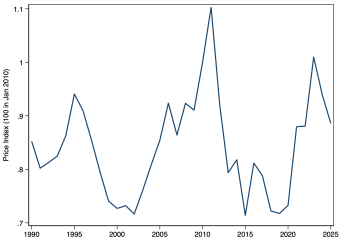
(i) Rice



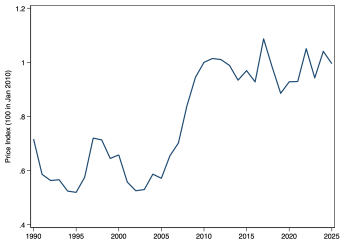
(j) Sorghum



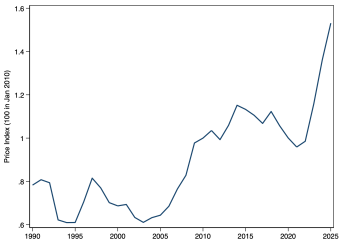
(k) Soybean



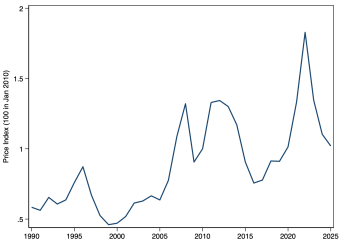
(l) Sugar



(m) Tea



(n) Tobacco



(o) Wheat

C.3.3. *First-stage results: effects on wealth and luminosity*

Table C11: Effect on wealth and income

	DHS data (cell level)		Cell-level data	
	Wealth quintiles	Wealth quintiles (sd)	Nightlights (mean)	Nightlights (sd)
	(1)	(2)	(3)	(4)
Cash-crop Price Index	0.0408**	0.0207*	0.0546***	0.0628***
	(0.0208)	(0.0115)	(0.0121)	(0.0128)
Mean Y	2.580	0.925	0.346	0.932
Cell and year fixed effects	Yes	Yes	Yes	Yes
R-squared	0.774	0.622	0.936	0.946
N	9158	9158	149984	149984

Note. Data: Demographic and Health Surveys, Food and Agriculture Organization (FAO), and Chiovelli et al. (2026). "Cash Crop Price Index" is the sum across fifteen crops (banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, wheat) of their price in t-1 (indexed to 1 in year 2010 for each crop) and their suitability. The index is standardized to have mean 0 and standard deviation 1 within the estimation sample, so coefficients are expressed for a one-standard-deviation increase. The outcome variable in column one is the average wealth quintile of the cell, averaged across all DHS clusters within that cell. In column 2, the dependent variable is the standard deviation of the wealth quintile variable computed among all individuals within a given cell. In column 3, the dependent variable is the average nightlights in the cell. In column 4, the dependent variable is the standard deviation of nightlights in the cell. Robust standard errors clustered at the cell level in parenthesis.

C.3.4. *Effect of cash-crop price index on different religions*

Table C12: Shocks to crops and conversion to/from different religions

	Respondent is:			
	New Christian	Catholic	Protestant	Traditional
	(1)	(2)	(3)	(4)
Cash Crop Price Index	0.0183***	-0.00989***	-0.0178***	-0.000637
	(0.00388)	(0.00267)	(0.00437)	(0.00145)
Mean Y	0.239	0.186	0.109	0.0211
R-squared	0.423	0.243	0.269	0.203
Controls	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	1862101	1862101	1862101	1862101

Note. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). "Cash Crop Price Index" is the sum across fifteen crops (banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, wheat) of their price in t-1 (indexed to 1 in year 2010 for each crop) weighted by their suitability. The index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. The outcome variable in column 1 uses the grouped definition: equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. The outcome variable equals Catholic in column 2, from a Protestant denomination in column 3, or from a Traditional religion in column 4. Controls include age, age squared, gender, education, labor market participation, marital status and urban/rural residence. Standard errors clustered at the cell level are in parenthesis.

Table C13: Shocks to crops and conversion to/from different religions, $25° \times 25°$ cells (Roessler et al., 2022)

	Respondent is:			
	New Christian	Catholic	Protestant	Traditional
	(1)	(2)	(3)	(4)
Cash Crop Price Index	0.0556***	-0.0283***	-0.0208***	-0.000262
	(0.00832)	(0.00419)	(0.00669)	(0.00377)
Mean Y	0.237	0.178	0.103	0.0215
R-squared	0.456	0.276	0.286	0.254
Basic controls	Yes	Yes	Yes	Yes
Full controls	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	1737891	1737891	1737891	1737891

Note. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). "Cash Crop Price Index" is the standardized sum across nine cash crops (coffee, cocoa, cotton, groundnuts, oil palm, sugarcane, tea, tobacco, and banana) of their price in t-1 (indexed to 1 in year 2010 for each crop) weighted by their suitability (ranging from 0 to 1). The index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. The outcome variable is an indicator that equals one if the respondent is from a new Christian church (or Protestant in surveys where new Christians and Protestants are grouped together) in column 1, Catholic in column 2, from a missionary Protestant denomination in column 3, or from a Traditional religion in column 4. Basic controls include age, age squared and gender. Full controls further include education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the country-year level appear in parenthesis.

C.3.5. Positive versus negative price shocks

Table C14: Positive vs. negative shocks to cash-crop prices and conversion to new Christian churches

	Respondent is New Christian		
	(1)	(2)	(3)
Positive shock	0.0479***	0.0463***	0.0458***
	(0.00672)	(0.00670)	(0.00693)
Negative shock	-0.00812	-0.00704	-0.00929
	(0.00707)	(0.00706)	(0.00730)
p-val: coef(pos)=-coef(neg)	0.000	0.000	0.000
Mean Y	0.243	0.243	0.239
R-squared	0.421	0.422	0.423
Basic controls	No	Yes	Yes
Full controls	No	No	Yes
Grid-cell FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	1966628	1962886	1862101

NOTE. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). We first compute the year-on-year change of the lagged cash-crop price index within each grid cell, $\Delta p_it = p_it - p_i, t-1$, and split it into the magnitudes of increases and decreases: $\max(\Delta p_it, 0)$ (0 in years the price fell) and $\max(-\Delta p_it, 0)$ (0 in years the price rose). Both magnitudes are then scaled by the sample standard deviation of Δp_it, so the coefficients on and report the effect on P(New Christian) of a one-standard-deviation price rise and a one-standard-deviation price fall, respectively. The row reports the p-value of the mirror-symmetry test $H_0 : \beta_pos = -\beta_neg$ (a rise of size Δ moves conversion by the same amount that a fall of size Δ moves it in the opposite direction). The outcome is an indicator equal to one if the respondent is from a new Christian church (or Protestant in surveys where new Christians and Protestants are grouped together). Basic controls include age, age squared and gender. Full controls further include education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the cell level are reported in parenthesis.

Table C15: Positive vs. negative cash-crop price shocks and conversion to new Christian churches

	Respondent is New Christian							
	Positive shock				Negative shock			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Price shock	0.0457***	0.0377***	0.0443***	0.0408***	-0.00894	-0.00807	-0.0145*	-0.0150**
	(0.00692)	(0.00691)	(0.00713)	(0.00716)	(0.00730)	(0.00733)	(0.00744)	(0.00746)
Female	0.0316***	0.0282***	0.0313***	0.0289***	0.0317***	0.0323***	0.0314***	0.0305***
	(0.00174)	(0.00179)	(0.00182)	(0.00182)	(0.00173)	(0.00183)	(0.00182)	(0.00184)
Price shock x Female		0.00967***		0.00450***		-0.00134		0.00122
		(0.00161)		(0.00160)		(0.00168)		(0.00170)
Price shock x witchcraft (cell)			-0.00182	-0.00544***			0.00814***	0.00905***
			(0.00161)	(0.00203)			(0.00278)	(0.00303)
Price shock x Female x witch.				0.00520***				-0.00120
				(0.00146)				(0.00198)
Mean Y	0.239	0.239	0.241	0.241	0.239	0.239	0.241	0.241
R-squared	0.423	0.424	0.429	0.429	0.423	0.423	0.428	0.429
Full controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1862101	1862101	1709683	1709683	1862101	1862101	1709683	1709683

NOTE. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). The cash-crop price index is the sum across fifteen crops (banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, wheat) of their price in t-1 (indexed to 1 in year 2010 for each crop) weighted by their suitability. We compute the year-on-year change of this index within each grid cell, $\Delta p_it = p_it - p_i, t-1$, and split it into the magnitudes of increases and decreases. In columns (1)–(4) the is the positive part, $\max(\Delta p_it, 0)$ (equal to zero in years the price did not rise); in columns (5)–(8) it is the negative part, $\max(-\Delta p_it, 0)$ (equal to zero in years the price did not fall). Both are scaled by the sample standard deviation of Δp_it, so each coefficient reports the effect on P(New Christian) of a one-standard-deviation price rise (columns 1–4) or fall (columns 5–8). Within each block, the four columns replicate the specifications of columns (3)–(6) of Table 5. "Witchcraft (cell)" is the standardized mean prevalence of witchcraft in a given grid cell; columns (3)–(4) and (7)–(8) restrict the sample to grid-cells for which it is computed from at least 50 observations. The outcome variable is an indicator that equals one if the respondent is from a new Christian church (or Protestant in surveys where new Christians and Protestants are grouped together). All columns include full controls: age, age squared, gender, education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the cell level are reported in parenthesis.

C.3.6. Robustness: different measures of new Christian

Table C16: Shocks to cash crops and conversion to New Christian churches, raw definition

	Respondent is New Christian					
	(1)	(2)	(3)	(4)	(5)	(6)
Cash Crop Price Index	0.0219***	0.0220***	0.0230***	0.0191***	0.0231***	0.0201***
	(0.00410)	(0.00411)	(0.00415)	(0.00418)	(0.00443)	(0.00455)
Female		0.0259***	0.0304***	0.0302***	0.0309***	0.0302***
		(0.00154)	(0.00168)	(0.00166)	(0.00178)	(0.00169)
Cash Crop Price Index x Female				0.00588***		0.00429***
				(0.000947)		(0.000925)
Cash-crop PI x witchcraft (cell)					0.00676***	0.000292
					(0.00121)	(0.00146)
Cash-crop PI x Female x witch.						0.00935***
						(0.00111)
Mean Y	0.199	0.199	0.197	0.197	0.205	0.205
R-squared	0.407	0.408	0.404	0.404	0.406	0.406
Basic controls	No	Yes	Yes	Yes	Yes	Yes
Full controls	No	No	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1966628	1962886	1862101	1862101	1709683	1709683

Note. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). "Cash Crop Price Index" is the sum across fifteen crops (banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, wheat) of their price in t-1 (indexed to 1 in year 2010 for each crop) weighted by their suitability. The index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. "Witchcraft (cell)" is the standardized mean prevalence of witchcraft in a given grid cell. The outcome variable uses the raw definition: equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination, using DHS categories as reported without recoding. Basic controls include age, age squared and gender. Full controls further include education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the cell level appear in parenthesis.

Table C17: Shocks to cash crops and conversion to New Christian churches, restricted definition

	Respondent is New Christian					
	(1)	(2)	(3)	(4)	(5)	(6)
Cash Crop Price Index	0.0117***	0.0115***	0.0122***	0.00688	0.0105**	0.00706
	(0.00423)	(0.00427)	(0.00440)	(0.00450)	(0.00477)	(0.00507)
Female		0.0292***	0.0314***	0.0314***	0.0320***	0.0312***
		(0.00198)	(0.00208)	(0.00206)	(0.00225)	(0.00209)
Cash Crop Price Index x Female				0.00760***		0.00505***
				(0.00120)		(0.00120)
Cash-crop PI x witchcraft (cell)					0.00241*	-0.00364**
					(0.00137)	(0.00180)
Cash-crop PI x Female x witch.						0.00867***
						(0.00119)
Mean Y	0.174	0.174	0.173	0.173	0.179	0.179
R-squared	0.342	0.344	0.344	0.344	0.345	0.346
Basic controls	No	Yes	Yes	Yes	Yes	Yes
Full controls	No	No	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1430839	1427097	1388328	1388328	1253748	1253748

Note. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). "Cash Crop Price Index" is the sum across fifteen crops (banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, wheat) of their price in t-1 (indexed to 1 in year 2010 for each crop) and their suitability. The index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. "Witchcraft (cell)" is the standardized mean prevalence of witchcraft among individuals with information on witchcraft in a given grid cell, assigned to everyone in the grid-cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable is an indicator that equals one if the respondent is from a new Christian church. Country-waves where new Christians cannot be distinguished from Protestant denominations are dropped. Basic controls include age, age2 and gender. Full controls further include education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the cell level in parenthesis.

C.3.7. Robustness: alternative standard errors

Table C18: Shocks to cash crops and conversion to new Christian churches — Conley standard errors

	Respondent is New Christian					
	(1)	(2)	(3)	(4)	(5)	(6)
Cash Crop Price Index	0.0168***	0.0168***	0.0184***	0.0151***	0.0182***	0.0163***
	(0.00553)	(0.00553)	(0.00553)	(0.00569)	(0.00578)	(0.00606)
Female		0.0271***	0.0334***	0.0333***	0.0330***	0.0325***
		(0.00306)	(0.00340)	(0.00339)	(0.00360)	(0.00331)
Cash Crop Price Index x Female				0.00484***		0.00271*
				(0.00153)		(0.00143)
Cash-crop PI x witchcraft (cell)					0.00492***	-0.000306
					(0.00152)	(0.00214)
witch_female						0.00797***
						(0.00278)
Cash-crop PI x Female x witch.						0.00751***
						(0.00156)
Mean Y	0.243	0.243	0.239	0.239	0.241	0.241
R-squared	0.421	0.422	0.423	0.423	0.429	0.429
Basic controls	No	Yes	Yes	Yes	Yes	Yes
Full controls	No	No	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1966628	1962886	1862101	1862101	1709683	1709683

Note. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). "Cash Crop Price Index" is the sum across fifteen crops (banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, wheat) of their price in t-1 (indexed to 1 in year 2010 for each crop) weighted by their suitability. The index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. "Witchcraft (cell)" is the standardized mean prevalence of witchcraft in a given grid cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable uses the grouped definition: equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. Basic controls include age, age squared and gender. Full controls further include education, labor market participation, marital status and urban/rural residence. Conley (1999) spatial HAC standard errors with a 100 km cutoff are reported in parenthesis, using DHS cluster centroids.

Table C19: Shocks to cash crops and conversion to new Christian churches, country-year clustering

	Respondent is New Christian					
	(1)	(2)	(3)	(4)	(5)	(6)
Cash Crop Price Index	0.0168**	0.0168**	0.0183**	0.0150*	0.0182**	0.0162**
	(0.00752)	(0.00754)	(0.00777)	(0.00785)	(0.00773)	(0.00787)
Female		0.0271***	0.0317***	0.0316***	0.0314***	0.0308***
		(0.00393)	(0.00421)	(0.00418)	(0.00436)	(0.00415)
Cash Crop Price Index x Female				0.00491		0.00280
				(0.00336)		(0.00321)
Cash-crop PI x witchcraft (cell)					0.00496*	-0.000271
					(0.00275)	(0.00319)
Cash-crop PI x Female x witch.						0.00751**
						(0.00304)
Mean Y	0.243	0.243	0.239	0.239	0.241	0.241
R-squared	0.421	0.422	0.423	0.423	0.429	0.429
Basic controls	No	Yes	Yes	Yes	Yes	Yes
Full controls	No	No	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1966628	1962886	1862101	1862101	1709683	1709683

Note. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). "Cash Crop Price Index" is the sum across fifteen crops (banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, wheat) of their price in t-1 (indexed to 1 in year 2010 for each crop) weighted by their suitability. The index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. "Witchcraft (cell)" is the standardized mean prevalence of witchcraft in a given grid cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable is an indicator that equals one if the respondent is from a new Christian church (or Protestant in surveys where new Christians and Protestants are grouped together). Basic controls include age, age squared and gender. Full controls further include education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the country-year level are reported in parenthesis.

C.3.8. *Robustness:* $0.25° \times 0.25°$ *cells, 9 main export crops (Roessler et al., 2022)*

Table C20: Shocks to cash crops and conversion to New Christian churches, $0.25° \times 0.25°$ cells (Roessler et al., 2022)

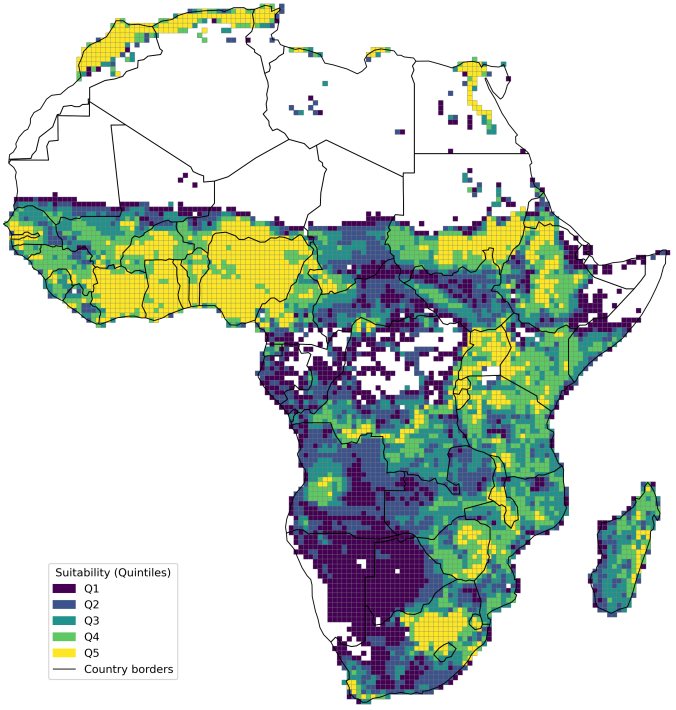
	Respondent is New Christian					
	(1)	(2)	(3)	(4)	(5)	(6)
Cash Crop Price Index	0.0576***	0.0569***	0.0556***	0.0510***	0.0511***	0.0481***
	(0.00498)	(0.00497)	(0.00500)	(0.00494)	(0.00548)	(0.00538)
Female		0.0250***	0.0297***	0.0295***	0.0296***	0.0287***
		(0.00133)	(0.00141)	(0.00137)	(0.00146)	(0.00150)
Cash Crop Price Index x Female				0.00680***		0.00458***
				(0.00117)		(0.00144)
Cash-crop PI x witchcraft (cell)					0.00355	0.00174
					(0.00232)	(0.00234)
Cash-crop PI x Female x witch.						0.00237**
						(0.00112)
Mean Y	0.237	0.238	0.237	0.237	0.243	0.243
R-squared	0.459	0.460	0.456	0.456	0.458	0.459
Basic controls	No	Yes	Yes	Yes	Yes	Yes
Full controls	No	No	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1822215	1818726	1737891	1737891	1627689	1627689

NOTE. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). "Cash Crop Price Index" is measured as in Roessler et al. (2022): the sum across nine cash crops (coffee, cocoa, cotton, groundnuts, oil palm, sugarcane, tea, tobacco, and banana) of their price in t-1 (indexed to 1 in year 2010 for each crop) weighted by their suitability. The index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. "Witchcraft (cell)" is the standardized mean prevalence of witchcraft in a given grid cell. The outcome variable uses the grouped definition: equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. Basic controls include age, age squared and gender. Full controls further include education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the cell level are reported in parenthesis.

C.3.9. *Robustness: alternative measure of economic opportunities*

The M3 crop database comes from Monfreda et al. (2008) and contains information on the estimated harvested area in hectares for 137 different crops for grid-cells of 5 arc minutes x 5 arc minutes resolution for the year 2000. In contrast with our baseline measure, M3-crop data report actual harvested areas rather than degrees of soil suitability. In this case, prices are weighted by the crop share of harvested area in 2000 in the cell (limiting attention to crops for which price data are available). Figure C5 shows the spatial distribution of this suitability measure.

Figure C5: Crop suitability, M3 database



Source: The figure shows the spatial distribution of cash crop suitability using the information from the M3 crop database (Monfreda et al., 2008). The unit of observation is a $0.5° \times 0.5°$ grid-cell. Suitability is measured as the sum of the estimated harvested area in hectares across the fifteen crops used in the analysis: banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, and wheat.

Table C21: Shocks to cash crops and conversion to New Christian churches, M3 crop database

	Respondent is New Christian					
	(1)	(2)	(3)	(4)	(5)	(6)
Cash Crop Price Index	0.0608***	0.0622***	0.0692***	0.0537***	0.0677***	0.0551***
	(0.00910)	(0.00915)	(0.00957)	(0.00920)	(0.0105)	(0.00997)
Female		0.0270***	0.0316***	0.00859**	0.0312***	0.0130***
		(0.00160)	(0.00174)	(0.00344)	(0.00182)	(0.00389)
Cash Crop Price Index x Female				0.0223***		0.0177***
				(0.00349)		(0.00398)
Cash-crop PI x witchcraft (cell)					0.0142***	0.00899*
					(0.00401)	(0.00462)
Cash-crop PI x Female x witch.						0.00741**
						(0.00360)
Mean Y	0.243	0.243	0.239	0.239	0.240	0.240
R-squared	0.423	0.425	0.426	0.426	0.430	0.430
Basic controls	No	Yes	Yes	Yes	Yes	Yes
Full controls	No	No	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1951664	1948014	1848754	1848754	1702959	1702959

Note. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). "Cash Crop Price Index" is computed using M3-crop instead of GAEZ data. "Witchcraft (cell)" is the standardized mean prevalence of witchcraft in a given grid cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable uses the grouped definition: equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. Basic controls include age, age squared and gender. Full controls further include education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the cell level appear in parenthesis.

C.3.10. Robustness: additional control interactions

Table C22: Cash crop price shocks and new Christian conversion: robustness to alternative baseline interactions

	Respondent is New Christian								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Cash-crop PI	0.0162***	0.0171***	0.0160***	0.0163***	0.0113**	0.0140***	0.0162***	0.0182***	0.0149***
	(0.00422)	(0.00405)	(0.00417)	(0.00424)	(0.00453)	(0.00405)	(0.00397)	(0.00418)	(0.00426)
Female	0.0308***	0.0306***	0.0306***	0.0307***	0.0308***	0.0306***	0.0307***	0.0308***	0.0306***
	(0.00171)	(0.00171)	(0.00170)	(0.00171)	(0.00171)	(0.00171)	(0.00172)	(0.00172)	(0.00173)
Cash-crop PI × Female	0.00280***	0.00366***	0.00262***	0.00283***	0.00273***	0.00317***	0.00360***	0.00356***	0.00294***
	(0.000940)	(0.000945)	(0.000911)	(0.000946)	(0.000926)	(0.000944)	(0.000867)	(0.000971)	(0.000874)
Cash-crop PI × Witchcraft (cell)	-0.000271	-0.00113	-0.000136	-0.000841	0.000932	-0.000798	0.00120	-0.00225	-0.00271
	(0.00144)	(0.00152)	(0.00143)	(0.00154)	(0.00147)	(0.00149)	(0.00142)	(0.00159)	(0.00172)
Cash-crop PI × Female × Witchcraft (cell)	0.00751***	0.00742***	0.00739***	0.00763***	0.00771***	0.00754***	0.00780***	0.00721***	0.00838***
	(0.00103)	(0.00104)	(0.00105)	(0.00110)	(0.00106)	(0.00104)	(0.00104)	(0.00103)	(0.00113)
Cash-crop PI × Education (cell)		0.0113***							0.0174***
		(0.00173)							(0.00261)
Cash-crop PI × Female × Education (cell)		0.000655							-0.00212
		(0.000807)							(0.00159)
Cash-crop PI × Urbanization (cell)			0.00101						0.00713***
			(0.00203)						(0.00215)
Cash-crop PI × Female × Urbanization (cell)			-0.00294***						-0.00187
			(0.00113)						(0.00125)
Cash-crop PI × Trad. religion (cell)				0.00171*					0.00338***
				(0.000930)					(0.00102)
Cash-crop PI × Female × Trad. religion (cell)				-0.000442					-0.000170
				(0.000696)					(0.000673)
Cash-crop PI × Employment (cell)					-0.00605***				-0.00728***
					(0.00198)				(0.00218)
Cash-crop PI × Female × Employment (cell)					-0.000278				0.000965
					(0.000994)				(0.00101)
Cash-crop PI × % children dead (cell)						-0.00581***			0.00318
						(0.00169)			(0.00193)
Cash-crop PI × Female × % children dead (cell)						-0.00265***			-0.00346***
						(0.000842)			(0.00123)
Cash-crop PI × modern contracepives (cell)							0.00751***		-0.00470*
							(0.00210)		(0.00285)
Cash-crop PI × Female × modern contracepives (cell)							0.00205*		0.00351**
							(0.00106)		(0.00173)
Cash-crop PI × Decision-making index (cell)								0.00989***	0.00378*
								(0.00200)	(0.00229)
Cash-crop PI × Female × Decision-making index (cell)								-0.000939	-0.00343***
								(0.000924)	(0.00129)
Mean Y	0.241	0.241	0.241	0.241	0.241	0.241	0.241	0.241	0.241
R-squared	0.429	0.429	0.429	0.429	0.429	0.429	0.429	0.429	0.429
Full controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1709683	1709683	1709683	1709683	1709683	1709683	1709683	1709683	1709683

NOTE. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). All specifications include full individual controls and grid-cell and year fixed effects. "Witchcraft (cell)" is the standardized mean prevalence of witchcraft in the grid cell; the sample is restricted to cells with at least 50 observations. The Cash Crop Price Index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. Column 1 replicates column 6 of Table 5. Columns 2–8 add to the column 1 specification the full set of interactions of the Cash Crop Price Index (and its interaction with gender) with the standardized cell-level mean of: education (column 2), urbanization (column 3), share in traditional religion (column 4), employment rate (column 5), share of children ever born who are dead (column 6), use of modern contraceptives (column 7), and a decision-making index (column 8). The decision-making index is the average of four indicators equal to one if the woman has some say (alone or jointly with her partner) in: how her earnings are used, healthcare decisions, large household purchases, and visiting relatives. Column 9 adds all eight sets simultaneously. Cell-level means are computed across all survey waves, mirroring the construction of "Witchcraft (cell)". The outcome uses the grouped new-Christian definition. Robust standard errors clustered at the cell level in parentheses.

C.3.11. Robustness: alternative measures of witchcraft beliefs

PEW measure. Our baseline indicator of witchcraft beliefs is built from the DHS question on whether HIV can be caused by witchcraft. As a robustness test, we construct an alternative measure from an entirely different source, the survey on religion and public life conducted by the Pew Research Center between December 2008 and April 2009 (Pew Research Center, 2009), which interviewed 25,091 respondents in 19 sub-Saharan African countries: Botswana, Cameroon, Chad, the Democratic Republic of the Congo, Djibouti, Ethiopia, Ghana, Guinea-Bissau, Kenya, Liberia, Mali, Mozambique, Nigeria, Rwanda, Senegal, South Africa, Tanzania, Uganda, and Zambia.

The survey includes a battery of items introduced by the question "Which, if any, of the following do you believe in?", to which respondents answer yes or no. Among these items, there are two that broadly refer to witchcraft beliefs. The fi

rst one asks directly whether a respondent believes in "witchcraft", without specifying what the latter means. The second one asks whether a respondent believes in "the evil eye, or that certain people can cast curses or spells that cause bad things to happen to someone". As noted by Gershman (2016), the first one is rather vague given the numerous variations of the superstition that exist across the continent including, among other things, the belief in good witchcraft. The second term, on the other hand, is a close operationalization of the standard definition of witchcraft in the anthropological literature (Evans-Pritchard, 1937; Geschiere, 2013) and captures precisely the definition we use in this paper (this measure is used in existing papers to measure witchcraft prevalence across different contexts — see, for example, Le Rossignol et al., 2025). This measure elicits the belief itself rather than relying on the respondent's own understanding of the *word* "witchcraft", a term whose connotations vary a great deal across languages, translations and religious traditions.

We use the second item and code it as one for a positive answer and zero for a negative one (respondents who do not know or refuse to answer are set to missing). Overall, 45 percent of the respondents answer positively. We then average individual answers within each of the 188 sub-national regions covered by the survey, and we assign to every DHS cluster the mean of the Pew region in which it falls. The resulting regional measure displays substantial spatial variation: its mean is 0.48 and its standard deviation 0.24 (the 10th and 90th percentiles are 0.18 and 0.82, respectively). Because the Pew survey covers 19 of the countries present in the DHS, this exercise restricts the estimation sample to about 1.1 million respondents.

Table C23: Shocks to cash crops and conversion to new Christian churches, witchcraft from PEW

	Respondent is New Christian					
	(1)	(2)	(3)	(4)	(5)	(6)
Cash Crop Price Index	0.0168***	0.0168***	0.0183***	0.0150***	0.0111***	0.0102**
	(0.00381)	(0.00382)	(0.00388)	(0.00393)	(0.00402)	(0.00399)
Female		0.0271***	0.0317***	0.0316***	0.0274***	0.0268***
		(0.00159)	(0.00173)	(0.00172)	(0.00208)	(0.00202)
Cash Crop Price Index x Female				0.00491***		0.00146
				(0.000980)		(0.000978)
Cash-crop PI x witchcraft (PEW)					0.000794	-0.000708
					(0.00180)	(0.00182)
Cash-crop PI x F x witch. (PEW)						0.00201**
						(0.000890)
Mean Y	0.243	0.243	0.239	0.239	0.268	0.268
R-squared	0.421	0.422	0.423	0.423	0.481	0.481
Basic controls	No	Yes	Yes	Yes	Yes	Yes
Full controls	No	No	Yes	Yes	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1966628	1962886	1862101	1862101	1120068	1120068

Note. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). "Cash Crop Price Index" is the sum across fifteen crops (banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, wheat) of their price in t-1 (indexed to 1 in year 2010 for each crop) weighted by their suitability. The index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. "Witchcraft (PEW)" is the (standardized) mean prevalence of witchcraft by region from the PEW data (19 countries). The outcome variable is an indicator that equals one if the respondent is from a new Christian church (or Protestant in surveys where new Christians and Protestants are grouped together). Basic controls include age, age squared and gender. Full controls further include education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the cell level are reported in parenthesis.

C.3.12. *Women's participation in cash-crops and new Christianity*

Table C24: Shocks to cash crops, women participation in cash crop cultivation and conversion

	Respondent is New Christian							
	Women sample				Men sample			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cash-crop price index	0.0205***	0.0206***	0.0234***	0.0239***	0.0123***	0.0126***	0.0127***	0.0121**
	(0.00417)	(0.00422)	(0.00439)	(0.00443)	(0.00462)	(0.00465)	(0.00466)	(0.00475)
CC PI x women in CC	0.00431***	0.00431***	0.00452***	0.00451***	0.00122	0.00123	0.00143	0.00174
	(0.00161)	(0.00161)	(0.00161)	(0.00163)	(0.00163)	(0.00163)	(0.00164)	(0.00173)
CC PI x witchcraft (cell)				0.00518***				0.00335**
				(0.00116)				(0.00136)
CC PI x women in CC x witch.				-0.00247*				-0.00229
				(0.00142)				(0.00151)
Mean Y	0.245	0.246	0.240	0.237	0.242	0.243	0.243	0.243
R-squared	0.427	0.427	0.429	0.435	0.409	0.410	0.410	0.418
Basic controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Full controls	No	No	Yes	Yes	No	No	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1280586	1279117	1198832	1152315	540582	538803	525981	506266

Note. Data: Demographic and Health Surveys and Food and Agriculture Organization (FAO). "Cash Crop Price Index" is the sum across fifteen crops (banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, wheat) of their price in t-1 (indexed to 1 in year 2010 for each crop) and their suitability. The index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. "Women in cash-crops" is measured at the grid-cell level and it is calculated as the standardized share of women who earn cash in their work and their main occupation is agriculture. "Witchcraft (cell)" is the standardized mean prevalence of witchcraft among individuals with information on witchcraft in a given grid cell, assigned to everyone in the grid-cell. The sample is restricted to grid-cells for which "Women in cash-crops" has been computed using at least 50 observations. In columns 4-7, the sample is restricted to grid-cells for which "Women in cash-crops" or "Witchcraft (cell)" have been computed using at least 50 observations. The outcome variable is an indicator that equals one if the respondent is from a new Christian church (or Protestant in surveys where new Christians and Protestants are grouped together). Basic controls include age, age2 and gender. Full controls further include education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the cell level in parenthesis.

C.3.13. *Heterogeneity by women's historical participation in agriculture*

Ethnographic Atlas measure. The Ethnographic Atlas (Murdock, 1967) is a database of 1,265 pre-industrial societies, each of them described by a set of variables coded from the ethnographic record as of the earliest date at which the society was observed — that is, before, or at the very beginning of, its integration into colonial and market structures (for sub-Saharan Africa, most observations refer to the late nineteenth century and the first half of the twentieth century). One of these variables, *sex differences: agriculture* (variable v54), records the gender division of labor in farming. It distinguishes between societies where agriculture is carried out by (i) males only, (ii) males appreciably more than females, (iii) both genders, in differentiated but equal amounts, (iv) both genders equally, (v) females appreciably more than males, and (vi) females only. A residual category groups the societies where agriculture is absent or unimportant.

Following Alesina et al. (2013), who operationalize the hypothesis of Boserup (1970)

on the historical division of labor in agriculture using this same variable, we construct a binary indicator of women's historical participation in agriculture, which we label *Women in agriculture (EA)*. It is equal to one for the societies falling in categories (v) and (vi), and to zero for those falling in categories (i) to (iv) as well as for the few societies where agriculture is absent or unimportant. It is set to missing when the variable is not coded for the society under concern.

Matching the Atlas with the DHS. Most — yet not all — DHS surveys record the ethnic group to which a respondent declares to belong. We therefore match them with the societies of the Ethnographic Atlas by name. Each respondent is then assigned the value taken by the Atlas variable for their group, so that the indicator varies across individuals *within* a grid-cell whenever the latter is ethnically mixed. The baseline sample used in Section 4 comprises 1,966,628 respondents (1,380,772 women and 585,856 men), living in 3,913 grid-cells, and coming from 114 country-waves in 33 countries. Requiring a match with the Atlas leaves us with 927,377 respondents (47 percent of the baseline sample), of whom 666,764 are women and 260,613 are men, distributed over 149 ethnic groups, 2,437 grid-cells, and 78 country-waves in 25 countries.[51] In the remaining sample, 60 of the 149 ethnic groups are classified as groups in which women historically dominated agriculture. They account for 32.5 percent of the respondents and 32.1 percent of the women.

Reassuringly, the historical measure lines up with the contemporaneous measure of women's involvement in cash-cropping used above, yet without being mechanically related to it: women who belong to a historically female-farming group live in grid-cells where the current share of women who earn cash and whose main occupation is agriculture is 0.35 standard deviations higher (standard error of 0.07, clustered at the cell level).

[51] Among the non-matched individuals, 526,722 (27 percent of the baseline sample) come from surveys that do not ask about ethnicity at all. Second, among the respondents whose ethnicity *is* recorded, 36 percent belong to groups that cannot be matched with an Atlas society, or to societies for which the agriculture variable is not coded.

Table C25: Shocks to cash crops, historical participation of women in agriculture and conversion

	Respondent is New Christian							
	Women sample				Men sample			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Cash-crop price index	0.0182***	0.0185***	0.0202***	0.0211***	0.0223***	0.0225***	0.0217***	0.0212***
	(0.00508)	(0.00515)	(0.00534)	(0.00570)	(0.00676)	(0.00677)	(0.00667)	(0.00684)
CC PI x women in agric. (EA)	0.0104***	0.0103***	0.0127***	0.0123***	0.00590	0.00588	0.00617	0.00816*
	(0.00351)	(0.00351)	(0.00355)	(0.00408)	(0.00394)	(0.00392)	(0.00393)	(0.00433)
CC PI x witchcraft (cell)				0.00944***				0.00623***
				(0.00144)				(0.00205)
CC PI x women in agric. x witch.				-0.00790***				-0.00702**
				(0.00260)				(0.00321)
Mean Y	0.229	0.229	0.222	0.225	0.247	0.247	0.249	0.251
R-squared	0.484	0.484	0.480	0.478	0.474	0.474	0.472	0.469
Basic controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Full controls	No	No	Yes	Yes	No	No	Yes	Yes
Grid-cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	666652	665865	630145	608804	260512	259512	252944	245073

Note. Data: Demographic and Health Surveys (DHS), Food and Agriculture Organization (FAO), and Ethnographic Atlas (Murdock, 1967). "Cash-crop price index" is the sum across fifteen crops (banana, barley, cocoa, coconut, coffee, cotton, maize, palm oil, rice, sorghum, soybean, sugar, tea, tobacco, wheat) of their price in t-1 (indexed to 1 in year 2010 for each crop) weighted by their suitability. The index is standardized to have mean 0 and standard deviation 1, so coefficients are expressed for a one-standard-deviation increase. "Women in agric. (EA)" is a pre-determined indicator equal to one if, in the ethnic group the respondent declares to belong to, women historically performed appreciably more agricultural work than men (or were the only ones to farm) according to the Ethnographic Atlas, and zero otherwise. It is missing for respondents whose ethnicity is not recorded in the survey or whose group cannot be matched with an Atlas society (see Appendix C.3.13 for details on the measure and on the remaining sample). "Witchcraft (cell)" is the standardized mean prevalence of witchcraft beliefs among individuals with information on witchcraft in a given grid-cell, assigned to everyone in the grid-cell. In columns 4 and 8, the sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable is an indicator that equals one if the respondent is from a new Christian church (or Protestant in surveys where new Christians and Protestants are grouped together). Basic controls include age and age squared. Full controls further include education, labor market participation, marital status and urban/rural residence. Robust standard errors clustered at the cell level in parenthesis.

C.4. Economic opportunity II: Development aid projects

C.4.1. Robustness: different measures of new Christian

Table C26: New Christians and World Bank gender-related projects, raw definition (55x55km)

	New Christian (raw def.)			
	(1)	(2)	(3)	(4)
Female	0.0253***	0.0274***	0.0278***	0.0310***
	(0.00172)	(0.00165)	(0.00173)	(0.00181)
Female x Witchcraft (cell)			0.00881***	0.00888***
			(0.00196)	(0.00190)
WB gender project	-0.0106	0.0281***	0.00566	0.00709
	(0.0227)	(0.00970)	(0.00776)	(0.00774)
WB gender project x Female	0.00868	0.00411	0.00136	0.000961
	(0.00759)	(0.00753)	(0.00441)	(0.00425)
WB gndr project x Witch. (cell)			0.0389***	0.0378***
			(0.00641)	(0.00612)
WB gndr x Female x Witch. (cell)			0.0139***	0.0154***
			(0.00497)	(0.00477)
Total number of WB projects	0.00834***	0.00162	0.00114	0.00130
	(0.00182)	(0.00115)	(0.00114)	(0.00115)
Mean Y	0.208	0.208	0.214	0.212
R-squared	0.259	0.418	0.422	0.418
Country and year FE	Yes	Yes	Yes	Yes
Grid-cell FE	No	Yes	Yes	Yes
Individual	No	No	No	Yes
N	1784632	1784632	1650755	1580922

Note. Data: Demographic and Health Surveys and AidData research lab (2020). "WB gender project" is an indicator that equals one if there is a WB gender-related project before the month of the interview in the grid-cell where the individual lives. "Witchcraft (cell)" is the (standardized) mean prevalence of witchcraft in a given grid cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable uses the raw definition: "New Christian" equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination, using DHS categories as reported without recoding. The total number of World Bank projects before the month of the interview in a given cell always included as a control. Individual controls include age, age squared, gender, education, labor market participation, marital status and urban/rural residence. Country and year fixed effects are always included. Robust standard errors clustered at the cell level appear in parenthesis.

Table C27: New Christians and World Bank gender-related projects, restricted definition (55x55km)

	New Christian (restricted)			
	(1)	(2)	(3)	(4)
Female	0.0318***	0.0316***	0.0316***	0.0317***
	(0.00218)	(0.00216)	(0.00225)	(0.00229)
Female x Witchcraft (cell)			0.0115***	0.0109***
			(0.00221)	(0.00219)
WB gender project	0.0266	0.0513***	0.0228**	0.0231**
	(0.0184)	(0.0143)	(0.00890)	(0.00916)
WB gender project x Female	0.0110	0.00800	0.00343	0.00289
	(0.00986)	(0.00995)	(0.00529)	(0.00531)
WB gndr project x Witch. (cell)			0.0343***	0.0337***
			(0.00669)	(0.00658)
WB gndr x Female x Witch. (cell)			0.0119**	0.0132***
			(0.00510)	(0.00508)
Total number of WB projects	0.00502***	-0.000967	-0.00195*	-0.00195*
	(0.00151)	(0.00111)	(0.00105)	(0.00107)
Mean Y	0.178	0.178	0.182	0.183
R-squared	0.266	0.350	0.354	0.354
Country and year FE	Yes	Yes	Yes	Yes
Grid-cell FE	No	Yes	Yes	Yes
Individual	No	No	No	Yes
N	1283132	1283132	1170404	1144365

Note. Data: Demographic and Health Surveys and AidData research lab (2020). "WB gender project" is an indicator that equals one if there is a WB gender-related project before the month of the interview in the grid-cell where the individual lives. "Witchcraft (cell)" is the (standardized) mean prevalence of witchcraft in a given grid cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable is an indicator that equals one if the respondent is from a new Christian church. Country-waves where new Christians and Protestants are grouped together are dropped. The total number of World Bank projects before the month of the interview in a given cell always included as a control. Individual controls include age, age squared, gender, education, labor market participation, marital status and urban/rural residence. Country and year fixed effects are always included. Robust standard errors clustered at the cell level appear in parenthesis.

C.4.2. Robustness: alternative standard errors

Table C28: New Christians and World Bank gender-related projects, country-year clustering

	New Christian (broad def.)			
	(1)	(2)	(3)	(4)
Female	0.0282***	0.0302***	0.0296***	0.0326***
	(0.00468)	(0.00446)	(0.00461)	(0.00461)
Female x Witchcraft (cell)			0.00882*	0.00896*
			(0.00488)	(0.00471)
WB gender project	-0.00898	0.0294*	0.00336	0.00509
	(0.0214)	(0.0163)	(0.00960)	(0.00990)
WB gender project x Female	0.00575	0.00134	-0.000403	-0.000781
	(0.00828)	(0.00786)	(0.00576)	(0.00582)
WB gndr project x Witch. (cell)			0.0423***	0.0412***
			(0.00683)	(0.00687)
WB gndr x Female x Witch. (cell)			0.0139**	0.0153***
			(0.00544)	(0.00554)
Total number of WB projects	0.00822***	-0.000429	-0.00104	-0.000941
	(0.00190)	(0.00110)	(0.00110)	(0.00115)
Mean Y	0.250	0.250	0.249	0.247
R-squared	0.287	0.434	0.444	0.441
Country and year FE	Yes	Yes	Yes	Yes
Grid-cell FE	No	Yes	Yes	Yes
Individual	No	No	No	Yes
N	1784632	1784632	1650755	1580922

Note. Data: Demographic and Health Surveys and AidData research lab (2020). "WB gender project" is an indicator that equals one if there is a WB gender-related project before the month of the interview in the grid-cell where the individual lives. "Witchcraft (cell)" is the (standardized) mean prevalence of witchcraft in a given grid cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable is an indicator that equals one if the respondent is from a new Christian church (or Protestant in surveys where new Christians and Protestants are grouped together). The total number of World Bank projects before the month of the interview in a given cell always included as a control. Individual controls include age, age squared, gender, education, labor market participation, marital status and urban/rural residence. Country and year fixed effects are always included. Robust standard errors clustered at the country-year level are reported in parenthesis.

Table C29: New Christians and World Bank gender-related projects — Conley standard errors

	New Christian (broad def.)			
	(1)	(2)	(3)	(4)
Female	0.0282***	0.0302***	0.0296***	0.0346***
	(0.00321)	(0.00306)	(0.00317)	(0.00352)
Female x Witchcraft (cell)			0.00882***	0.00895***
			(0.00328)	(0.00318)
WB gender project	-0.00904	0.0293**	0.00307	0.00544
	(0.0373)	(0.0140)	(0.00916)	(0.00915)
WB gender project x Female	0.00577	0.00135	-0.000364	-0.00125
	(0.0107)	(0.0107)	(0.00599)	(0.00595)
WB gndr project x Witch. (cell)			0.0423***	0.0408***
			(0.00676)	(0.00661)
WB gndr x Female x Witch. (cell)			0.0139**	0.0158**
			(0.00642)	(0.00646)
Total number of WB projects	0.00822***	-0.000424	-0.00104	-0.000943
	(0.00260)	(0.00109)	(0.00105)	(0.00107)
Mean Y	0.250	0.250	0.249	0.247
R-squared	0.287	0.434	0.444	0.441
Country and year FE	Yes	Yes	Yes	Yes
Grid-cell FE	No	Yes	Yes	Yes
Individual	No	No	No	Yes
N	1784632	1784632	1650755	1580922

Note. Data: Demographic and Health Surveys and AidData research lab (2020). "WB gender project" is an indicator that equals one if there is a WB gender-related project before the month of the interview in the grid-cell where the individual lives. "Witchcraft (cell)" is the (standardized) mean prevalence of witchcraft in a given grid cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable is an indicator that equals one if the respondent is from a new Christian church (or Protestant in surveys where new Christians and Protestants are grouped together). The total number of World Bank projects before the month of the interview in a given cell always included as a control. Individual controls include age, age squared, gender, education, labor market participation, marital status and urban/rural residence. Country and year fixed effects are always included. Conley (1999) spatial HAC standard errors with a 100 km cutoff are reported in parenthesis, using DHS cluster centroids.

C.4.3. Robustness: witchcraft beliefs from the Pew Research Center

Here we replace the DHS-based measure of witchcraft beliefs by the regional-level measure constructed from the Pew Research Center survey, whose definition and merits are detailed in Appendix C.3.11.

Table C30: New Christians and World Bank gender-related projects, witchcraft from PEW

	New Christian (broad def.)			
	(1)	(2)	(3)	(4)
Female	0.0282***	0.0302***	0.0224***	0.0281***
	(0.00176)	(0.00169)	(0.00171)	(0.00194)
Female x Witchcraft (PEW)			-0.00754***	-0.00737***
			(0.00147)	(0.00141)
WB gender project	-0.00898	0.0294***	-0.0173	-0.0130
	(0.0228)	(0.0103)	(0.0124)	(0.0126)
WB gender project x Female	0.00575	0.00134	0.00701	0.00451
	(0.00759)	(0.00754)	(0.0120)	(0.0120)
WB gndr project x Witch. (PEW)			0.00554	0.00590
			(0.0140)	(0.0143)
WB gndr x Female x Witch. (PEW)			0.0293*	0.0282*
			(0.0155)	(0.0154)
Total number of WB projects	0.00822***	-0.000429	0.000169	0.000380
	(0.00181)	(0.000985)	(0.00108)	(0.00107)
Mean Y	0.250	0.250	0.270	0.268
R-squared	0.287	0.434	0.488	0.486
Country and year FE	Yes	Yes	Yes	Yes
Grid-cell FE	No	Yes	Yes	Yes
Individual	No	No	No	Yes
N	1784632	1784632	1166086	1104443

Note. Data: Demographic and Health Surveys and AidData research lab (2020). "WB gender project" is an indicator that equals one if there is a WB gender-related project before the month of the interview in the grid-cell where the individual lives. "Witchcraft (PEW)" is the mean prevalence of witchcraft by region from the PEW data (19 countries). The outcome variable is an indicator that equals one if the respondent is from a new Christian church (or Protestant in surveys where new Christians and Protestants are grouped together). The total number of World Bank projects before the month of the interview in a given cell always included as a control. Individual controls include age, age squared, gender, education, labor market participation, marital status and urban/rural residence. Country and year fixed effects are always included. Robust standard errors clustered at the cell level are reported in parenthesis.

C.4.4. Robustness: $0.25^\circ \times 0.25^\circ$ *cells*

Table C31: New Christians and World Bank gender-related projects, 28x28km

	New Christian (broad def.)			
	(1)	(2)	(3)	(4)
Female	0.0281***	0.0301***	0.0302***	0.0320***
	(0.00140)	(0.00135)	(0.00150)	(0.00154)
Female x Witchcraft (cell)			0.00914***	0.00942***
			(0.00171)	(0.00167)
WB gender project	0.000790	0.0175	0.00309	0.00268
	(0.0241)	(0.0131)	(0.0108)	(0.0109)
WB gender project x Female	0.0138	0.0114	0.000299	-0.000110
	(0.0124)	(0.0124)	(0.00642)	(0.00632)
WB gndr project x Witch. (cell)			0.0265***	0.0275***
			(0.00949)	(0.00936)
WB gndr x Female x Witch. (cell)			0.0198**	0.0199**
			(0.00842)	(0.00825)
Total number of WB projects	0.00776***	-0.000476	-0.00126	-0.00109
	(0.00237)	(0.00150)	(0.00151)	(0.00151)
Mean Y	0.250	0.250	0.256	0.254
R-squared	0.285	0.456	0.462	0.459
Country and year FE	Yes	Yes	Yes	Yes
Grid-cell FE	No	Yes	Yes	Yes
Individual	No	No	No	Yes
N	1786646	1786645	1476964	1414070

Note. Data: Demographic and Health Surveys and AidData research lab (2020). "WB gender project" is an indicator that equals one if there is a WB gender-related project before the month of the interview in the grid-cell where the individual lives. "Witchcraft (cell)" is the (standardized) mean prevalence of witchcraft in a given grid cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable uses the grouped definition: "New Christian" equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. The total number of World Bank projects before the month of the interview in a given cell always included as a control. Individual controls include age, age squared, gender, education, labor market participation, marital status and urban/rural residence. Country and year fixed effects are always included. Robust standard errors clustered at the cell level are reported in parenthesis.

C.4.5. Alternative measure of gender-related project: description, validation and results

Relying on a project's main theme to measure its pro-women orientation may be too restrictive. We therefore broaden our definition to include projects that provide income-earning opportunities for women, even if their main theme is not officially classified as

"gender." Towards that purpose, we inspected project documents made available by the World Bank. More specifically, we checked all projects whose main theme falls in the following categories: "Social protection and risk management", "Social dev/gender/inclusion", "Rural development", "Other human development", "Micro, Small and Medium Enterprise support", "Other financial and private sector development", "Land administration and management" and "Other environment and natural resources management".[52] This procedure enabled us to classify 344 additional project sites as gender-focused. Based on this broader definition, 7% of the respondents (130,000 in 107 grid-cells) have been exposed to a woman-focused project.[53]

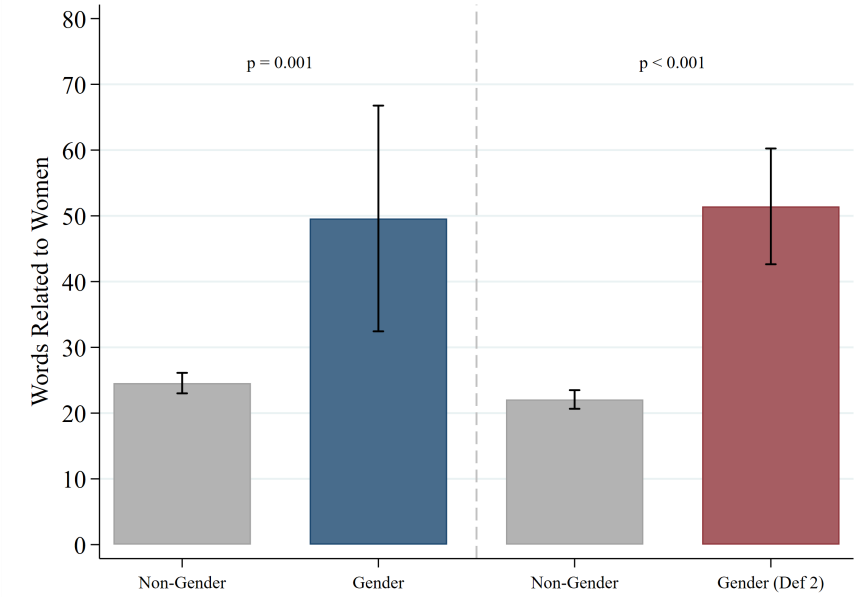
Validation. To verify that the broader measure captures a genuine pro-women orientation, we compare the average number of words associated with women in the descriptions of projects classified as gender-related versus the rest. Figure C6 shows that project descriptions classified as gender-related — whether under the narrow (main-theme) or the broader definition — contain significantly more women-related words than other projects, confirming that our alternative measure is meaningful.

Results. Table C33 reports the estimates obtained with this broader definition. The direct effect of exposure to a woman-focused project on women's new Christian affiliation vanishes: there is no significant increase in new Christian church membership when female respondents were exposed to a project on average. What is striking, however, is that the differential effect remains strong in areas where witchcraft beliefs prevail: there, exposure to a project increases women's likelihood of having joined a new Christian church by more than 4 pp (column 4). This finding reinforces the interpretation that new Christian communities play a special role in combating the perceived intervention of malevolent supernatural forces.

[52] Within these broad categories, we excluded projects explicitly mentioning the following words: electrification, HIV, roads/transport, river basin, research and training, petroleum, emergency reconstruction, trade reform, street children, energy, sanitation, infrastructure renewal, infrastructure rehabilitation, water sector, infrastructure project, communications infrastructure, national health, family planning, education, water development, hydropower, transmission hub, agricultural research, governance, budget support, financial sector, rural infrastructure, institutional reform, city management, biodiversity, cultural heritage, mining sector capacity building, nutrition enhancement, municipal development, coastal environment management, mineral resource, distance learning, school access, local government, avian influenza control, hydroelectric, structural adjustment credit.

[53] The proportion goes up to 13% (244,000 in 138 grid-cells) when larger grid-cells are used.

Figure C6: Women-related words by type of project, different definitions



Notes: The figure illustrates the difference in the average number of words associated with women in the descriptions of 368 gender-related and non-gender-related World Bank projects. A gender-related project is defined in two ways: 1) a project whose main theme is gender, and 2) a project whose main theme is gender or that focuses on providing income-generating activities for women. Bars represent standard errors, and p-values from t-tests are reported above each figure.

Table C32: Women-related words and World Bank projects

	(Log) Number of women-related worlds			
	(1)	(2)	(3)	(4)
Gender Project	0.580**		0.758***	
	(0.233)		(0.238)	
Gender Project (Def 2)		0.790***		1.107***
		(0.196)		(0.155)
Country FE	No	No	Yes	Yes
Controls	No	No	Yes	Yes
R-squared	0.008	0.047	0.439	0.499
N	4401	4401	4400	4400

NOTE. Unit of observation: project-location site. "Gender Project" is defined as a project whose main theme is gender. "Gender Project (Def 2") also includes projects focusing on providing income-generating activities for women. Fixed effects include country and start year of the project. Controls include the (log) cost of the project and project status (active or closed). Standard errors clustered at project level appear in parentheses.

Table C33: New Christians and World Bank gender-related projects, alternative WB gender project measure (55x55km)

	New Christian (broad def.)			
	(1)	(2)	(3)	(4)
Female	0.0270***	0.0300***	0.0295***	0.0325***
	(0.00182)	(0.00173)	(0.00182)	(0.00189)
Female x Witchcraft (cell)			0.00810***	0.00822***
			(0.00209)	(0.00203)
WB gender project	-0.0279*	0.00169	-0.00253	-0.00181
	(0.0154)	(0.00856)	(0.00615)	(0.00628)
WB gender project x Female	0.0103*	0.00245	0.00186	0.00142
	(0.00569)	(0.00529)	(0.00372)	(0.00362)
WB gndr project x Witch. (cell)			0.0256***	0.0257***
			(0.00608)	(0.00596)
WB gndr x Female x Witch. (cell)			0.0125***	0.0137***
			(0.00394)	(0.00382)
Total number of WB projects	0.00857***	-0.000355	-0.000890	-0.000824
	(0.00189)	(0.00102)	(0.000982)	(0.001000)
Mean Y	0.250	0.250	0.249	0.247
R-squared	0.287	0.434	0.444	0.441
Country and year FE	Yes	Yes	Yes	Yes
Grid-cell FE	No	Yes	Yes	Yes
Individual	No	No	No	Yes
N	1784632	1784632	1650755	1580922

Note. Data: Demographic and Health Surveys and AidData research lab (2020). "WB gender project" is an indicator that equals one if there is a WB gender-related project before the month of the interview in the grid-cell where the individual lives. "Witchcraft (cell)" is the (standardized) mean prevalence of witchcraft in a given grid cell. The sample is restricted to grid-cells for which "Witchcraft (cell)" has been computed using at least 50 observations. The outcome variable uses the grouped definition: "New Christian" equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. The total number of World Bank projects before the month of the interview in a given cell always included as a control. Individual controls include age, age squared, gender, education, labor market participation, marital status and urban/rural residence. Country and year fixed effects are always included. Robust standard errors clustered at the cell level appear in parenthesis.

C.5. *New Christian churches and decision-making*

Table C34: New Christian churches, labor and reproductive outcomes, and decision-making, raw definition

	Labor market	Reproductive			Decision-making		
					At home	In the church	
	Working status	Contraceptives	Ideal nr of children	Children ever born	Decision-making Index	Respect for women	Female leadership
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: New Christians compared to all other denominations							
New Christian	0.0574***	0.0513***	-0.633***	-0.444***	0.0734***	0.0968***	0.185***
	(0.00214)	(0.00141)	(0.0127)	(0.0151)	(0.00188)	(0.00570)	(0.00686)
R-squared	0.0922	0.111	0.244	0.156	0.271	0.0859	0.197
Panel B: New Christians (and non-Christians) compared to missionary churches							
New Christian	0.0161***	0.0124***	-0.126***	-0.104***	0.0233***	0.0234***	0.0571***
	(0.00213)	(0.00139)	(0.0103)	(0.0147)	(0.00176)	(0.00795)	(0.00975)
Non Christian	-0.0960***	-0.0928***	1.241***	0.786***	-0.111***	-0.113***	-0.197***
	(0.00233)	(0.00132)	(0.0132)	(0.0151)	(0.00192)	(0.00917)	(0.0107)
Country-round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y	0.765	0.217	5.155	5.562	0.578	0.791	0.460
R-squared	0.0980	0.117	0.269	0.165	0.279	0.0921	0.210
N	626816	1554236	1392334	443637	965130	24605	23839

Note. Data: All available DHS in columns 1-5, and PEW for columns 6 and 7. The sample is restricted to women. In column 1, the sample is restricted to women over 30 years old and in column 4 to women over 35 years old to approximate completed fertility. The table reports OLS estimates. "New Christian" uses the raw definition: it equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination, using DHS categories as reported without recoding. Outcome variables are: probability to work (column 1), use of contraceptives at the time of the survey (column 2), ideal number of children (column 3), total number of children ever born (column 4), decision-making index (column 5). This index is calculated as the average of all non-missing responses to four yes/no questions (indicators) that equal one if the respondent has some say – either the woman decides together with her partner/husband or she decides alone – in the following matters: (1) how a woman's earning are used, (2) healthcare, (3) large household purchases, and (4) visiting relatives. In column 6, the outcome variable is an indicator that equals one if the respondent associates attitudes of respect for women with Christianity. In column 7, the outcome variable is an indicator equal to one if the respondent thinks that women should be allowed to serve in religious leadership roles, such as pastor, priest or imam. Country-wave fixed effects always included. Controls include age, age squared, and rural/urban residence. Robust standard errors clustered at the DHS cluster level are in parentheses in the DHS data.

Table C35: New Christian churches, labor and reproductive outcomes, and decision-making, restricted definition

	Labor market	Reproductive			Decision-making		
					At home	In the church	
	Working status	Contraceptives	Ideal nr of children	Children ever born	Decision-making Index	Respect for women	Female leadership
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: New Christians compared to all other denominations							
New Christian	0.0323***	0.0259***	-0.163***	-0.0573***	0.0239***	0.0968***	0.185***
	(0.00239)	(0.00161)	(0.0106)	(0.0156)	(0.00176)	(0.00570)	(0.00686)
R-squared	0.0905	0.115	0.269	0.165	0.293	0.0859	0.197
Panel B: New Christians (and non-Christians) compared to missionary churches							
New Christian	0.0105***	0.00424***	0.0869***	0.109***	-0.000121	0.0234***	0.0571***
	(0.00237)	(0.00159)	(0.0101)	(0.0158)	(0.00180)	(0.00795)	(0.00975)
Non Christian	-0.0720***	-0.0758***	0.905***	0.558***	-0.0753***	-0.113***	-0.197***
	(0.00257)	(0.00156)	(0.0125)	(0.0165)	(0.00202)	(0.00917)	(0.0107)
Country-round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y	0.766	0.221	5.072	5.521	0.565	0.791	0.460
R-squared	0.0937	0.119	0.282	0.169	0.297	0.0921	0.210
N	482044	1171691	1063596	331292	753293	24605	23839

Note. Data: All available DHS in columns 1-5, and PEW for columns 6 and 7. The sample is restricted to women. In column 1, the sample is restricted to women over 30 years old and in column 4 to women over 35 years old to approximate completed fertility. The table reports OLS estimates. "New Christian" is an indicator that equals one if the respondent reports one of the new Christian religions as their main religion in the raw DHS data. Outcome variables are: probability to work (column 1), use of contraceptives at the time of the survey (column 2), ideal number of children (column 3), total number of children ever born (column 4), decision-making index (column 5). This index is calculated as the average of all non-missing responses to four yes/no questions (indicators) that equal one if the respondent has some say – either the woman decides together with her partner/husband or she decides alone – in the following matters: (1) how a woman's earning are used, (2) healthcare, (3) large household purchases, and (4) visiting relatives. In column 6, the outcome variable is an indicator that equals one if the respondent associates attitudes of respect for women with Christianity. In column 7, the outcome variable is an indicator equal to one if the respondent thinks that women should be allowed to serve in religious leadership roles, such as pastor, priest or imam. Country-wave fixed effects always included. Controls include age, age squared, and rural/urban residence. Robust standard errors clustered at the DHS cluster level are in parentheses in the DHS data.

Table C36: New Christian churches and women's participation in decision-making, different components

	Woman makes decisions about:				
	Healthcare	Large purchases	Visits to relatives	Use of earnings	Index
	(1)	**(2)**	**(3)**	**(4)**	**(5)**
Panel A: New Christians compared to everyone else					
New Christian	0.0833***	0.0842***	0.0738***	-0.000874	0.0693***
	(0.00222)	(0.00220)	(0.00218)	(0.00171)	(0.00178)
Country-round FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Mean Y	0.504	0.478	0.584	0.886	0.578
R-squared	0.210	0.203	0.168	0.0639	0.270
N	901844	902853	903689	478906	965130
Panel B: New Christians (and non-Christians) as compared to missionary churches					
New Christian	0.0263***	0.0247***	0.0216***	-0.00126	0.0215***
	(0.00208)	(0.00207)	(0.00204)	(0.00182)	(0.00166)
Non Christian	-0.132***	-0.138***	-0.121***	-0.000890	-0.112***
	(0.00241)	(0.00239)	(0.00248)	(0.00184)	(0.00192)
Country-round FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Mean Y	0.504	0.478	0.584	0.886	0.578
R-squared	0.217	0.211	0.174	0.0639	0.279
N	901844	902853	903689	478906	965130

Note. Data: DHS. The sample is restricted to women. "New Christian" uses the grouped definition: it equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. Outcome variables are indicators equal to one if the respondent has some say in decisions (either the woman decides together with her partner/husband or she decides alone) regarding (1) healthcare, (2) large household purchases, (3) family visits, (4) use of a woman's earnings. The index is the average of all non-missing responses. Country-round FE are always included. Individual controls include: age, age squared and rural/urban place of residence. Standard errors are clustered at the DHS cluster level.

Table C37: New Christian churches and women's participation in decision-making, 1-3 outcome

	The woman makes decisions about:				
	Use of earnings	Healthcare	Large purchases	Family visits	Index
	(1)	(2)	(3)	(4)	(5)
New Christian	-0.0243***	0.0352***	0.0296***	0.0317***	0.0243***
	(0.00382)	(0.00309)	(0.00285)	(0.00303)	(0.00240)
Non Christian	0.0830***	-0.170***	-0.163***	-0.145***	-0.134***
	(0.00426)	(0.00342)	(0.00322)	(0.00363)	(0.00277)
Country-round FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Mean Y	2.498	1.691	1.609	1.771	1.822
R-squared	0.134	0.185	0.169	0.123	0.263
N	478906	901844	902853	903689	965130

Note. Data: DHS. The sample is restricted to women. "New Christian" uses the grouped definition: it equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. Outcome variables: women were asked who usually decides on (1) how a woman's earnings are used, (2) healthcare, (3) large household purchases, and (4) family visits. The response options have been rescaled so as to obtain a 1 to 3 categorical variable defined as follows: the partner or another person makes the decision (value 1), the respondent and the partner jointly decide (value 2), and the respondent makes the decision alone (value 3). The index presented in column 5 is calculated as the average of all non-missing responses. Country-round FE always are included. Individual controls include: age, age squared and rural/urban place of residence. Standard errors are clustered at the DHS cluster level.

Table C38: New Christianity and decisions about the use of women's earnings

	The woman makes decisions regarding earnings:			
	Categorical [1-3]	Alone or w/ husband	Alone	With husband
	(1)	(2)	(3)	(4)
New Christian	-0.0243***	-0.00126	-0.0231***	0.0218***
	(0.00382)	(0.00182)	(0.00266)	(0.00249)
Non Christian	0.0830***	-0.000890	0.0839***	-0.0848***
	(0.00426)	(0.00184)	(0.00303)	(0.00264)
Country-round FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Mean Y	2.498	0.886	0.613	0.273
R-squared	0.134	0.0639	0.173	0.156
N	478906	478906	478906	478906

Note. Data: DHS. The sample is restricted to women. "New Christian" uses the grouped definition: it equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. The outcome variable in column 1 is a categorical variable with values from 1 to 3: 1 if the partner or another person decides, 2 if the respondent and her partner decide jointly, and 3 if the respondent decides alone. In column 2, the outcome variable is a indicator equal to one if the woman decides alone or together with her husband. In column 3, the indicator is equal to one if the woman decides alone. In column 4, it is equal to one if both spouses decide together. Country-round FE are always included. Individual controls include: age, age squared and rural/urban place of residence. Standard errors are clustered at the DHS cluster level.

C.6. *Prohibitions in new Christian churches: alcohol and polygyny*

Table C39: New Christian churches and alcohol consumption

	DHS data		First-hand data Benin
	Partner/husband drinks alcohol	Man drinks alcohol	Respondent drinks trad. beverage
	(1)	(2)	(3)
New Christian	-0.0671***	-0.147***	-0.0330*
	(0.00392)	(0.0323)	(0.0201)
Non Christian	-0.290***	0.0629***	0.0781***
	(0.00405)	(0.0201)	(0.0288)
Mean Y	0.353	0.428	0.156
R-squared	0.186	0.128	0.0370
Controls	Yes	Yes	Yes
Country-Wave FE	Yes	Yes	No
N	255115	12049	1480

Note. Data: DHS and first-hand data from Benin. The sample is restricted to women in column 1, to men in column 2, and to both men and women in column 3. The outcome variable in column 1 is a indicator that equals one if the woman reports that her husband or partner drinks alcohol. The outcome variable in column 2 is a indicator that equals one if the man reports drinking alcohol. The outcome variable in column 3 is a indicator that equals one if the respondent drinks sodabi (traditional alcoholic drink in Benin). "New Christian" uses the raw definition: it equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination, using DHS categories as reported without recoding. Controls include age, age squared, education, occupation, marital status and rural/urban place of residence in the DHS data; and age, education, marital status, and electricity at home in the first-hand data. Standard errors are clustered at the DHS-cluster level when using DHS data.

Table C40: New Christian churches and polygny

	In polygamous union			
	Unconditional		Married at the time of the survey	
	(1)	(2)	(3)	(4)
New Christian	-0.00560***	-0.00535***	-0.00523***	-0.00505***
	(0.00191)	(0.00193)	(0.00179)	(0.00180)
Non Christian	0.157***	0.159***	0.157***	0.159***
	(0.00206)	(0.00206)	(0.00207)	(0.00206)
Mean Y	0.334	0.338	0.334	0.338
R-squared	0.203	0.204	0.203	0.204
Controls	Yes	Yes	Yes	Yes
Country and year FE	Yes	Yes	Yes	Yes
N	1242133	1014472	1242133	1014472

Note. Data: Demographic and Health Surveys. The outcome variable is an indicator equal to one if the respondent is engaged in a polygamous union. In columns 2 and 4, attention is restricted to women who have ever been married. Columns 1–2 use the raw definition: "New Christian" equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination, using DHS categories as reported without recoding. Columns 3–4 use the grouped definition: "New Christian" equals one if the respondent belongs to an Evangelical, Pentecostal, Apostolic, Charismatic, or related denomination; in country-waves where these denominations are not separately coded from mainstream Protestants, all respondents in that bundled category are coded as new Christian. Country and year FE and individual controls include: age, age squared, gender and urban/rural place of residence. Standard errors clustered at the DHS cluster level are in parenthesis.

C.7. New Christian churches vis-à-vis traditional beliefs and practices

Table C41: Traditional beliefs and gender in SSA

	Rituals	Sacrifices	Initiation	Trad. healer
	(1)	(2)	(3)	(4)
Female	-1.273**	-2.564***	-3.204***	-1.222*
	(0.549)	(0.577)	(0.554)	(0.628)
Controls	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
Mean Y	0.262	0.329	0.267	0.423
R-squared	0.109	0.165	0.104	0.103
N	23537	22816	23474	22964

Note. Data: PEW Research Center. The table reports OLS estimates. The outcome variable is a dummy equal to one if the respondent has ever participated in an African traditional ceremony or performed rituals to honor ancestors (column 1), if s/he believes that sacrifices to ancestors prevents unfortunate events from happening (column 2), if s/he has ever participated in an initiation ritual (column 3), and if s/he has ever consulted with a traditional religious healer (column 4). Individual controls include age, urban/rural place of residence, marital status, education, and whether the respondent has recently lacked money. Robust standard errors are reported in parenthesis.

Table C42: Traditional beliefs or practices and gender in South Benin

	Traditional beliefs and practices			
	Vodoun religion	power of ancestors	power of wizards	power of magic potion
	(1)	(2)	(3)	(4)
Female/100	-7.295***	-14.69***	-5.392**	-9.690***
	(1.798)	(2.735)	(2.339)	(2.271)
Controls	Yes	Yes	Yes	Yes
Mean Y	0.103	0.367	0.156	0.146
R-squared	0.0194	0.0305	0.0322	0.0273
N	1703	1705	1480	1383

Note. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is a dummy equal to one if the respondent's religion is Vodoun (column 1), if s/he believes that the spirits of her/his ancestors influence her/his life (column 2), if s/he has consulted with a wizard (the bokono, in local parlance) or used a fetish to protect against coronavirus (column 3), and if s/he has drunk the traditional alcoholic beverage (sodabi) to protect against coronavirus (column 4). Individual controls include age, marital status, whether the respondent attended school, cultivated pineapple in 2020, and had electricity at home in 2020. Robust standard errors are in parenthesis.

Table C43: New Christians and attitudes toward tradition and supernatural power, by gender

	Traditional beliefs and practices				Beliefs in supernatural power		
	Rituals	Sacrifices	Initiation	Trad. healing	Miracles	Exorcism	God-ordained prosperity
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**
Panel A: New Christians compared to all other denominations							
New Christian	-0.0564***	-0.0821***	-0.0615***	-0.0710***	0.193***	0.167***	0.0580***
	(0.00787)	(0.00830)	(0.00802)	(0.00898)	(0.00905)	(0.00901)	(0.00916)
Female	-0.0123	-0.00562	-0.0321***	0.00353	0.0135	0.0244***	0.0176*
	(0.00795)	(0.00838)	(0.00805)	(0.00873)	(0.00879)	(0.00866)	(0.00905)
New Christian x Female	0.0157	-0.0113	0.0175	-0.00834	0.0252**	0.0119	-0.00682
	(0.0107)	(0.0113)	(0.0109)	(0.0123)	(0.0124)	(0.0125)	(0.0125)
R-squared	0.108	0.164	0.101	0.104	0.0749	0.0687	0.0718
Panel B: New Christians (and non-Christians) as compared to missionary churches							
New Christian	-0.0508***	-0.0647***	-0.0388***	-0.0359***	0.141***	0.101***	0.0557***
	(0.0102)	(0.0109)	(0.0104)	(0.0114)	(0.0116)	(0.0117)	(0.0118)
Female	-0.0121	-0.00510	-0.0314***	0.00461	0.0121	0.0225***	0.0175*
	(0.00795)	(0.00838)	(0.00805)	(0.00873)	(0.00876)	(0.00862)	(0.00906)
New Christian x Female	0.0155	-0.0121	0.0165	-0.00991	0.0274**	0.0147	-0.00673
	(0.0107)	(0.0113)	(0.0109)	(0.0123)	(0.0124)	(0.0125)	(0.0125)
Non Christian	0.00846	0.0260**	0.0339***	0.0525***	-0.0784***	-0.0996***	-0.00339
	(0.00979)	(0.0104)	(0.00997)	(0.0108)	(0.0109)	(0.0109)	(0.0113)
Mean Y	0.263	0.329	0.267	0.424	0.477	0.420	0.571
R-squared	0.108	0.165	0.102	0.105	0.0768	0.0719	0.0718
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	24204	23458	24132	23587	24369	23694	23562

Note. Data: PEW Research Center. The table reports OLS estimates. "New Christian" is an indicator equal to one if the respondent reports one of the new Christian religions as their main religion. The outcome variable, also an indicator, is equal to one if the respondent has ever participated in an African traditional ceremony or performed rituals to honor ancestors (column 1), if s/he believes that sacrifices to ancestors can prevent disturbances from happening (column 2), if s/he has ever participated in an initiation ritual (column 3), if s/he has ever visited a traditional healer (column 4). In column 5, the binary variable is equal to one if the respondent answered "yes" to the question "have you ever experienced or witnessed a divine healing of an illness or injury?". In column 6, value one is obtained equal if s/he answered "yes" to the question "have you ever experienced or witnessed a situation where the devil or evil spirits were driven out of a person?". In column 7, the outcome indicator equals one if the respondent thinks that "God will grant wealth and good health to all believers who have enough faith in Him" rather than "God doesn't always give wealth and good health even to believers who have deep faith in Him". Individual controls include age, urban/rural place of residence, marital status, education, and whether the respondent has recently experienced a money shortage. Robust standard errors are reported in parentheses.

Table C44: New Christianity and trust in traditional leaders and the family

	Trad. leaders (1)	Family (2)	Trad. leaders (3)	Family (4)
Panel A: New Christians as compared to all other denominations				
New Christian	-0.0932***	-0.0693***	-0.0755***	-0.0612***
	(0.00554)	(0.00566)	(0.00751)	(0.00759)
Female	0.000422	-0.0132***	0.0120**	-0.00812
	(0.00462)	(0.00452)	(0.00558)	(0.00523)
New Christian x Female			-0.0350***	-0.0159
			(0.00989)	(0.0101)
R-squared	0.128	0.122	0.128	0.122
Panel B: New Christians (and non-Christians) as compared to missionary churches				
New Christian	-0.0345***	-0.0335***	-0.0130	-0.0233***
	(0.00616)	(0.00619)	(0.00802)	(0.00802)
Female	0.00518	-0.0101**	0.0190***	-0.00381
	(0.00462)	(0.00452)	(0.00557)	(0.00523)
Non Christian	0.154***	0.0984***	0.155***	0.0988***
	(0.00707)	(0.00678)	(0.00707)	(0.00678)
New Christian x Female			-0.0416***	-0.0198**
			(0.00989)	(0.0101)
Mean Y	1.838	2.361	1.838	2.361
R-squared	0.130	0.123	0.130	0.123
Controls	Yes	Yes	Yes	Yes
Country-Wave FE	Yes	Yes	Yes	Yes
N	197268	141462	197268	141462

Note. Data: Afrobarometer Surveys 3,4,5 and 9. The outcome variables in columns 1 and 3 and columns 2 and 4 show the degree of trust in traditional leaders or in the family, respectively. They take value 0 (not at all), 1 (a little), 2 (a lot) and 3 (very much). The explanatory variable is an indicator that equals one if the respondent's religion is New Christian. Controls include age, age squared, and whether the respondent lives in a rural place. Country and Afrobarometer round FE always included. Robust standard errors are reported in parentheses.

Table C45: Treatment effects on protection against witchcraft, by NCC conversion

	Protection from husband (2025)		Protection from church (2025)		Protection from family (2025)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	-0.0636*	-0.0807**	-0.0634**	-0.0885***	-0.0710**	-0.0844**
	(0.0330)	(0.0375)	(0.0276)	(0.0315)	(0.0348)	(0.0403)
Treat × Conv. to NCC (2024)		0.0950		0.114*		0.0985
		(0.0835)		(0.0685)		(0.0851)
Mean control	0.655	0.650	0.820	0.817	0.509	0.506
p-val: Treat + Treat x Conv = 0		0.848		0.676		0.851
R-squared	0.167	0.166	0.159	0.168	0.116	0.117
N	836	820	836	820	836	820

NOTE. Data: First-hand survey data from southern Benin. The sample is restricted to married women. The outcome variables are indicators equal to one if the respondent reports being protected against witchcraft by her husband (columns 1–2), by her church (columns 3–4), or by her family — either her own family of origin or her husband's family — (columns 5–6). Odd columns report average treatment effects (ITT). Even columns interact treatment with an indicator for having ever converted to a new Christian church. The row *p-val: Treat + Treat x Conv = 0* reports the p-value of the linear-combination test that the total treatment effect for women who converted to a new Christian church equals zero. The number of observations differs between odd and even columns because the outcomes are measured in the 2025 follow-up survey, while the interaction variable (ever converted to a new Christian church) is measured in the 2024 endline survey. Controls include age, education at baseline, NCC membership at baseline, marital status, and baseline isolation. Department fixed effects are included. Robust standard errors in parenthesis.